Agenda

- SCS Sequences
- SCS Search Sequence based on Planning Parameters
- SCS Optimization capabilities

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SCS Planning Sequence

- Data delete and loading
- WIP Planning
- Demand Planning
- Safety Stock Planning
- Replanning
- Excess Clean-up *
- RCA
- Pegging

* Excess cleanup can happen after every phase of planning WIP/Demand/SS/Replan and is configurable in SCS.No carry violations are also handled during this phase.

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Tenant Setup

- We should have a tenant setup with reference model config available in the course.
- For this, Download the config available in the course into your local system.
- Once the config is downloaded, port this config into the target tenant.
  Go to Tenants→ Restore→ Upload backup Zip File→ Submit.
- After successfully porting the config, next step is to change the tenant setting.For this go to Deployment→ Tenant System Settings and search for the following flags:
  - PropagateNullValues = True.
  - SelfServiceMode = True.
  - Btree = inmem.
  - LockFreeMemberCrud = true.
  - FilterScopeForMemberEditReplan = True.

- Now your tenant is ready with reference model config for using.

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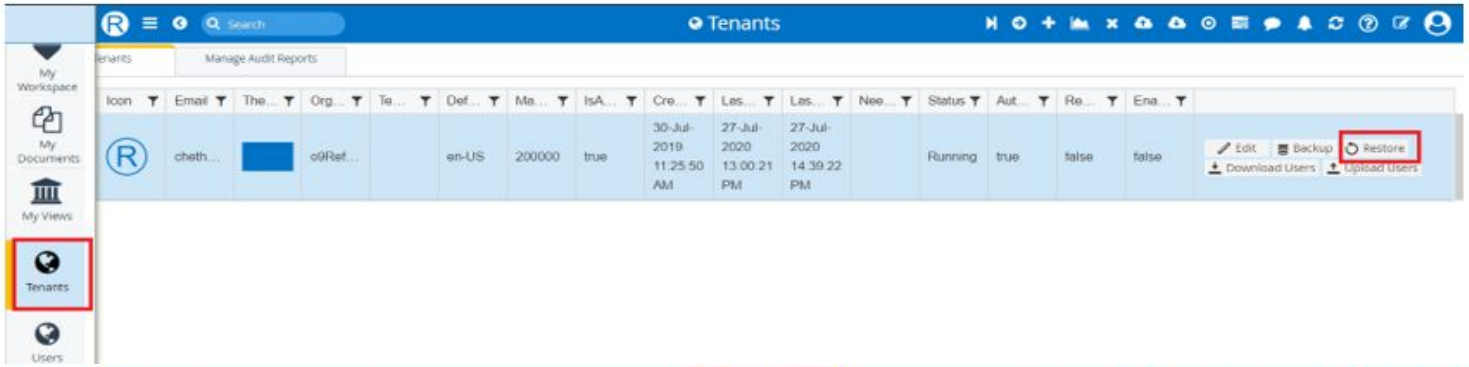
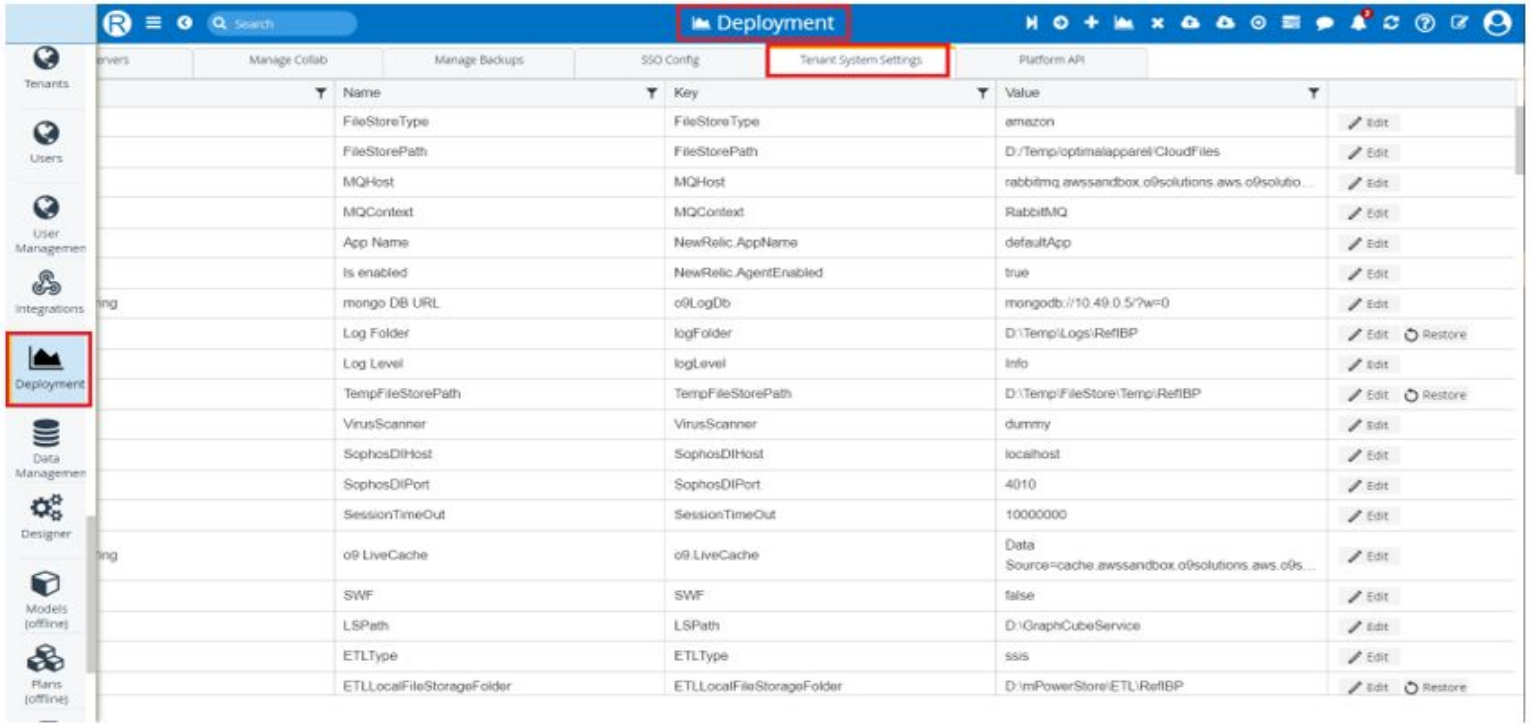
Tenant Setup (Demo)





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Initial Setup for Plan Run

- Download Data Folder and Unzip it.
- Upload the Dimension files from Basic folder.
- Upload the Fact files from the Basic folder.
- Run Create Network
- Run Demand Netting
- Run Demand Prioritization

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Initial Setup for Plan Run(Demo)

- Download the data files available in the course and unzip it.
- Upload all the dimension files from the folder basic into your tenant.
- Upload all the fact files from the folder basic into your tenant.
- Go to W Network Management -> Supply Chain Network -> Network page.
- Run create network using the action button with empire symbol, This will show you the supply chain network.
- Go to W Netting and Prioritization -> Forecast Netting at Month -> Netting Outputs -> Demand Netting Output.
- Run demand netting using the action button Run Demand Netting (Month) in the bottom grid.
- Go to Demand Prioritization -> Setup -> Priority Group Rule Details.
- Run demand Prioritization using the action button Run Advanced Demand Prioritization.

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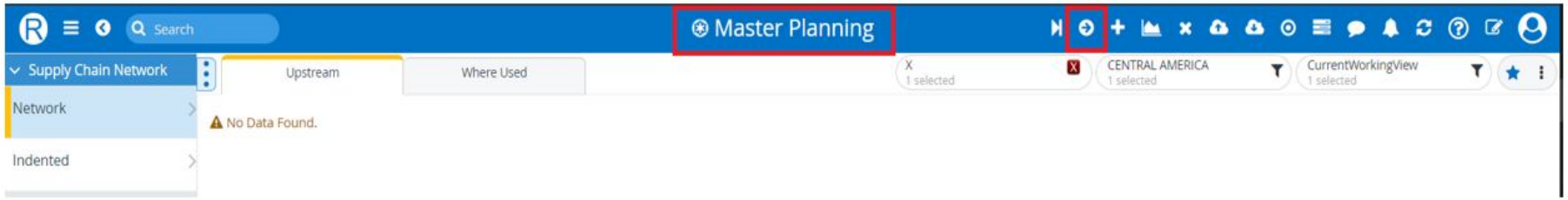
Plan Run (Demo)

- Go to Master Planning Workspace and execute action button Run Plan Weekly as shown below.



- Go to Master Planning Workspace
- Run the plan using the action button Run Plan Weekly.

- Go to Logs, To see the logs for deleting entire output tables and loading the input tables as shown in the next slides.
- After Delete Data and Load Data, SCS Plans for the WIP's.

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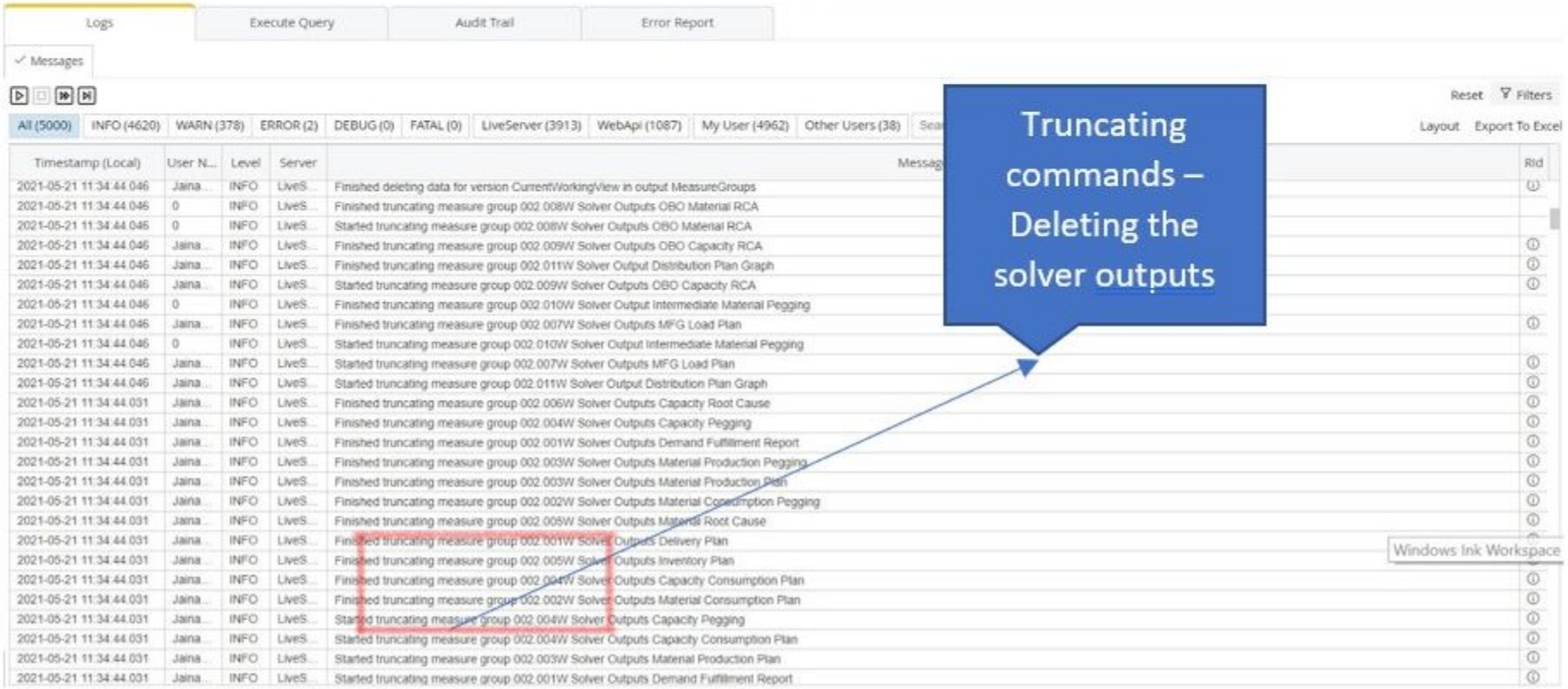
Data Delete and loading

- SCS does a full delete of existing output tables (Delivery Plan, Inventory Plan, Flow Plans, Peggings etc).
- SCS reloads all the input data tables (Network, Plan Parameters, Demand, Supply)



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Data Delete and loading

✓ Messages

Reset | Filters

All (5000) | INFO (4620) | WARN (378) | ERROR (2) | DEBUG (0) | FATAL (0) | LiveServer (3913) | WebApi (1087) | My User (4962) | Other Users (38)

Layout | Export To

Search in Messages

Timestamp (Local)	User N...	Level	Server	Messa	Rid
2021-05-21 11:34:44.343	Jaina...	INFO	LiveS...	[Segment : 0]. Finished segregating demands.	
2021-05-21 11:34:44.343	Jaina...	INFO	LiveS...	[Segment : 0]. Started segregating demands.	
2021-05-21 11:34:44.343	Jaina...	INFO	LiveS...	Depth of Supply Chain is : 4	
2021-05-21 11:34:44.343	Jaina...	INFO	LiveS...	[Segment : 0]. Finished sorting supply chain.	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	[Segment : 0]. Started sorting supply chain.	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	Memory utilization of plugin instance WSupplyChainSolver : 568 MB	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	[Segment : 0]. Number of Storage Consumption Edges : 0	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	[Segment : 0]. Number of Capacity Consumption Edges : 1	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	[Segment : 0]. Number of Material Consumption Edges : 3	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	[Segment : 0]. Number of Material Production Edges : 3	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	[Segment : 0]. Number of Storage nodes : 0	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	[Segment : 0]. Number of Capacity nodes : 1	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	[Segment : 0]. Number of Activity nodes : 5	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	[Segment : 0]. Number of Material nodes : 4	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	[Segment : 0]. Number of Orders : 1	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	[Segment : 0]. Number of Demands : 1	
2021-05-21 11:34:44.328	Jaina...	INFO	LiveS...	[Segment : 0]. Finished loading Data	
2021-05-21 11:34:44.265	Jaina...	INFO	LiveS...	Finished reading 0 records from Fact001.015WSolverInputsResourceGroupPriority	

Loading data to the SCS Solver

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WIP Planning

Steps

- Solves from upstream to downstream i.e upstream WIP will be solved first then downstream WIP.
- Consumes upstream Material/Capacity similar to Demands and plans for the WIP.
- During the upstream traversal checks for feasibility.
- Feasible WIPs will be converted into inventory.
- Converted inventory is available for consumption for downstream WIPs and Demands.

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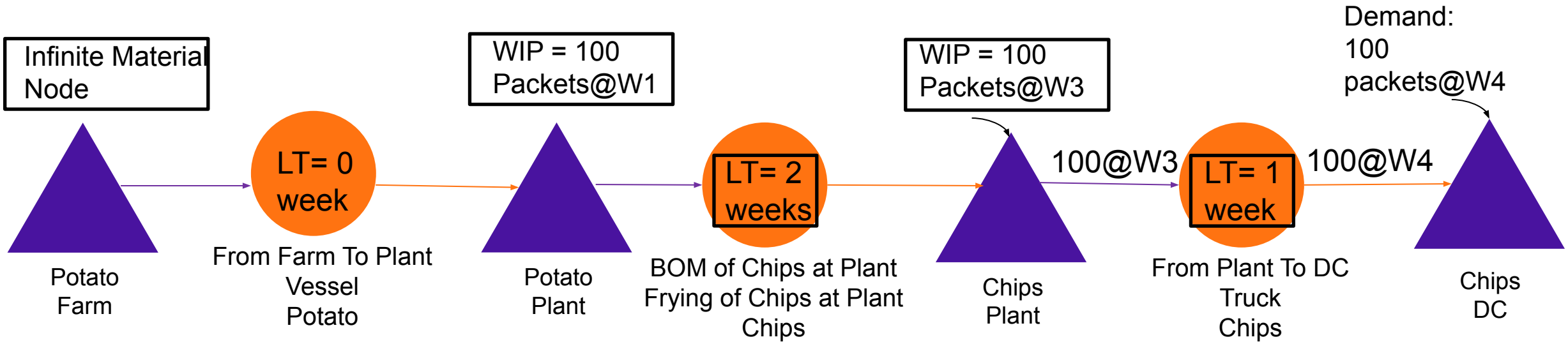
WIP Planning

Consider the below Network

- WIPs will be solved from upstream to downstream, so Potato@Plant WIP of 100 units will be planned first.
- Potato@Plant WIP is feasible wrt both LTs and material(upstream Potato@farm marked as infinite).
- WIP at Chips@W3 will be planned next, it will consume the inventory available at Potato@Plant which was the result of the WIP Planning of Potato@Plant
- After the WIP Planning, Inventory of 100 is available at Chips@Plant@W3 for downstream consumption of demand.
- As shown in below network picture, we have 2 WIP's for which SCS will plan.



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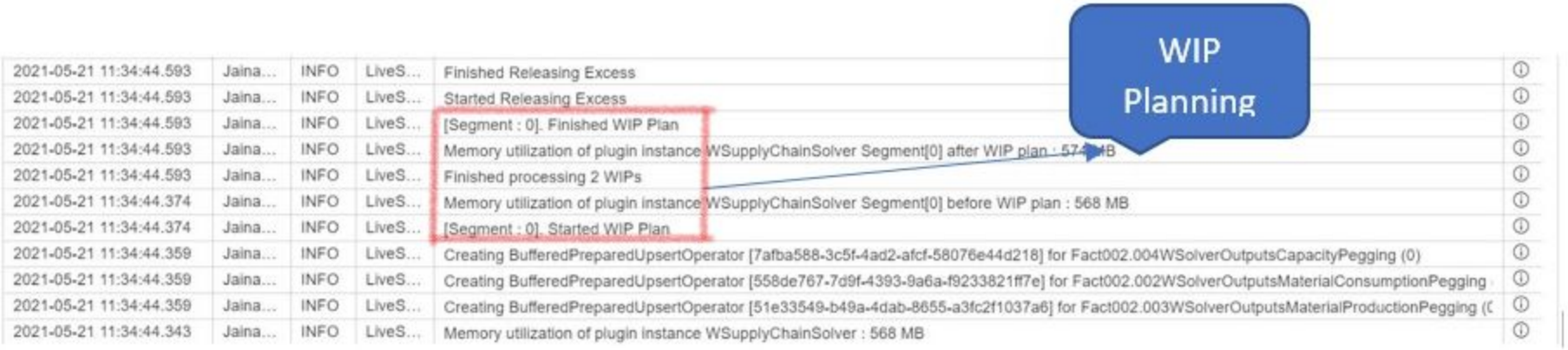
WIP Planning

Go to logs to see the logs for WIP Planning as shown in the screenshot.



2021-05-21 11:34:44.593	Jaina...	INFO	LiveS...	Finished Releasing Excess
2021-05-21 11:34:44.593	Jaina...	INFO	LiveS...	Started Releasing Excess
2021-05-21 11:34:44.593	Jaina...	INFO	LiveS...	[Segment : 0]. Finished WIP Plan
2021-05-21 11:34:44.593	Jaina...	INFO	LiveS...	Memory utilization of plugin instance WSupplyChainSolver Segment[0] after WIP plan : 574 MB
2021-05-21 11:34:44.593	Jaina...	INFO	LiveS...	Finished processing 2 WIPs
2021-05-21 11:34:44.374	Jaina...	INFO	LiveS...	Memory utilization of plugin instance WSupplyChainSolver Segment[0] before WIP plan : 568 MB
2021-05-21 11:34:44.374	Jaina...	INFO	LiveS...	[Segment : 0]. Started WIP Plan
2021-05-21 11:34:44.359	Jaina...	INFO	LiveS...	Creating BufferedPreparedUpsertOperator [7afba588-3c5f-4ad2-afcf-58076e44d218] for Fact002.004WSolverOutputsCapacityPegging (0)
2021-05-21 11:34:44.359	Jaina...	INFO	LiveS...	Creating BufferedPreparedUpsertOperator [558de767-7d9f-4393-9a6a-f9233821ff7e] for Fact002.002WSolverOutputsMaterialConsumptionPegging
2021-05-21 11:34:44.359	Jaina...	INFO	LiveS...	Creating BufferedPreparedUpsertOperator [51e33549-b49a-4dab-8655-a3fc2f1037a6] for Fact002.003WSolverOutputsMaterialProductionPegging (((
2021-05-21 11:34:44.343	Jaina...	INFO	LiveS...	Memory utilization of plugin instance WSupplyChainSolver : 568 MB

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Demand Planning

Next the solver will plan for the demands.
As we have only one demand, we can see in the logs that solver planned for that demand.

2021-05-21 11:34:44.703	Jaina...	INFO	LiveS...	Finished Releasing Excess
2021-05-21 11:34:44.687	Jaina...	INFO	LiveS...	Started Releasing Excess
2021-05-21 11:34:44.687	Jaina...	INFO	LiveS...	[Segment : 0]. Finished plan generation
2021-05-21 11:34:44.687	Jaina...	INFO	LiveS...	Memory utilization of plugin instance WSupplyChainSolver Segment[0] after plan genera
2021-05-21 11:34:44.687	Jaina...	INFO	LiveS...	Planned 1 demands out of 1 demands for Plan Generation Process.
2021-05-21 11:34:44.657	Jaina...	INFO	WebApi	WebApi served the request: /api/uiusage/usage in : 17ms with status : NoContent
2021-05-21 11:34:44.657	Jaina...	INFO	WebApi	WebApi served the request: /api/crud/tenants/9914/ls/systemsettings/settingName/UIUsageStatsEnabled in : 8ms with status : OK
2021-05-21 11:34:44.641	Jaina...	INFO	WebApi	WebApi received the request: GET /api/crud/tenants/9914/ls/systemsettings/settingName/UIUsageStatsEnabled
2021-05-21 11:34:44.641	Jaina...	INFO	WebApi	WebApi received the request: POST /api/uiusage/usage
2021-05-21 11:34:44.610	Jaina...	INFO	LiveS...	Memory utilization of plugin instance WSupplyChainSolver Segment[0] before plan generation : 574 MB
2021-05-21 11:34:44.610	Jaina...	INFO	LiveS...	[Segment : 0]. Started plan generation

Demand Planning

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Demand Planning

- When SCS is invoked, it follows the following sort order while solving for demands
  a. Demand Priority
     - User supplied input or prioritization number arrived based on prioritization schema.
     - Lower the number, higher the priority
  b. Demand Due Date
     - In case of a priority tie or NULL value, sort the order by due date in ascending order
     - Earlier due date demand gets planned first
  c. Demand Quantity
     - In case of a tie-on priority and due date, sort the order by quantity in ascending order
     - Lower quantity demand gets planned first. This increases the possibility of meeting more demands in full. For e.g., consider 2 demands of quantity 50 and 100 with same priority(or NULL priority) with same due date, the demand quantity 50 will be solved first then 100.
  d. If all the above are same, then SCS follows an internal sort order for tie breaking the demands.

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Demand Planning(Demo)

- Upload the fact files from the folder Demand Planning Sequence.
- Go to W Netting and Prioritization ->Forecast Netting at Month ->Netting Outputs ->Demand Netting Output.
- Run demand netting using the action button Run Demand Netting (Month) in the bottom grid.
- Go to Demand Prioritization ->Setup ->Priority Group Rule Details.
- Run demand Prioritization using the action button Run Advanced Demand Prioritization.
- Run the plan again using the action button run plan weekly in master planning workspace.
- As we have enabled the demand debug flag (Demand Trace), We can see the order of demands in which the solver plans in the logs.
- Switch to logs to see the planning sequence as shown in the next slides.

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Demand Planning

Outputs | Debugging

All : Item 2 selected | Customer_1 1 selected | Planning Horizon 1 selected | CurrentWorkingView 1 selected

Showing 4 rows of data.

Filters ▾ Layout Local Edit

Actions	Item	Location	Customer Group	Demand Type	DemandID	Week	Demand Quantity	W Demand Priority R Output	W Demand Priority R Output Normalised	W Demand Priority Rules Used
•••	Chips	DC1	Customer_1	UNF_ORDER	Order_2	W04-2020	50	104,799	1	|Group-1,Rule-1,Rule-8
•••					Order_1	W04-2020	100	104,799	1	|Group-1,Rule-1,Rule-8
•••					Order_3	W04-2020	100	104,799	1	|Group-1,Rule-1,Rule-8
•••					Order_4	W05-2020	100	105,799	2	|Group-1,Rule-1,Rule-8

- Consider Order 1 and 2
- As their demand priorities are same, solver will check for demand due date.
- As demand due date is also same, solver looks for quantity and plans for order 2 first as its quantity is 50 which is less than 100.
- Also, as all three demand priority, demand due date and quantity are same for order 1 and order 3, SCS follows an internal sort order.So, SCS planned for order 1 first and then order 3.
- Sequence of Planning for demands are => Order 2, Order 1, Order 3, Order 4

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Demand Planning Sequence

Sequence of demand planning

- ○JIT
- ○Build Ahead
- ○Build Late

- ●JIT
  - ○Use BOM/Lanes for upstream traversal by offsetting the LT. Look for supply at the immediate upstream node
  - ○Check for current capacity availability in that time bucket for that activity node
  - ○If material is available, provide the material and update the inventory profile
  - ○Else recurse upstream asking capacity and material till most upstream node is reached
  - ○Objective is to meet demand on the requested date

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Demand Planning Sequence

- Build Ahead
  - If JIT bucket is not feasible (steps above), check for capacity in earlier buckets (up to build ahead limit) and follow the steps above
  - Explore primary path (based on Activity Priority) before checking alternates.

- Build Late
  - In case demand is not satisfied on time or completely, it builds late up to the buckets specified (one at a time)
  - Objective here is to minimize delay. Least late supplies among all late supplies is selected
  - For each bucket, it will search for alternate activity (based on Priority) for reducing delay
  - If supply is not available in any of the lateness bucket, demand will be shorted

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Demand Planning Sequence(Demo)

```
Completed Planning Demand of quantity 50]
    Produced (50) units of (Chips, DC1) @ Bucket (W04-2020)]
        Produced (50) units at (LANE_From_Plant1_To_DC1_Using_TRUCK, TRUCK, Chips) @ Bucket (W04-2020) . isMatConstrained = False, return = False]
            Produced (50) units of (Chips, Plant1) @ Bucket (W03-2020)]
            [Requested (50) units of (Chips, Plant1) @ Bucket (W03-2020)
        [Requested (50) units at (LANE_From_Plant1_To_DC1_Using_TRUCK, TRUCK, Chips) @ Bucket (W04-2020)
    [Requested (50) units of (Chips, DC1) @ Bucket (W04-2020)
[Planning Demand : (Order_2 for Material Chips, DC1 and Quantity 50 with Priority 1 @ Bucket W04-2020)
```

- As Demand Trace is enabled we can see the trace of the demand in the logs.
- In the above logs we can see the solver planning for the demand Order_2.
- For Order_2, solver is requesting for 50 units of Chips at DC1 @ W4 and which got produced in W4.
- Demand is satisfied in the JIT Bucket.

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Demand Planning Sequence

```
Completed Planning Demand of quantity 100]
    Produced (50) units of (Chips, DC1) @ Bucket (W05-2020)]
        Produced (50) units at (LANE_From_Plant1_To_DC1_Using_TRUCK, TRUCK, Chips) @ Bucket (W05-2020) . isMatConstrained = False, return = False]
            Produced (50) units of (Chips, Plant1) @ Bucket (W04-2020)]
                Produced (50) units at (BOM-V1.1-for-Chips-at-Plant1, Routing-for-Brand_F_Line_1-at-Plant1, Chips) @ Bucket (W04-2020) . isMatConstrained = False, return = False]
                    Produced (50) units of (Potato, Plant1) @ Bucket (W02-2020)]
                    [Requested (50) units of (Potato, Plant1) @ Bucket (W02-2020)
                [Requested (50) units at (BOM-V1.1-for-Chips-at-Plant1, Routing-for-Brand_F_Line_1-at-Plant1, Chips) @ Bucket (W04-2020)
            [Requested (50) units of (Chips, Plant1) @ Bucket (W04-2020)
        [Requested (50) units at (LANE_From_Plant1_To_DC1_Using_TRUCK, TRUCK, Chips) @ Bucket (W05-2020)
    [Requested (50) units of (Chips, DC1) @ Bucket (W05-2020)
    Produced (50) units of (Chips, DC1) @ Bucket (W04-2020)]
        Produced (50) units at (LANE_From_Plant1_To_DC1_Using_TRUCK, TRUCK, Chips) @ Bucket (W04-2020) . isMatConstrained = True, return = True]
            Produced (50) units of (Chips, Plant1) @ Bucket (W03-2020)]
                Produced (50) units at (BOM-V1.1-for-Chips-at-Plant1, Routing-for-Brand_F_Line_1-at-Plant1, Chips) @ Bucket (W03-2020) . isMatConstrained = True, return = True]
                    Produced (50) units of (Potato, Plant1) @ Bucket (W01-2020)]
                        Produced (0) units at (LANE_From_Supplier_1_To_Plant1_Using_VESSEL, VESSEL, Potato) @ Bucket (W01-2020) . isMatConstrained = True, return = True]
                        [Requested (50) units at (LANE_From_Supplier_1_To_Plant1_Using_VESSEL, VESSEL, Potato) @ Bucket (W01-2020)
                    [Requested (100) units of (Potato, Plant1) @ Bucket (W01-2020)
                [Requested (100) units at (BOM-V1.1-for-Chips-at-Plant1, Routing-for-Brand_F_Line_1-at-Plant1, Chips) @ Bucket (W03-2020)
            [Requested (100) units of (Chips, Plant1) @ Bucket (W03-2020)
        [Requested (100) units at (LANE_From_Plant1_To_DC1_Using_TRUCK, TRUCK, Chips) @ Bucket (W04-2020)
    [Requested (100) units of (Chips, DC1) @ Bucket (W04-2020)
[Planning Demand : (Order_3 for Material Chips, DC1 and Quantity 100 with Priority 1 @ Bucket W04-2020)
```

- In the above logs we can see the solver planning for the demand Order_3.
- For Order_3, solver is requesting for 100 units of Chips at DC1 @ W4 and which produced 50 units in W4 and 50 units in W5.
- Demand is satisfied by 50 units in JIT Bucket and 50 units in late bucket as there is no supply in previous buckets.

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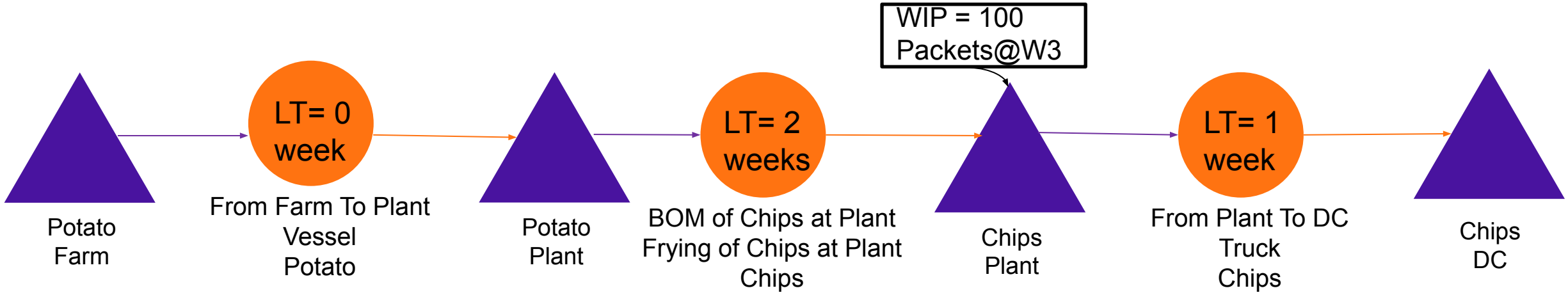
Demand Planning (Fair Share)

- Based on the prioritization schema, if two demands are assigned the same priority number and there is a supply constraint or capacity constraint then SCS has a fair share mechanism by which it can allocate the material/capacity in a fair share manner.
- Consider the below network and Input Demand.
- Fair share factor/Slice factor is set to 2.

DemandID	Item	Location	Customer	Week	Demand Quantity	Demand Priority
Order1	Chips	DC	C1	W4	100	1
Order2	Chips	DC	C1	W4	80	1



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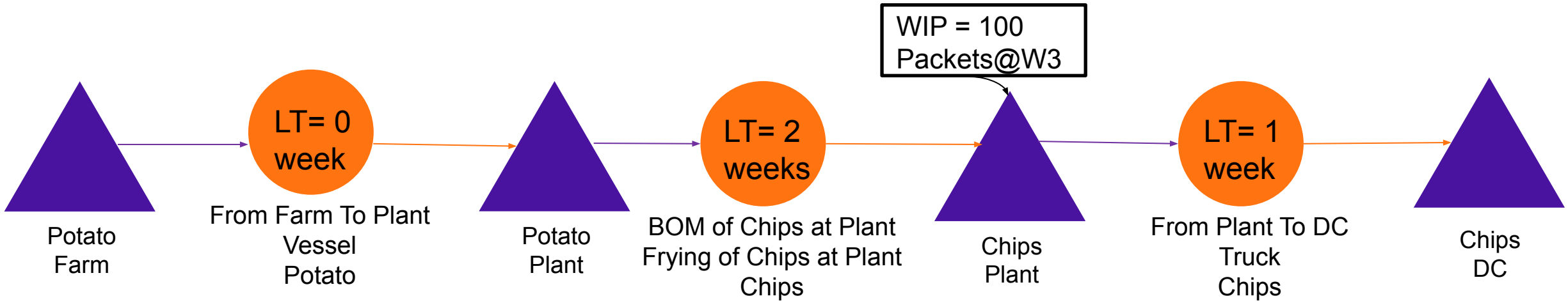
Demand Planning (Fair Share)

- SCS will slice the demand to 2 parts (Order1 - 50 and 50, Order2 - 40 and 40).
- Lower quantity slice will be picked up first to be solved. So the solve sequence will be 40,50,40,50.
- Final delivery plan output will be

DemandID	Item	Location	Customer	Week	Demand Quantity	Demand Priority	Met OnTime	Short	Late
Order1	Chips	DC	C1	W4	100	1	50	50	
Order2	Chips	DC	C1	W4	80	1	50	30	



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Demand Planning (Fair Share) (Demo)

- Create a Global scenario using the plus action button from any workspace with the name DemandFairShare.
- Upload the files from the folder Demand Fair Share.
- In this we have set the demand fair share as 2 using Fact.1SDPlanParameters file.
- Demand Inputs for understanding demand fair share functionality.

Outputs | Debugging

All 1 selected | Customer_1 1 selected | Planning Horizon 1 selected | DemandFairShare 1 selected

Showing 4 rows of data.

Filters | Layout | Local Edit

Actions	Item	Location	Customer Group	Demand Type	DemandID	Week	Demand Quantity	W Demand Priority R Output	W Demand Priority R Output Normalised	W Demand Priority Rules Used
•••	Chips	DC1	Customer_1	UNF_ORDER	Order_2	W04-2020	50	104,799	1	|Group-1,Rule-1,Rule-8
•••					Order_3	W04-2020	80	104,799	1	|Group-1,Rule-1,Rule-8
•••					Order_1	W04-2020	100	104,799	1	|Group-1,Rule-1,Rule-8
•••					Order_4	W05-2020	100	105,799	2	|Group-1,Rule-1,Rule-8

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Demand Fair Share

- Now the outputs are as follows for the demands shown above
- Go to W Plan Validations->Plan Outputs->Fulfillment Report Page to see the below report

Fulfillment Report

Chips 1 selected | DC1 1 selected | All : to.[Sales Domain].[Cust... 2 selected | DemandFairShare 1 selected

Showing 5 rows of data.

Download | Filters | Layout

Actions	DemandID	Demand Type	Demand Item	Demand Location	Customer Group	Supply Week	Demand Week	Demand Fulfilled Qty
•••	Order_1	UNF_ORDER	Chips	DC1	Customer_1	W04-2020	W04-2020	70
•••						W05-2020	W04-2020	30
•••	Order_2	UNF_ORDER	Chips	DC1	Customer_1	W04-2020	W04-2020	50
•••	Order_3	UNF_ORDER	Chips	DC1	Customer_1	W04-2020	W04-2020	80
•••	Order_4	UNF_ORDER	Chips	DC1	Customer_1	W05-2020	W05-2020	70

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Demand Planning (Search Sequence)

- Search sequence during demand planning is affected by various plan parameters, some of the parameters are listed below
  - Minimize Inventory Build Ahead
  - Use Bands To Minimize Build Ahead. Bands are basically flex capacity bands over and above the base capacity for eg
    - Base Capacity = 1 Shift
    - Flex Band1 Capacity = 1 additional shift over and above the Base Capacity
    - Flex Band2 Capacity = 1 more additional shift over and above Base + Flex Band1

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Demand Planning (Search Sequence)

Minimize Inventory Build Ahead	Use Bands To Minimize Build Ahead	Search sequence
True	True	JIT Primary Base □ JIT Flex □ JIT Alternate Base □ JIT Alternate Flex □ JIT -n Primary □ JIT-n Flex Capacity □ JIT-n Alternate Base□JIT-n Alternate Flex and so on
True	False	JIT Primary Base □ JIT Alternate Base □ JIT Primary Flex □ JIT Alternate Flex □ JIT-n Primary Base □ JIT-n Alternate Base □ JIT-n Primary Flex □ JIT-n Alternate Flex and so on
False	True	JIT Primary Base □ JIT Primary Flex □ JIT-n Primary Base □ JIT-n Primary Flex □ JIT Alternate Base □ JIT-n Alternate Base □ JIT Alternate Flex □ JIT-n Alternate Flex
False	False	JIT Primary Base □ JIT-n Primary Base □ JIT Alternate Base □ JIT-n Alternate Base □ JIT Primary Flex □ JIT-n Primary Flex □ JIT Alternate Flex □ JIT-n Alternate Flex

-n Primary -- Primary supply available ‘n’ time buckets before base time bucket. This is to ensure build ahead happens

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Safety Stock Planning

- Steps
  - Input is either BOS(Buckets of Supply) or Quantity for a material node.
  - SS planning happens after Demand planning.
  - SS Planning will happen based on Material Node SS Priority defined.
    - If no priority is defined then SCS does SS Planning from downstream to upstream.
    - Within the same level, the SCS follows an internal sort order
  - Pulls in Production/Consumption flow plans based on upstream material/capacity.

- Inventory Plan Type Parameter
  - Quantity
  - WOS
  - Quantity and WOS
  - Disabled

- Inventory Plan Policy Parameter
  - Target Level Based
  - Min Max Based

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Safety Stock Planning Parameters

Navigate to W Supply Admin -> Plan Inputs -> Plan Parameters -> Global Plan Parameters
Below shown are the different solver parameters for Safety Stock Planning

Plan Inputs
Plan Parameters
Bulk Merge Inputs

Global Plan Parameters | Constrained Plan Paramet...

CurrentWorkingView
1 selected

Data successfully saved.

Filters | Layout | Local Edit

Version Name	Data	Values
CurrentWorkingView	Plan Start Bucket	W01-2020
	Plan Current Bucket	W01-2020
	Plan End Bucket	W50-2020
	Material Constrained	TRUE
	Capacity Constrained	TRUE
	Solver Delete Data	TRUE
	Solver Demand Slice Factor	1
	Solver Demand Slice Threshold	1
	Solver Inventory Plan Bands	1
	Solver Flag Inventory Plan Policy	Target Level Based
	Solver Flag Inventory Plan Strategy	MOVE_IN
	Solver Flag Inventory Plan Type	Time Bucket Cover Based
	Solver Flag Avoid Build Ahead In Inventory Plan	☐
	Solver Flag Respect Freeze Window For WIPs	☐
	Solver Flag Minimize Inventory Build Ahead	FALSE
	Solver Flag Try Material Reallocation In RCA	☑

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Replanning (Optimization)

- In some cases, because of SS Planning or Excess Cleanup there are possibilities of additional unutilized supplies available in the supply chain. The Replanning phase is triggered to utilize these supplies and fulfill more demand quantity.
- Also called as Replan at Delivery Node or Optimize Demand Supportability
- In OBO solver, some short/late demands can be re-planned at the end and could possible get met on time/late
- This is performance intensive as the solver unplans the late/short demands and re-plans it by traversing the entire supply chain
- It is controlled by a flag with options
  - None (Default)
  - Replan At Delivery Node
  - Replan Short Late Demands (Recommended)

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Excess Cleanup (Optimization)

- Any excesses produced during WIP/Demand/SS Planning will be cleaned up during the excess clean up phase.
- Excesses can be generated during any phase of planning(WIP/Demand/SS). Some of the conditions which will result in excesses are -
  - Non chronological demand prioritization is with upstream lot size activity.
  - If a WIP consumes primary and alternate paths and the alternate has lotsize.
  - Excess generated during SS planning

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Pegging

- Capability to see the path of fulfillment for an order in the supply chain
- Provides the details of what supply and capacity is utilized on what buckets to satisfy the demand
- Flow Plan & Pegging
  - Flow Plan provides information about aggregate flow of goods across material nodes
  - Pegging provides order level information with material and capacity consumption by buckets
- Pegging is a post-processing that is computed after planning
- It follows a FIFO(First In First Out) pegging to match producers to consumers on any node
- Any excess or unutilized/excess supply produced is pegged to a default demand “Inventory Demand”.
- Reasons for Pegging to be post planning and not during planning will be taken up in the next slides.

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Material Profile and Material Reallocation (Optimization)

- Why is pegging done after planning in SCS?
  - Pegging is done after planning because SCS does smart material reallocation. If pegging was done during planning(also known as hard pegging) then overall plan quality will get impacted.

- What is Material Profile?
  - Material profile is the available supply at particular Material Node across the time horizon.
  - Material profile is the sum of available supply(OnHand, ERs, WIPs) + any new flow plans generated by SCS.
  - As elaborated in previous modules, SCS does backward propagation of demand.
  - While doing backward propagation, SCS is only looking at the material profile and based on the availability of material it does either fulfills the demands or propagates upstream to fetch new supplies.

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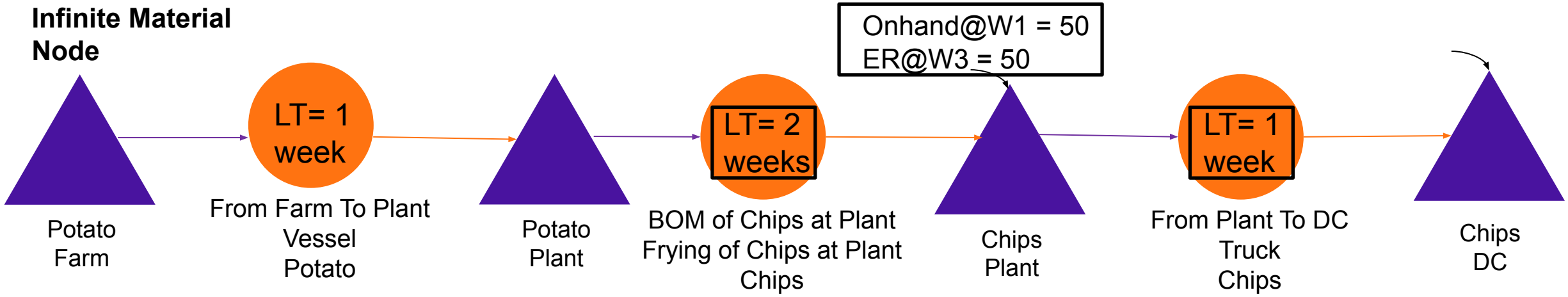
Material Profile and Material Reallocation (Optimization)

Consider the below supply chain and demands. Both the demands are orders but the demand priorities are non-chronological.

What is non-chronological prioritization?

Demand Priority numbers are arrived based on business rules. Based on the business rules it is possible that some orders which are in the later buckets can get a higher priority than some other demands which are in the nearer buckets. In this case, because C1 is a strategic customer Order 1 is given a higher priority than Order 2 even if Order 1 is later(W6) than Order 2(W4).

Demand Type	Demand ID	Item	Location	Customer	Demand Week	Demand Quantity	Demand Priority
Order	Order 1	Chips	DC	C1	W6	100	1
Order	Order 2	Chips	DC	C2	W4	100	2



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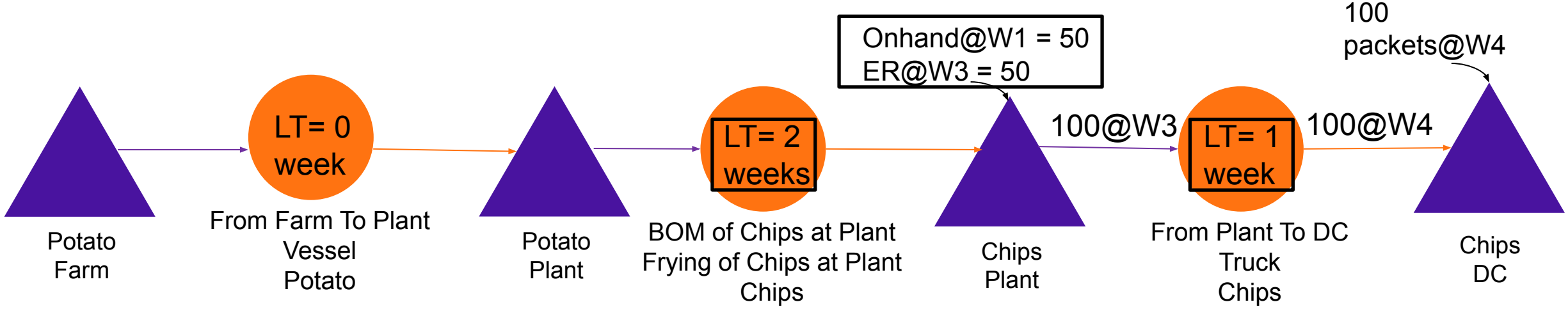
Material Profile and Material Reallocation (Optimization)

Consider the below Supply Chain
In this case, SCS while traversing upstream looks at the material profile of Chips@Plant and fulfils the demand.
Material profile of Chips@Plant before planning the demand

W1	W2	W3	W4	W5	W6	W7	W8
50	50	100	100	100	100	100	100

Material Profile of Chips@Plant after planning the demand

W1	W2	W3	W4	W5	W6	W7	W8
50	50	0	0	0	0	0	0



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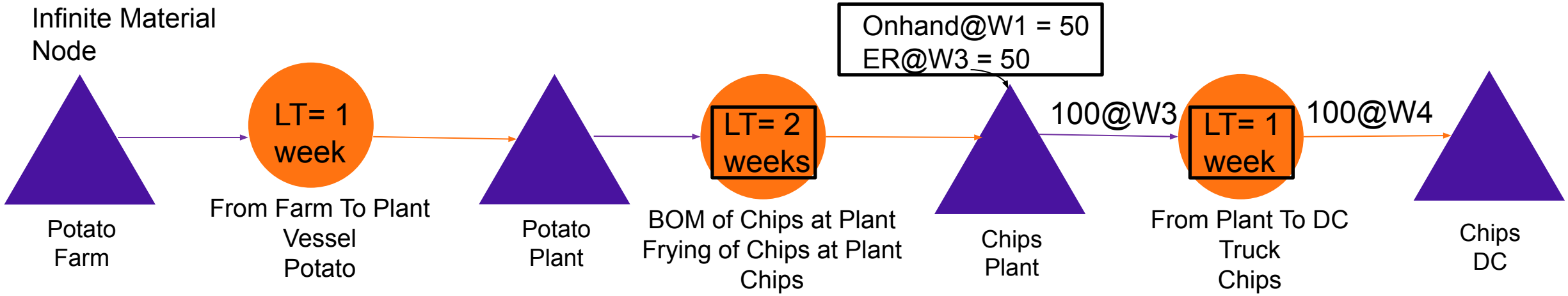
Material Profile and Material Reallocation (Optimization)

If SCS did Planning without Material Reallocation

- During Planning SCS will plan Order 1(W6) before Order 2(W4). For Order1, JIT bucket at Chips@Plant is W5. It has enough material(100) to satisfy Order 1.
- While Planning for Order 2(W4), JIT bucket at Chips@Plant is W3, there is not enough material at W3 because the higher priority Order 1 consumed the supply at Chips@Plant.
- SCS traverses upstream for fulfilling Order 2 but hits a lead time constraint at Potato@Plant(W1) considering W1 as current bucket.
- If SCS followed above planning logic and hard pegs supplies to demand after planning then we will end up in suboptimal plan. Planning result will be Order 1 would be fully met and Order 2 will be fully short.



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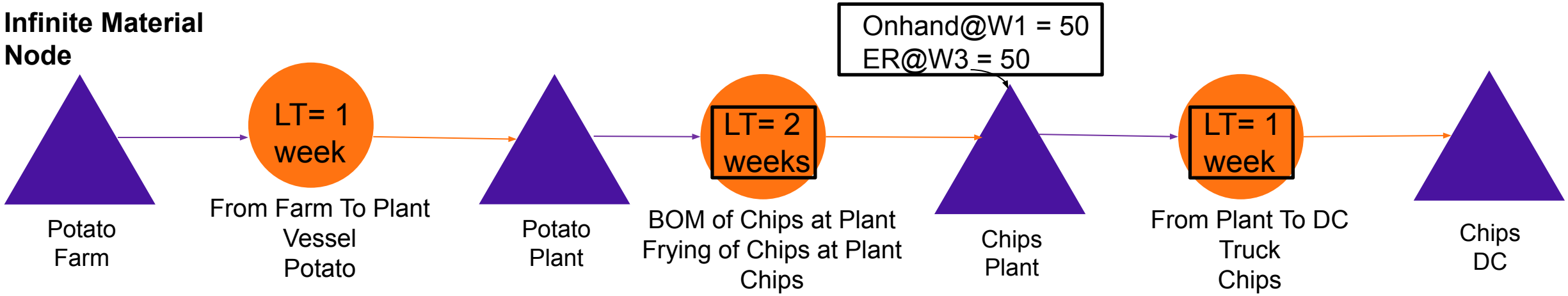
Material Profile and Material Reallocation (Optimization)

If SCS did Planning with Material Reallocation

- During Planning SCS will plan Order 1(W6) before Order 2(W4). For Order1, JIT bucket at Chips@Plant is W5. It has enough material(100) to satisfy Order 1.
- While Planning for Order 2(W4), JIT bucket at Chips@Plant is W3, there is not enough material at W3 because the higher priority Order 1 consumed the supply at Chips@Plant.
- SCS will check if there is a possibility of material reallocation. In this case because Order 1 due date is further out in the horizon it is possible to do the reallocation. SCS will check if Order 1 can source material from the upstream nodes. In this case, it is possible and within LT for Order 1 to source it from the most upstream.
- SCS will reallocate Inventory available at Chips@Plant to Order 2 and source new material from Potato@Farm for Order. This will ensure that both Order 1 and Order 2 are satisfied.



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Setup for Plan Run

- Download Data Folder and Unzip it.
- Upload the Dimension files from Basic folder.
- Upload the Fact files from the Basic folder.
- Upload the Files from “Material Profile and Material Reallocation (Optimization)” folder & override with Null ticked.
- Run Create Network
- Run Demand Netting
- Run Demand Prioritization
- Go To : W Netting and Prioritization >> Demand Prioritization >> Output
- Change the demand priority of Order_2 to 2
- Run Plan

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Material Profile and Material Reallocation (Optimization)

Reallocation Log

0) | DEBUG (0) | FATAL (0) | LiveServer (2052) | WebApi (2948) | My User (4656) | Other Users (344) | Search in Messages

rver	Message
eS...	[Segment : 0]. Finished plan generation
eS...	Memory utilization of plugin instance WSupplyChainSolver Segment[0] after plan generation : 424 MB
eS...	Planned 2 demands out of 2 demands for Plan Generation Process.
eS...	Completed Planning Demand of quantity 100]
eS...	Produced (100) units of (Chips, DC1) @ Bucket (W04-2020)]
eS...	Produced (100) units at [LANE_From_Plant1_To_DC1_Using_TRUCK, TRUCK, Chips] @ Bucket (W04-2020) . isMatConstrained = False, return = False]
eS...	Produced (100) units of (Chips, Plant1) @ Bucket (W03-2020)]
eS...	ReAllocated (100) units of (Chips, Plant1) @ Bucket (W05-2020)
eS...	Produced (100) units at [BOM-V1.1-for-Chips-at-Plant1, Routing-for-Brand_F_Line_1-at-Plant1, Chips] @ Bucket (W05-2020) . isMatConstrained = False, return = False]
eS...	Produced (100) units of (Potato, Plant1) @ Bucket (W03-2020)]
eS...	Produced (100) units at [LANE_From_Supplier_1_To_Plant1_Using_VESSEL, VESSEL, Potato] @ Bucket (W03-2020) . isMatConstrained = False, return = False]
eS...	Produced (100) units of (Potato, Supplier_1) @ Bucket (W03-2020)]
eS...	[Requested (100) units of (Potato, Supplier_1) @ Bucket (W03-2020)
eS...	[Requested (100) units at [LANE_From_Supplier_1_To_Plant1_Using_VESSEL, VESSEL, Potato] @ Bucket (W03-2020)
eS...	[Requested (100) units of (Potato, Plant1) @ Bucket (W03-2020)
eS...	[Requested (100) units at [BOM-V1.1-for-Chips-at-Plant1, Routing-for-Brand_F_Line_1-at-Plant1, Chips] @ Bucket (W05-2020)
eS...	Trying to ReAllocate (100) units of (Chips, Plant1) @ Bucket (W05-2020)
eS...	[Requested (100) units of (Chips, Plant1) @ Bucket (W03-2020)
eS...	[Requested (100) units at [LANE_From_Plant1_To_DC1_Using_TRUCK, TRUCK, Chips] @ Bucket (W04-2020)
eS...	[Requested (100) units of (Chips, DC1) @ Bucket (W04-2020)
eS...	[Planning Demand : (Order_2 for Material Chips, DC1 and Quantity 100 with Priority 2 @ Bucket W04-2020)
eS...	Completed Planning Demand of quantity 100]
eS...	Produced (100) units of (Chips, DC1) @ Bucket (W06-2020)]
eS...	Produced (100) units at [LANE_From_Plant1_To_DC1_Using_TRUCK, TRUCK, Chips] @ Bucket (W06-2020) . isMatConstrained = False, return = False]

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Material Profile and Material Reallocation (Optimization)

Pegging Log

```
Finished pegging [Order_2, UNF_ORDER, Customer_1, W04-2020, Chips, DC1]
    Finished pegging at [Chips, DC1]. Remaining ToPeg qty = 0]
        Pegged 100 units Flow on [Production Edge : LANE_From_Plant1_To_DC1_Using_TRUCK, TRUCK, Chips, Chips, DC1] at W04-2020. lotId 000SCS_LOT]
            Finished pegging at [Chips, Plant1]. Remaining ToPeg qty = 0]
                Pegged 100 units WIP on [Production Edge : BOM-V1.1-for-Chips-at-Plant1, Routing-for-Brand_F_Line_1-at-Plant1, Chips, Chips, Plant1] at W03-2020. lotId PRO1]
                    Finished pegging at [Potato, Plant1]. Remaining ToPeg qty = 0]
                        Pegged 100 units ER at W01-2020. lotId PO_2]
                        [Pegging 100 units ER at W01-2020. lotId PO_2. q2s 0.
                    [Trying to ToPeg 100 ToSkip 0 at [Potato, Plant1] from bucket [W01-2020] and supply code [000SCS_LOT]
                [Pegging 100 units WIP on [Production Edge : BOM-V1.1-for-Chips-at-Plant1, Routing-for-Brand_F_Line_1-at-Plant1, Chips, Chips, Plant1] at W03-2020. lotId PRO1. q2s 0.
            [Trying to ToPeg 100 ToSkip 0 at [Chips, Plant1] from bucket [W03-2020] and supply code [000SCS_LOT]
        [Pegging 100 units Flow on [Production Edge : LANE_From_Plant1_To_DC1_Using_TRUCK, TRUCK, Chips, Chips, DC1] at W04-2020. lotId 000SCS_LOT. q2s 0.
    [Trying to ToPeg 100 ToSkip 0 at [Chips, DC1] from bucket [W04-2020] and supply code [000SCS_LOT]
Started pegging [Order_2, UNF_ORDER, Customer_1, W04-2020, Chips, DC1]
Finished pegging [Order_1, UNF_ORDER, Customer_1, W06-2020, Chips, DC1]
```

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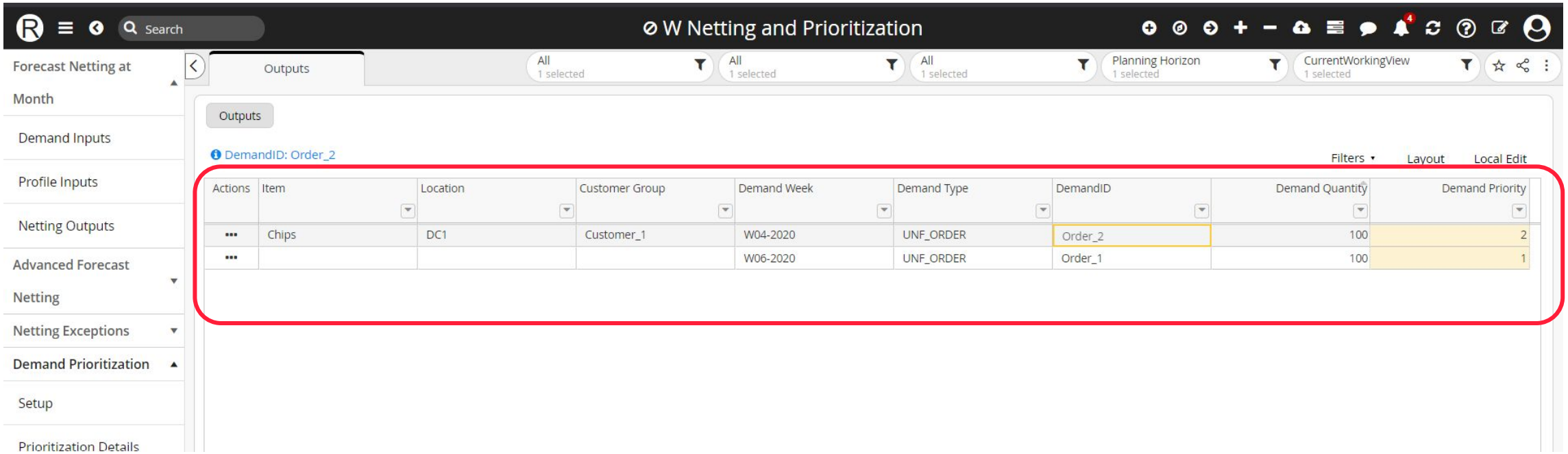
Material Profile and Material Reallocation (Optimization)

Non-chronologically prioritized orders



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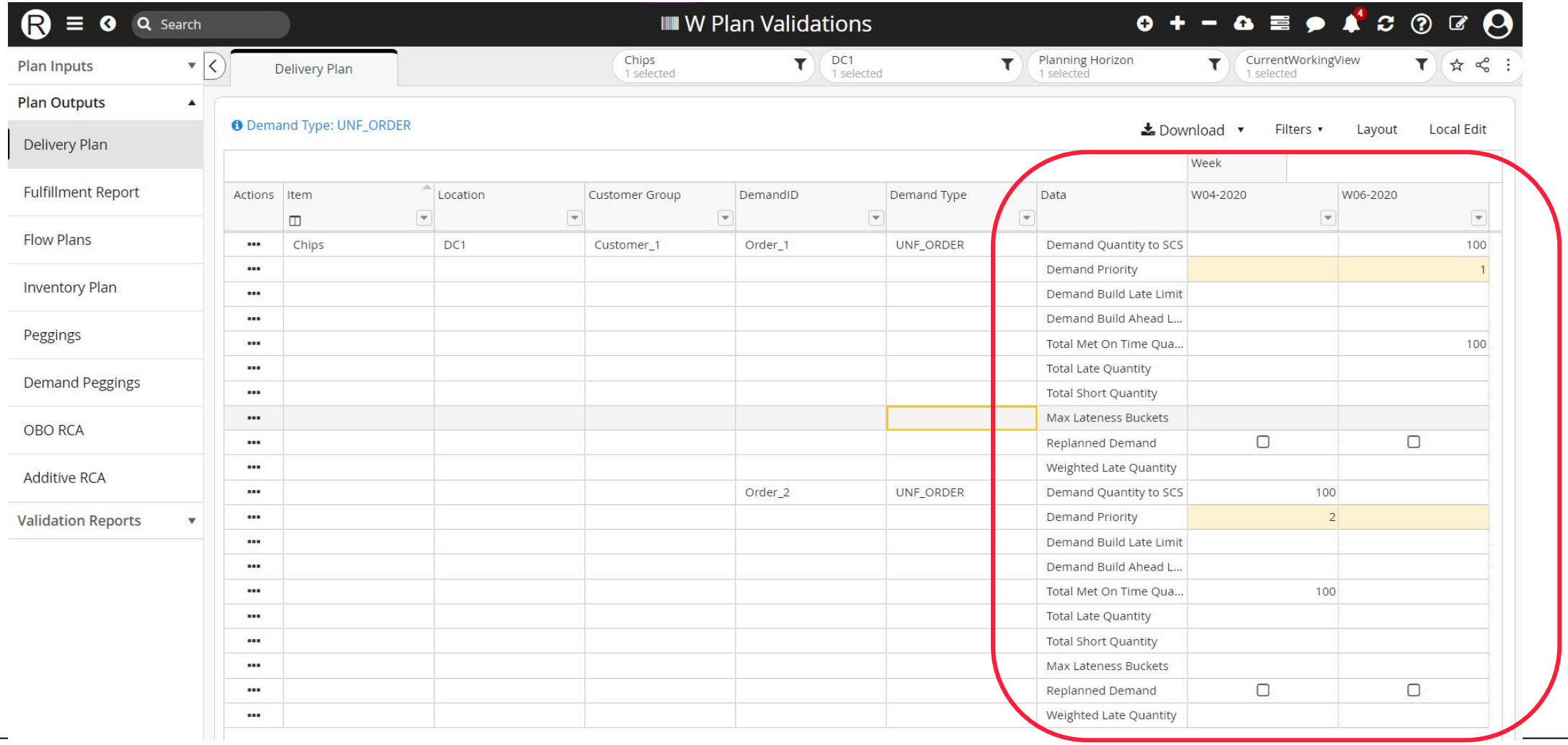
Material Profile and Material Reallocation (Optimization)

Ontime order fulfillment due to Material Reallocation



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Constraints Modelled in the Supply Chain

- Various constraints can be modelled in supply chain and SCS respects them while planning and also reports the constraints(root cause) for the same.
  - Some common Material Constraints reported by SCS
    - Lead Time
    - Frozen Period – Modelled to respect the fixed plan
    - Effectivity – For example, a lane is not active during specific season or a specific product is reaching its EOL(End of Life)
    - Holiday (Shipping, Receiving, Activity)
    - Broken Networks
  - Capacity Constraints reported by SCS
    - Capacity Short meaning both base + flex is short.
    - Regular Capacity Short meaning base capacity Short
  - Storage Constraints

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Material Profile and Material Reallocation (Optimization)

Points to be noted regarding Material Reallocation

- Material Reallocation results in a higher demand supportability.
- Material Reallocation also makes sure that inventory available at a material node is used up more efficiently.
- Material Reallocation will never happen if a higher priority demand would get shorted in order to fulfill lower priority demand.
- In some specific cases, Material reallocation will not happen and will result in hard pegging of supply with demands. These demands fall under special classes and are handled in feature call class based peggings.
- Material Reallocation is the reason why Pegging happens after Planning.

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Different Types of RCA

- SCS supports various types of RCA. Each RCA should be used based on its business use case.
  - **Additive RCA**
    - As the name suggests, Additive RCA is additive in nature.
    - Additive RCA results will be used to resolve constraints of multiple demands at one shot. For example, resolving constraint of one demand will probably resolve another related demand.
    - During Additive RCA phase, SCS does unconstrained planning and reports constraints based on when the material profile becomes negative. In other words, when SCS hits a constraint it generates an artificial supply/capacity and fulfills the demands.
    - While doing the RCA planning, SCS takes by default only the primary path and fulfills the demands in the JIT bucket.
    - Additive RCA will respect any lotsize constraints in the activity.
  - **When to use Additive RCA**
    - Additive RCA is used to resolve a set of demands and not one demand.
    - This is more tuned to a supply side planner like a material planner or a capacity planner.
    - Planner would get to know the aggregate material or capacity constraint at a particular material node or resource node in a bucket.

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Different Types of RCA

- **OBO RCA**
  - As the name suggests, OBO is order by order RCA.
  - Every demand will have its own RCA and will not depend on the other demand.
  - During OBO RCA phase, SCS does unconstrained planning and reports constraints based on when the material profile becomes negative. In other words, when SCS hits a constraint it generates an artificial supply/capacity and fulfills the demands.
  - While doing the RCA planning, SCS takes by default only the primary path and tries to fulfill the demands in the JIT bucket.

- **When to use OBO RCA**
  - OBO RCA is used to resolve one demand at a time. For example, a single high priority customer order needs to be fulfilled to meet the financial goal for this quarter.

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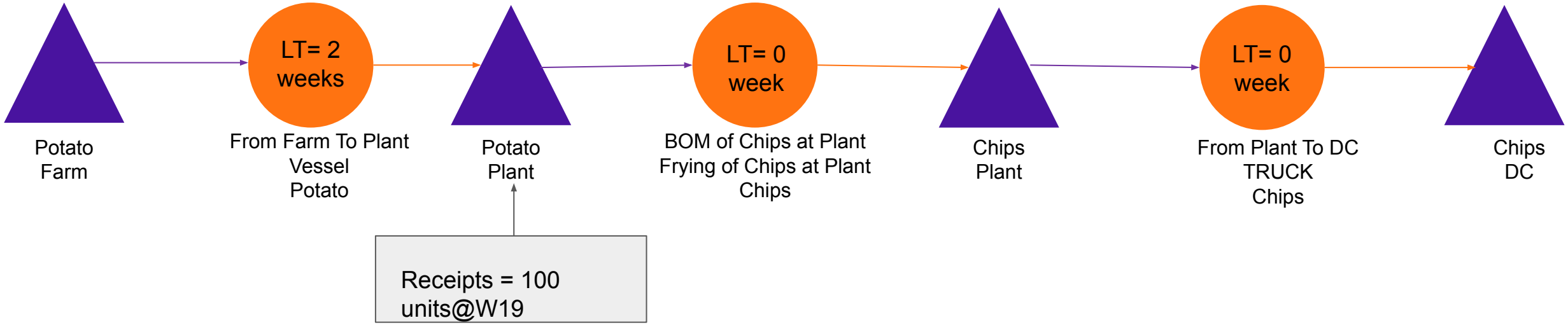
OBO vs Additive RCA - Late RCA

Consider the below network with the following details :- Current Bucket = W18

Demand Item	Demand Location	Demand Week	Demand ID	Customer ID	Demand Quantity	Demand Priority	BLL
Chips	DC	W18-2020	Order_1	Customer1	100	1	1
Chips	DC	W19-2020	Order_2	Customer1	100	2	0



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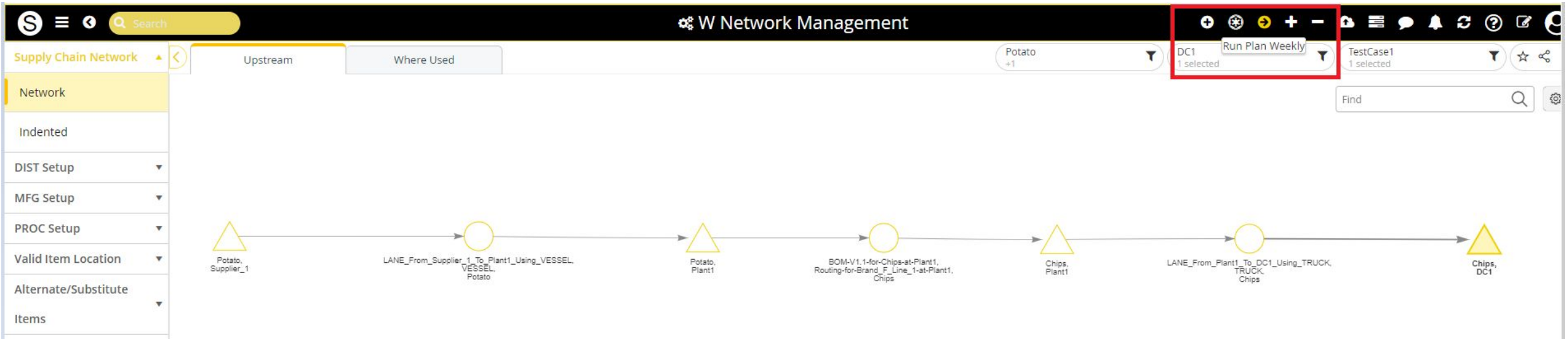
Triggering the solver run

Trigger the solver run using the “Run Basic Plan” action button in the “Network” page in the “W Network Management” workspace.



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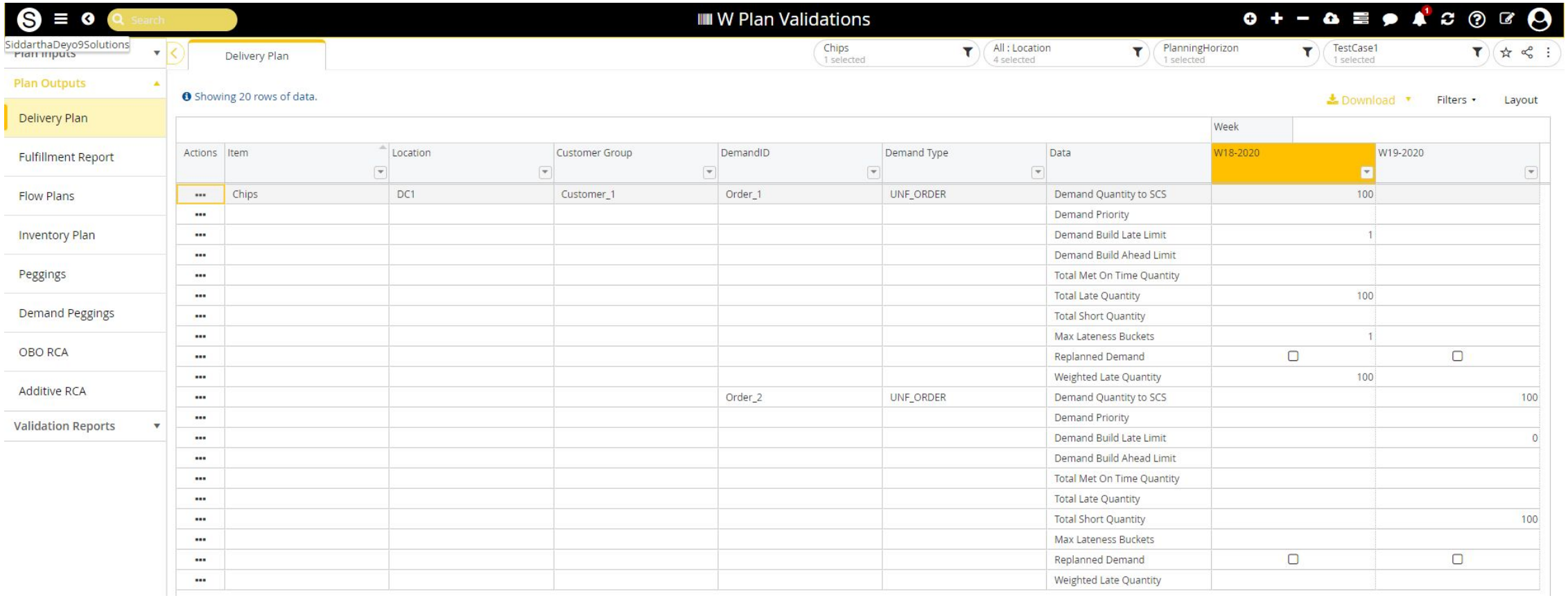
Post Solver Run Results

Once the solver run completes, navigate to the “Delivery Plan” page in W Plan Validation -> Plan Outputs _> Delivery Plan



Actions	Item	Location	Customer Group	DemandID	Demand Type	Data	Week W18-2020	W19-2020
•••	Chips	DC1	Customer_1	Order_1	UNF_ORDER	Demand Quantity to SCS	100	
•••						Demand Priority		
•••						Demand Build Late Limit	1	
•••						Demand Build Ahead Limit		
•••						Total Met On Time Quantity		
•••						Total Late Quantity	100	
•••						Total Short Quantity		
•••						Max Lateness Buckets	1	
•••						Replanned Demand	☐	☐
•••						Weighted Late Quantity	100	
•••				Order_2	UNF_ORDER	Demand Quantity to SCS		100
•••						Demand Priority		
•••						Demand Build Late Limit		0
•••						Demand Build Ahead Limit		
•••						Total Met On Time Quantity		
•••						Total Late Quantity		
•••						Total Short Quantity		100
•••						Max Lateness Buckets		
•••						Replanned Demand	☐	☐
•••						Weighted Late Quantity		

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OBO vs Additive RCA - Late RCA

- **Demand fulfillment**

Demand item	Demand Location	Demand Week	Customer ID	Demand ID	Demand Quantity	Met On Time	Late	Short
Chips	DC	W18-2020	Customer1	Order_1	100	0	100	0
Chips	DC	W19-2020	Customer1	Order_2	100	0	0	100

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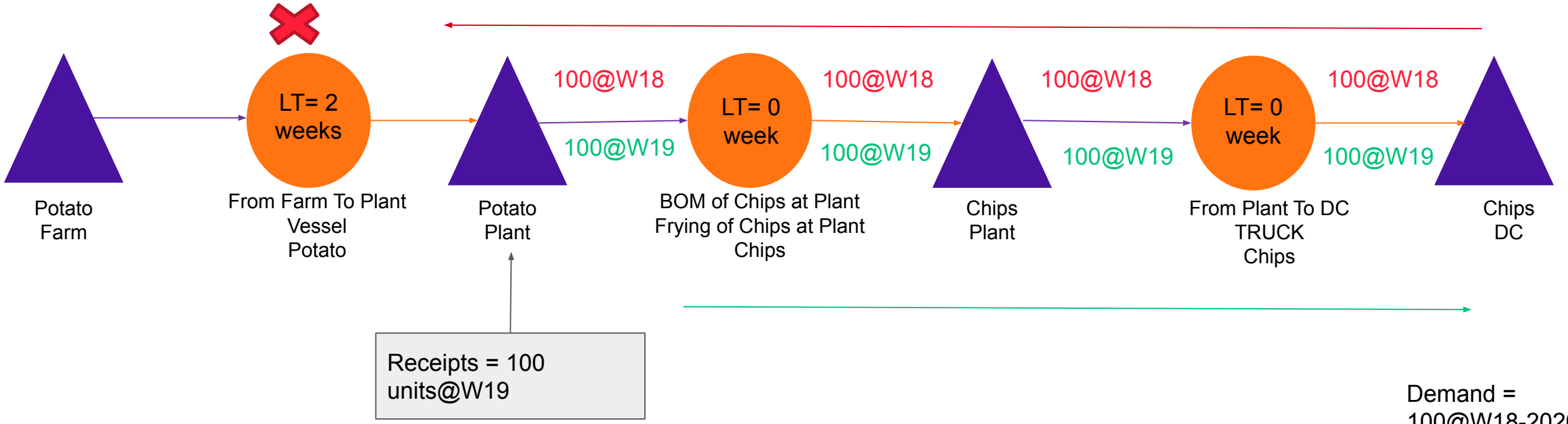
Demand Planning for Order_1

During the demand planning phase the solver does the following for Order_1:-

- The demand on W18-2020 i.e. Order_1 gets picked up first by the solver. Since there is no stock available and the total lead time from supplier to DC is greater than lead time + plan current it does not plan in JIT bucket.
- Since the build late demand is 1 week for this demand. It again gets propagated backwards from W19-2020. Since the plant has an Expected Receipts of 100 units in W19-2020. It is able to plan this demand (one week late).



Demand = 100@W18-2020

BLL = 1 week

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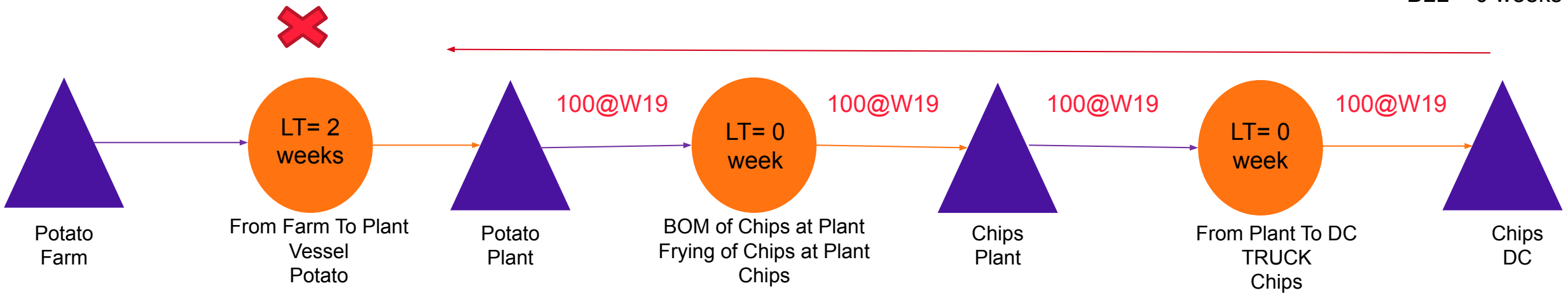
Demand Planning for Order_2

Demand = 100@W19-2020

BLL = 0 weeks



During the demand planning phase the solver does the following for Order_2:-

- The demand on W19-2020 i.e. Order_2 gets picked up next by the solver. Since there is no stock available and the total lead time from supplier to DC is greater than lead time + plan current it does not plan in JIT bucket. And since build late limit is 0 weeks, this demand gets shorted.

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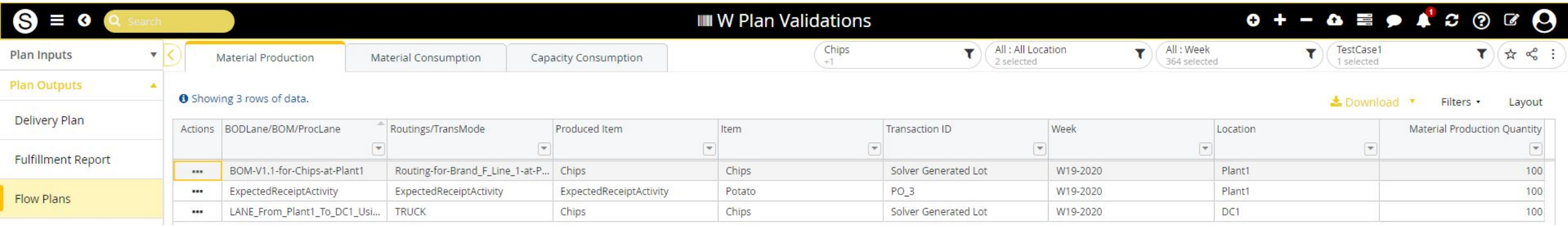
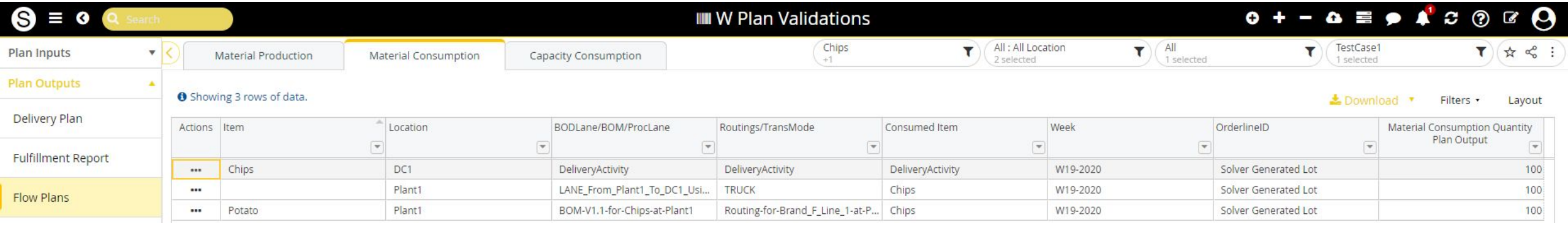
Material Constraint RCA - Example 1

The production/consumption flow plans can be seen in the page “Flow Plans” in W Plan Validations -> Plan Outputs -> Flow Plans



Actions	BODLane/BOM/ProcLane	Routings/TransMode	Produced Item	Item	Transaction ID	Week	Location	Material Production Quantity
•••	BOM-V1.1-for-Chips-at-Plant1	Routing-for-Brand_F_Line_1-at-P...	Chips	Chips	Solver Generated Lot	W19-2020	Plant1	100
•••	ExpectedReceiptActivity	ExpectedReceiptActivity	ExpectedReceiptActivity	Potato	PO_3	W19-2020	Plant1	100
•••	LANE_From_Plant1_To_DC1_Usi...	TRUCK	Chips	Chips	Solver Generated Lot	W19-2020	DC1	100



Actions	Item	Location	BODLane/BOM/ProcLane	Routings/TransMode	Consumed Item	Week	OrderlineID	Material Consumption Quantity Plan Output
•••	Chips	DC1	DeliveryActivity	DeliveryActivity	DeliveryActivity	W19-2020	Solver Generated Lot	100
•••		Plant1	LANE_From_Plant1_To_DC1_Usi...	TRUCK	Chips	W19-2020	Solver Generated Lot	100
•••	Potato	Plant1	BOM-V1.1-for-Chips-at-Plant1	Routing-for-Brand_F_Line_1-at-P...	Chips	W19-2020	Solver Generated Lot	100

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RCA Output Data

- OBO RCA output

Constrain Item	Constraint Location	Constrai nt Week	Demand Item	Demand Location	Demand Week	Customer	Demand ID	Material Constraint Qty	Problem Type
Potato	Plant1	W18	Chips	DC	W18-2020	Customer_1	Order_1	100	Lead Time
Potato	Plant1	W19	Chips	DC	W19-2020	Customer_1	Order_2	100	Lead Time

- Additive RCA Output

Constrain t Item	Constraint Location	Constr aint Week	Dema nd Item	Demand Location	Demand Week	Customer	Demand ID	Material Constraint Qty	Problem Type
Potato	Plant1	W18	Chips	DC	W18-2020	Customer_1	Order_1	100	Lead Time
Potato	Plant1	W19	Chips	DC	W19-2020	Customer_1	Order_2	0	Lead Time

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OBO vs Additive RCA behaviour (material constraint)

However looking at the RCA data we can see that there is a mismatch in the constraint values shown in the OBO vs Additive RCA.

- The OBO RCA shows a constraint of 200 units for Potato@Plant1(100 for each demand).

Actions	BOMID/ProcLane/BOI	RoutingID/TransMode	Activity3	From Item	From Location	Problem Type	From Week	To Item	To Location	Customer Group	Demand ID	To Week	W OBO Material Constraint Quantity
•••	LANE_From_Suppl...	VESSEL	Potato	Potato	Plant1	LeadtimePlanBefo...	W18-2020	Chips	DC1	Customer_1	Order_1	W18-2020	100
•••							W19-2020	Chips	DC1	Customer_1	Order_2	W19-2020	100

- However the Additive RCA shows a constraint of only 100 units (for the demand on W18-2020).

Showing 2 rows of data.

Download Filters Layout

Actions	ProcLane/BOMID/BODL	RoutingID/TransMode	Activity3	Consumed Item	Consumed Location	Consumed Week	DemandID	Demand Type	Produced Item	Produced Location	Produced Week	Material Constraint Qty
•••	LANE_From_Supplie...	VESSEL	Potato	Potato	Plant1	W18-2020	Order_1	UNF_ORDER	Chips	DC1	W18-2020	100
•••						W19-2020	Order_2	UNF_ORDER	Chips	DC1	W19-2020	0

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OBO vs Additive RCA behaviour (material constraint)

The reason we see this difference in RCA values is because of the difference in behaviour between OBO and Additive RCA.
During the OBO RCA :-

1. While planning for the demand on W18-2020 it runs into a material constraint at W18-2020 and that gets recorded as as a constraint.
2. Similarly for the demand on W19-2020 it runs into a material constraint at W19-2020 and that gets recorded as as a constraint.

During the additive RCA :-

1. Unplanning of late demands happens, in this case Order_1 gets unplanned and material profile is back to **(W18-0,W19-100)** at Potato@Plant1.
2. While planning for Order_1 on W18-2020 it runs into a material constraint at W18-2020 and that gets recorded as as a constraint.
3. The unplanned supply of Order_1 gets used up by Order_2 and hence no RCA is reported.

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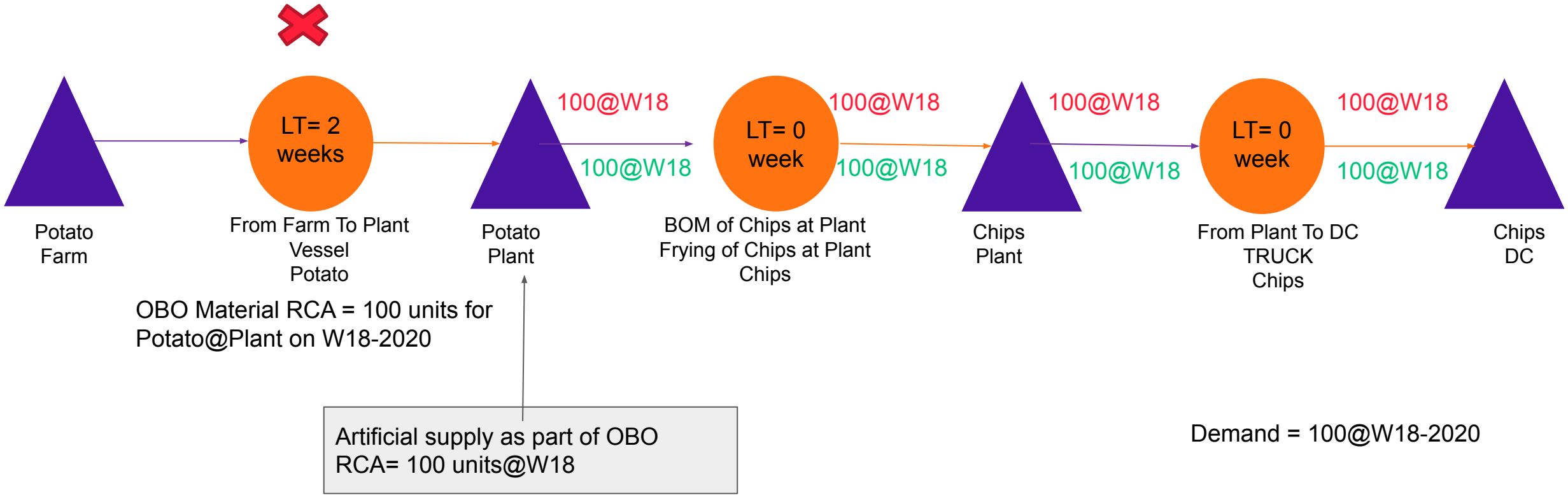
OBO RCA - Order_1

While planning for the demand on W18-2020 it runs into a material constraint at W18-2020 and that gets recorded as as a constraint.



Demand = 100@W18-2020

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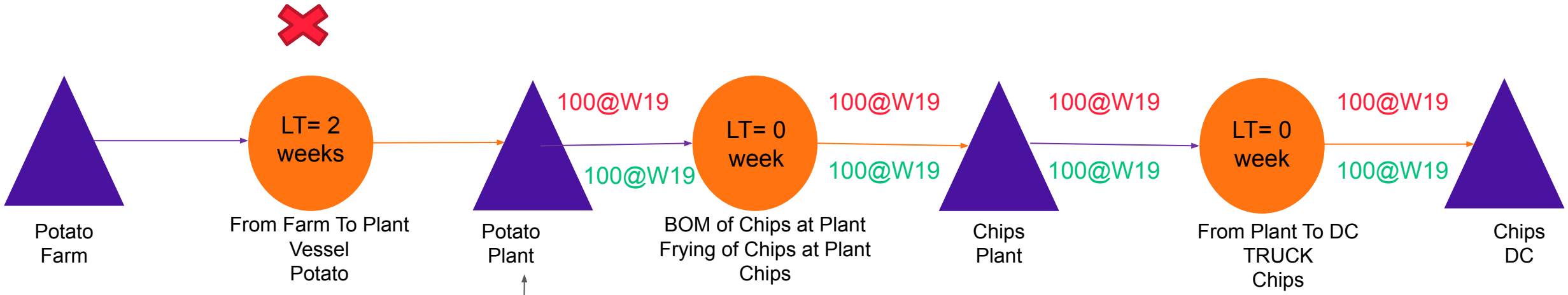
OBO RCA - Order_2

Similarly for the demand on W19-2020 it runs into a material constraint at W19-2020, since the material profile for Potato@Plant is 0 as the late demand did which had consumed the expected receipts did not get unplanned,hence that gets recorded as as a constraint.



OBO Material RCA = 100 units for Potato@Plant on W19-2020

Artificial supply as part of OBO RCA= 100 units@W19

Demand = 100@W19-2020

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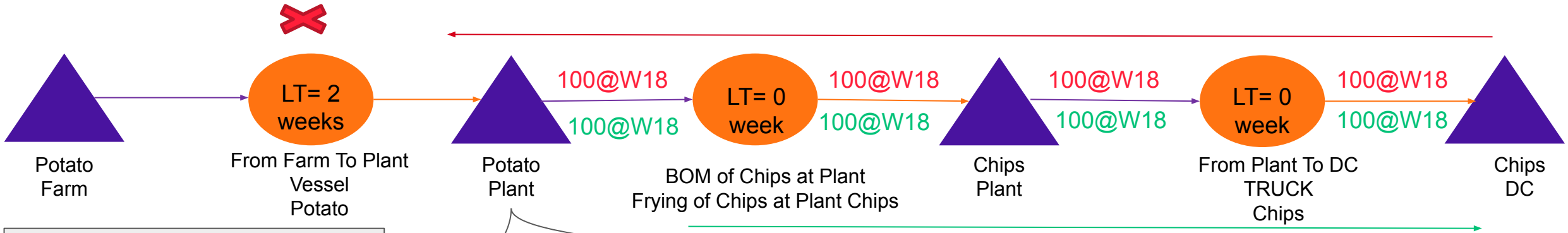
Additive RCA - Order_1

1. While planning for the demand on W18-2020 it runs into a material constraint at W18-2020 and that gets recorded as as a constraint.
2. The unplanning of this demand results in putting back the supply at W19 which can be consumed for future demands.



Receipts = 100 units@W19.Since the additive RCA unplans all late demands

Artificial supply as part of Additive RCA = 100 units@W18

Demand = 100@W18-2020

Additive RCA = 100@W18-2020 Potato@Plant

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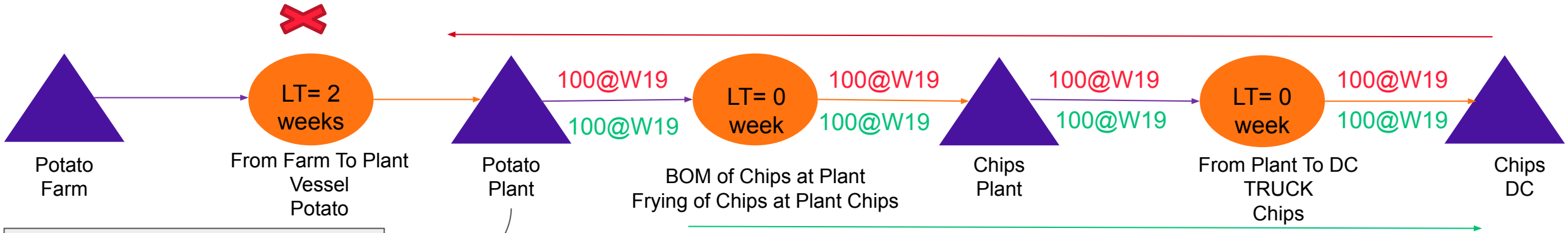
Additive RCA - Order_2

The Order_2 utilizes the unplanned supplies from Order_1 and hence a 0 quantity additive RCA is reported.



Receipts = 100 units@W19.Since the additive RCA unplans all late demands, the receipts qty of 100 units can be used in the RCA plase

Demand = 100@W19-2020

Additive RCA = 0@W19-2020 Potato@Plant

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Additive RCA vs OBO RCA explanation :-

The OBO RCA does not unplan the late demands. Hence the Receipts Qty of 100 units on W19-2020 can be reused in the RCA planning phase which leads to different values in OBO & Additive RCA.

During the OBO RCA, the solver performs the following steps:-

1. Replan for Order_1, since there is a material constraint of 100 units, an artificial supply of 100 units is created and a constraint for 100 units is recorded for W18-2020.
2. Replan for Order_2, since there is a material constraint of 100 units, an artificial supply of 100 units is created and a constraint for 100 units is recorded for W19-2020.

During the Additive RCA, the solver performs the following steps:-

1. Replan for Order_1, since there is a material constraint of 100 units, an artificial supply of 100 units is created and a constraint for 100 units is recorded for W18-2020.
2. Replan for Order_2, but since there is a receipt of 100 units available in W19-2020 due to the unplanned late demand Order_1, this 100 units are used and no constraint is recorded. Which is why the OBO and Additive RCA values are different.

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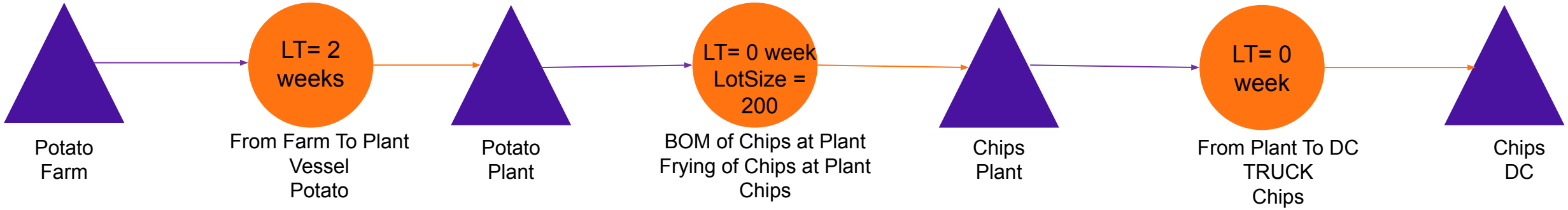
OBO vs Additive RCA - Lot size

Consider the below network with the following details :-

Demand Item	Demand Location	Demand Week	Demand ID	Customer ID	Demand Priority	Demand Quantity	BLL
Chips	DC	W18-2020	Order_1	Customer1	1	100	0
Chips	DC	W18-2020	Order_2	Customer1	2	100	0



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Post Solver Run Results

							Week
Actions	Item	Location	Customer Group	DemandID	Demand Type	Data	W18-2020
•••	Chips	DC1	Customer_1	Order_1	UNF_ORDER	Demand Quantity to SCS	100
•••						Demand Priority	1
•••						Demand Build Late Limit	0
•••						Demand Build Ahead Limit	
•••						Total Met On Time Quantity	
•••						Total Late Quantity	
•••						Total Short Quantity	100
•••						Max Lateness Buckets	
•••						Replanned Demand	☐
•••						Weighted Late Quantity	
•••				Order_2	UNF_ORDER	Demand Quantity to SCS	100
•••						Demand Priority	2
•••						Demand Build Late Limit	0
•••						Demand Build Ahead Limit	
•••						Total Met On Time Quantity	
•••						Total Late Quantity	
•••						Total Short Quantity	100
•••						Max Lateness Buckets	
•••						Replanned Demand	☐
•••						Weighted Late Quantity	

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OBO vs Additive RCA - Lot size

Demand Fulfillment

Demand item	Demand Location	Demand Week	Customer ID	Demand ID	Demand Priority	Demand Quantity	Demand On Time	Demand Late	Demand Short
Chips	DC	W18-2020	Customer1	Order_1	1	100	0	0	100
Chips	DC	W18-2020	Customer1	Order_2	2	100	0	0	100

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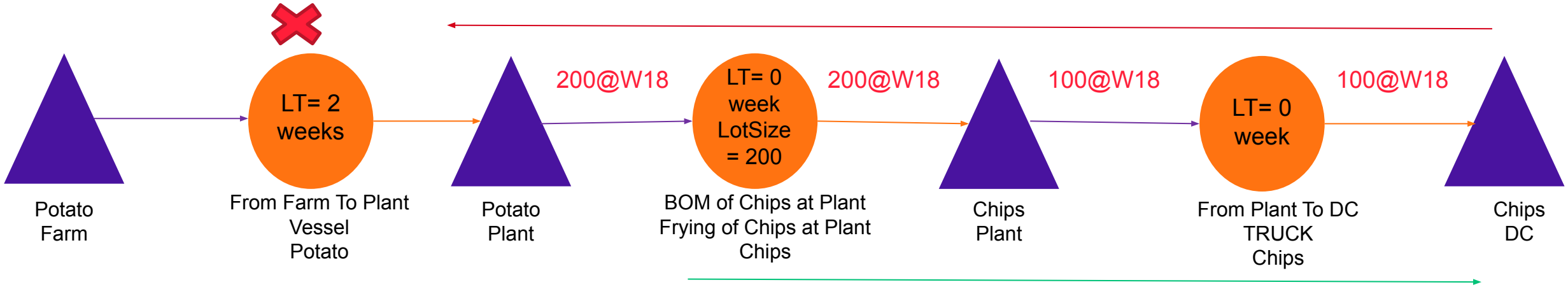
Demand Planning for Order_1

Demand = 100@W18-2020
BLL = 0 week



During the demand planning phase the solver does the following for Order_1:-

- The demand on W18-2020 i.e. Order_1 gets picked up first by the solver since it has a higher priority. Since there is no stock available and the total lead time from supplier to DC is greater than lead time + plan current it cannot plan in JIT bucket.
- Since the build late limit for this demand is 0 weeks. This demand gets shorted.

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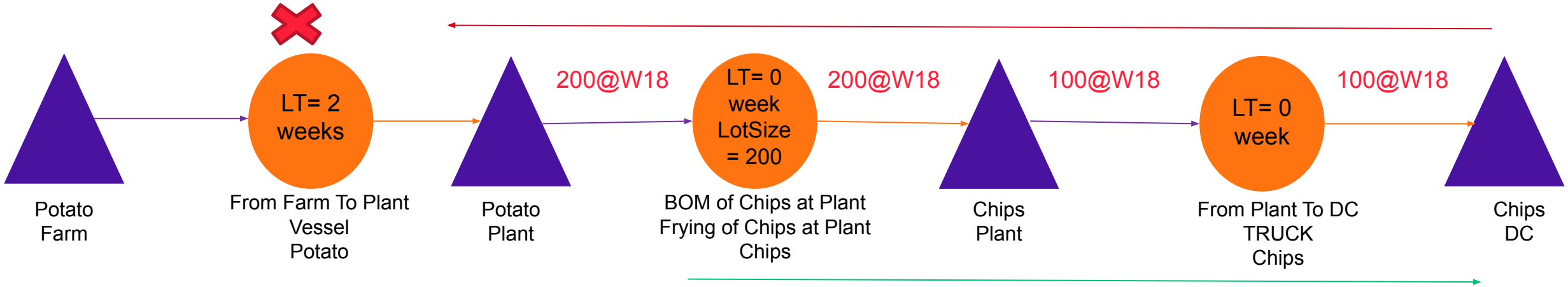
Demand Planning for Order_2

Demand = 100@W18-2020
BLL = 0 week



During the demand planning phase the solver does the following for Order_2:-

- The second demand on W18-2020 i.e. Order_2. This gets picked up next since it has a lower priority. Since there is no stock available and the total lead time from supplier to DC is greater than lead time + plan current it cannot plan in JIT bucket.
- Since the build late limit for this demand is 0 weeks. This demand gets shorted.

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RCA Output Data

- OBO RCA output

Constraint Item	Constraint Location	Constraint Week	Demand Item	Demand Location	Demand Week	Customer	Demand ID	Material Constraint Qty
Potato	Plant1	W18	Chips	DC	W18-2020	Customer_1	Order_1	200
Potato	Plant1	W18	Chips	DC	W18-2020	Customer_1	Order_2	200

- Additive RCA Output

Constraint Item	Constraint Location	Constraint Week	Demand Item	Demand Location	Demand Week	Customer	Demand ID	Material Constraint Qty
Potato	Plant1	W18	Chips	DC	W18-2020	Customer_1	Order_1	200
Potato	Plant1	W18	Chips	DC	W18-2020	Customer_1	Order_2	0

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OBO vs Additive RCA behaviour (material constraint)

There is a difference between constraints between OBO and Additive.

- The OBO RCA shows a constraint of 400 units for Potato@Plant1(200 for each demand).

Actions	BOMID/ProcLane/BOI	RoutingID/TransMode	Activity3	From Item	From Location	Problem Type	From Week	To Item	To Location	Customer Group	Demand ID	To Week	W OBO Material Constraint Quantity
•••	LANE_From_Suppl...	VESSEL	Potato	Potato	Plant1	LeadtimePlanBefo...	W18-2020	Chips	DC1	Customer_1	Order_1	W18-2020	200
•••											Order_2	W18-2020	200

- However the Additive RCA shows a constraint of only 200 units (i.e. Order_1).

Actions	ProcLane/BOMID/BOI	RoutingID/TransMode	Activity3	Consumed Item	Consumed Location	Consumed Week	DemandID	Demand Type	Produced Item	Produced Location	Produced Week	from.[RootCause].[Problem Type]	Material Constraint Qty
•••	LANE_From_Suppl...	VESSEL	Potato	Potato	Plant1	W18-2020	Order_1	UNF_ORDER	Chips	DC1	W18-2020	LeadtimePlanBefo...	200
•••							Order_2	UNF_ORDER	Chips	DC1	W18-2020	LeadtimePlanBefo...	0

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OBO vs Additive RCA behaviour (material constraint with lotsize)

The reason we see this difference in RCA values is because of the difference in behaviour between OBO and Additive RCA :-

1. During OBO RCA every demand gets planned separately without considering the impact of other demands i.e the flow plans created during the RCA planning is discarded after it is planned.
2. While planning for the demand on W18-2020 Order_1 it runs into a material constraint at W18-2020 of 200 units due to the lotsize in the manufacturing activity and that gets recorded as a constraint.
3. Similarly for the second demand on W18-2020 i.e. Order_2 it runs into a material constraint at W18-2020 of 200 units due to the lotsize at the manufacturing activity and that gets recorded as as another constraint.

The Additive RCA on the other hand retains the temporary flow plans for prior demands that were planned as part of RCA :-

1. While planning for the demand on W18-2020 Order_1 it runs into a material constraint at W18-2020 of 200 units due to the lotsize at the manufacturing activity and that gets recorded as as a constraint.
2. During additive RCA, the solver retains the flow plans which are planned by the previous demands.
3. Since the solver retains the temporary flow plans, the 200 units produced for Potato@Plant as part of the RCA planning for Order_1 is sufficient to meet both demands. So the solver does not record a constraint for Order_2 in Additive RCA.

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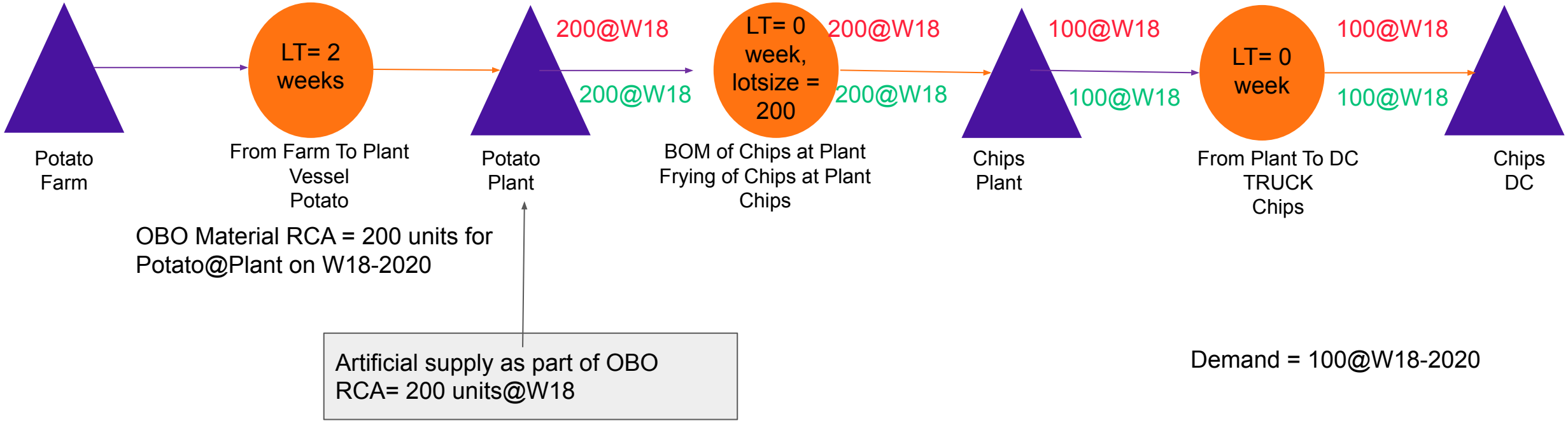
OBO RCA Replanning - Order_1

While planning for the demand on W18-2020 i.e. Order_1 it runs into a material constraint of 200 units at W18-2020 for Potato@Plant due to lotsize at manufacturing operation and that gets recorded as as a constraint.



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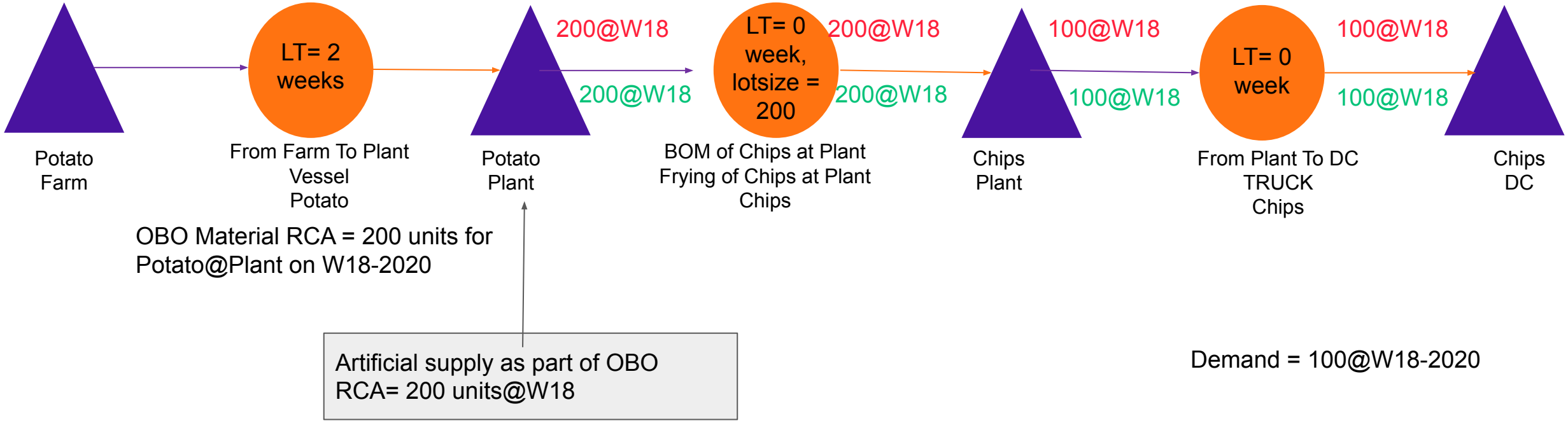
OBO RCA Replanning - Order_2

While planning for the second demand on W18-2020 i.e. Order_2 it runs into a material constraint of 200 units at W18-2020 for Potato@Plant due to lotsize at manufacturing operation and that gets recorded as as a constraint.



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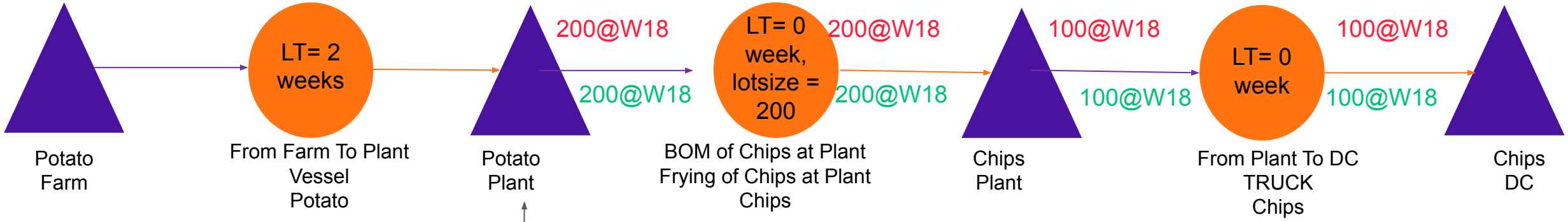
Additive RCA Replanning - Order_1

While planning for the first demand on W18-2020 i.e. Order_1 it runs into a material constraint of 200 units at W18-2020 for Potato@Plant due to lotsize at manufacturing operation and that gets recorded as as a constraint.



Additive Material RCA = 200 units for Potato@Plant on W18-2020 for Order_1

Artificial supply as part of Additive RCA= 200 units@W18 for Order_1

Demand = 100@W18-2020

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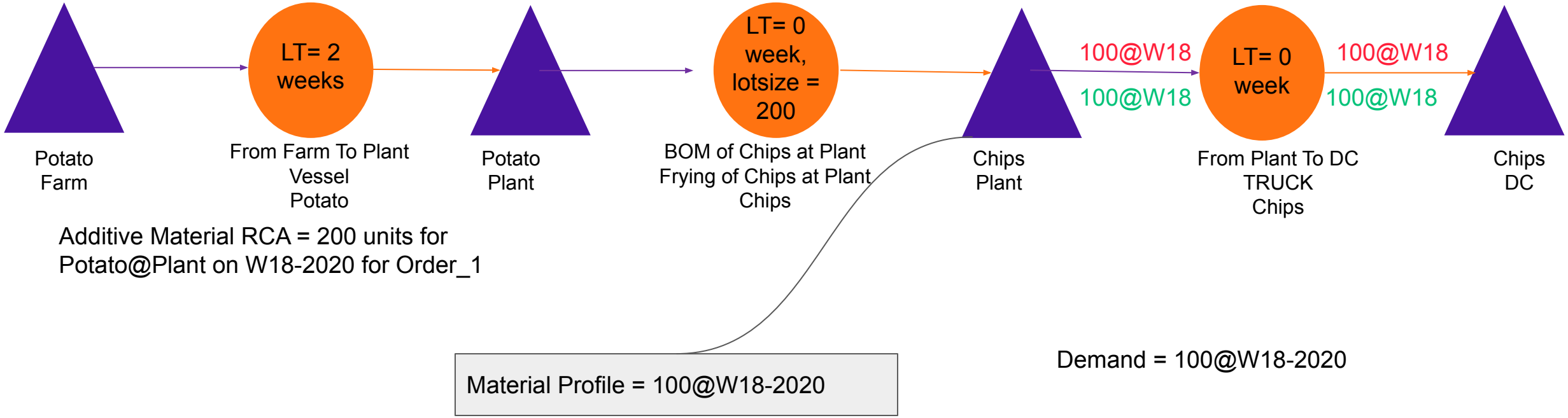
Additive RCA Replanning - Order_2

Since the Additive RCA retains the temporary flow plans created during the replanning previous demands, therefore Chips@Plant still has a material profile of 100 units which is sufficient to meet the demands for Order_2. Hence no material constraint gets recorded for Order_2



Additive Material RCA = 200 units for Potato@Plant on W18-2020 for Order_1

Demand = 100@W18-2020

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Additive RCA vs OBO RCA explanation :-

The OBO RCA does not retain the temporary flow plans while the Additive RCA does. This behaviour causes the difference in the OBO & Additive RCA values that we see in Case #2

During the OBO RCA, the solver performs the following steps:-

1. Replan for Order_1. Since there is a material constraint of 200 units, an artificial supply of 200 units is created and a constraint for 200 units is recorded for W18-2020 for Order_1.
2. Replan for Order_2. Since there is a material constraint of 200 units, an artificial supply of 200 units is created and a constraint for 200 units is recorded for W18-2020 for Order_2.

During the Additive RCA, the solver performs the following steps:-

1. Replan for Order_1. Since there is a material constraint of 200 units, an artificial supply of 200 units is created and a constraint for 200 units is recorded for W18-2020 for Order_1.
2. Replan for Order_2. The total demand for Chips@Plant is 200 units. Since the lotsize for the manufacturing activity is 200 units, 200 units were produced while planning for Order_1 out of which 100 units were used to meet the demand and the remaining 100 were in stock. Since temporary flow plans are retained in Additive RCA, the remaining 100 units can be used to meet Order_2. This causes a difference a difference in the OBO & Additive RCA values for material constraint.

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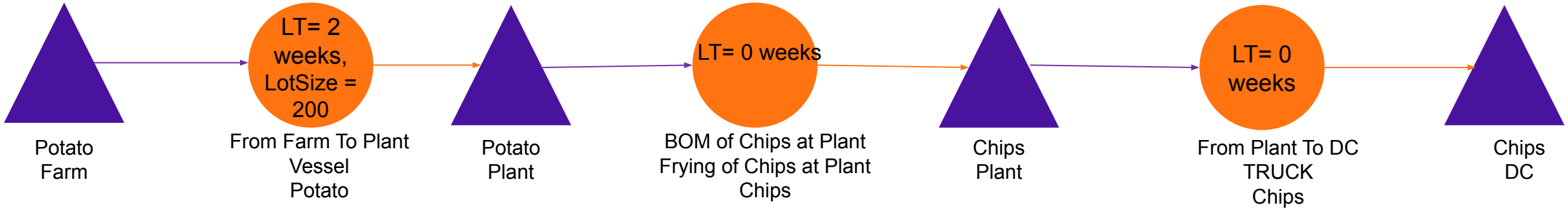
OBO vs Additive RCA - Late RCA

Consider the below network with the following details :-

Demand Item	Demand Location	Demand Week	Demand Priority	Demand ID	Customer ID	Demand Quantity	BLL
Chips	DC	W18-2020	1	Order_1	Customer1	100	0
Chips	DC	W18-2020	2	Order_2	Customer1	100	0



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Post Solver Planning Results

							Week
Actions	Item	Location	Customer Group	DemandID	Demand Type	Data	W18-2020
•••	Chips	DC1	Customer_1	Order_1	UNF_ORDER	Demand Quantity to SCS	100
•••						Demand Priority	1
•••						Demand Build Late Limit	0
•••						Demand Build Ahead Limit	
•••						Total Met On Time Quantity	
•••						Total Late Quantity	
•••						Total Short Quantity	100
•••						Max Lateness Buckets	
•••						Replanned Demand	☐
•••						Weighted Late Quantity	
•••				Order_2	UNF_ORDER	Demand Quantity to SCS	100
•••						Demand Priority	2
•••						Demand Build Late Limit	0
•••						Demand Build Ahead Limit	
•••						Total Met On Time Quantity	
•••						Total Late Quantity	
•••						Total Short Quantity	100
•••						Max Lateness Buckets	
•••						Replanned Demand	☐
•••						Weighted Late Quantity	

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Post Plan Demand Details

Demand item	Demand Location	Demand Week	Customer ID	Demand ID	Demand Priority	Demand Quantity	Demand Met On Time	Demand Late	Demand Short
Chips	DC	W18-2020	Customer1	Order_1	1	100	0	0	100
Chips	DC	W19-2020	Customer1	Order_2	2	100	0	0	100

Upon completion of the solver run we see that both the demands are being short planned.

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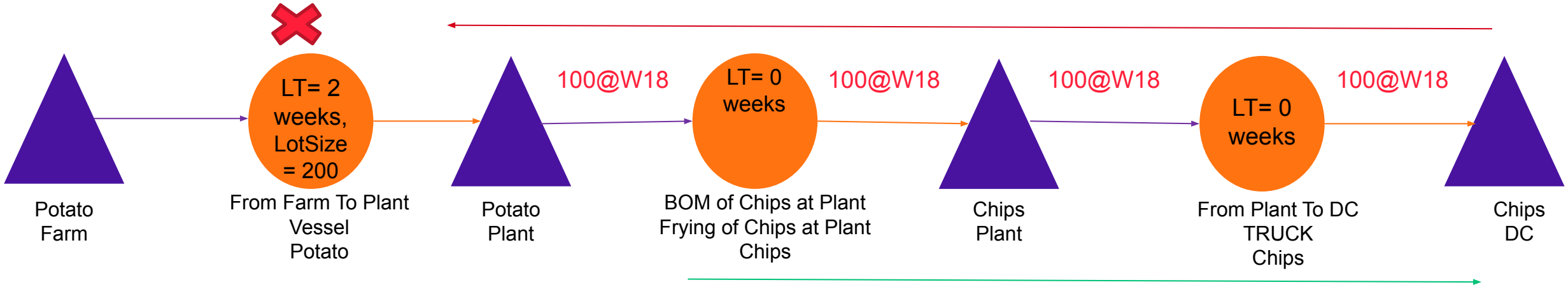
Demand Planning for Order_1

Demand = 100@W18-2020
BLL = 0 week



During the demand planning phase the solver does the following for Order_1:-

- The demand on W18-2020 i.e. Order_1 gets picked up first by the solver since it has a higher priority. Since there is no stock available and the total lead time from supplier to DC is greater than lead time + plan current it does not plan in JIT bucket.
- Since the build late limit for this demand is 0 weeks. This demand gets shorted.
- Since the requested quantity for Potato@Plant is 100 units for Potato@Plant, a material constraint of 100 units is generated at Potato@Plant.

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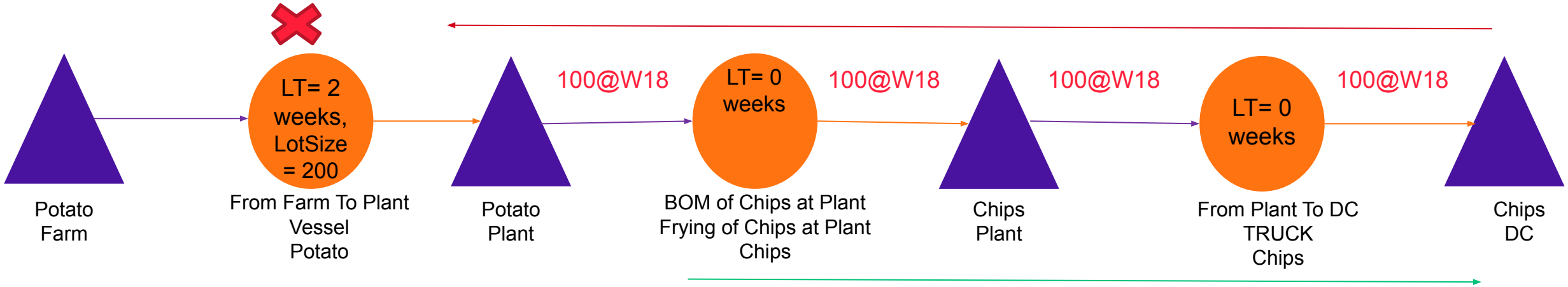
Demand Planning for Order_2

Demand = 100@W18-2020
BLL = 0 week



During the demand planning phase the solver does the following for Order_2:-

- The demand on W18-2020 i.e. Order_2 gets picked up next by the solver since it has a lower priority. Since there is no stock available and the total lead time from supplier to DC is greater than lead time + plan current it does not plan in JIT bucket.
- Since the build late limit for this demand is 0 weeks. This demand gets shorted.
- Since the requested quantity for Potato@Plant is 100 units for Potato@Plant, a material constraint of 100 units is generated at Potato@Plant.

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RCA Output Data

- OBO RCA output :-

From Item	From Location	To Item	To Location	To Week	Customer	Demand ID	Material Constraint Qty
Potato	Plant1	Chips	DC	W18-2020	Customer_1	Order_1	100
Potato	Plant1	Chips	DC	W18-2020	Customer_1	Order_2	100

- Additive RCA Output :-

From Item	From Location	To Item	To Location	To Week	Customer	Demand ID	Material Constraint Qty
Potato	Plant1	Chips	DC	W18-2020	Customer_1	Order_1	100
Potato	Plant1	Chips	DC	W18-2020	Customer_1	Order_2	100

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OBO vs Additive RCA behaviour (material constraint)

- The OBO RCA shows a constraint of 400 units for Potato@Plant1(200 for each demand).

Actions	BOMID/ProcLane/BOI	RoutingID/TransMode	Activity3	From Item	From Location	Problem Type	From Week	To Item	To Location	Customer Group	Demand ID	To Week	W OBO Material Constraint Quantity
•••	LANE_From_Suppl...	VESSEL	Potato	Potato	Plant1	LeadtimePlanBefo...	W18-2020	Chips	DC1	Customer_1	Order_1	W18-2020	100
•••											Order_2	W18-2020	100

- However the Additive RCA shows a constraint of only 200 units (i.e. Order_1).

Actions	ProcLane/BOMID/BOI	RoutingID/TransMode	Activity3	Consumed Item	Consumed Location	Consumed Week	DemandID	Demand Type	Produced Item	Produced Location	Produced Week	from.[RootCause].[Problem Type]	Material Constraint Qty
•••	LANE_From_Suppl...	VESSEL	Potato	Potato	Plant1	W18-2020	Order_1	UNF_ORDER	Chips	DC1	W18-2020	LeadtimePlanBefo...	100
•••							Order_2	UNF_ORDER	Chips	DC1	W18-2020	LeadtimePlanBefo...	100

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OBO vs Additive RCA behaviour (material constraint with lotsize)

1. While replanning for the demand on W18-2020 for Order_1 & Order_2 as part of a constraint of 100 units is recorded for each of the demands. Since the solver cannot access the activity where we have defined the lotsize due to leadtime constraints therefore the lotsize is not even considered during RCA replanning.
2. Similarly while Additive RCA also the solver cannot access the activity where we have defined the lotsize due to leadtime constraints therefore the lotsize is not even considered during RCA replanning. This results in the OBO & Additive RCA having the same output for this SC

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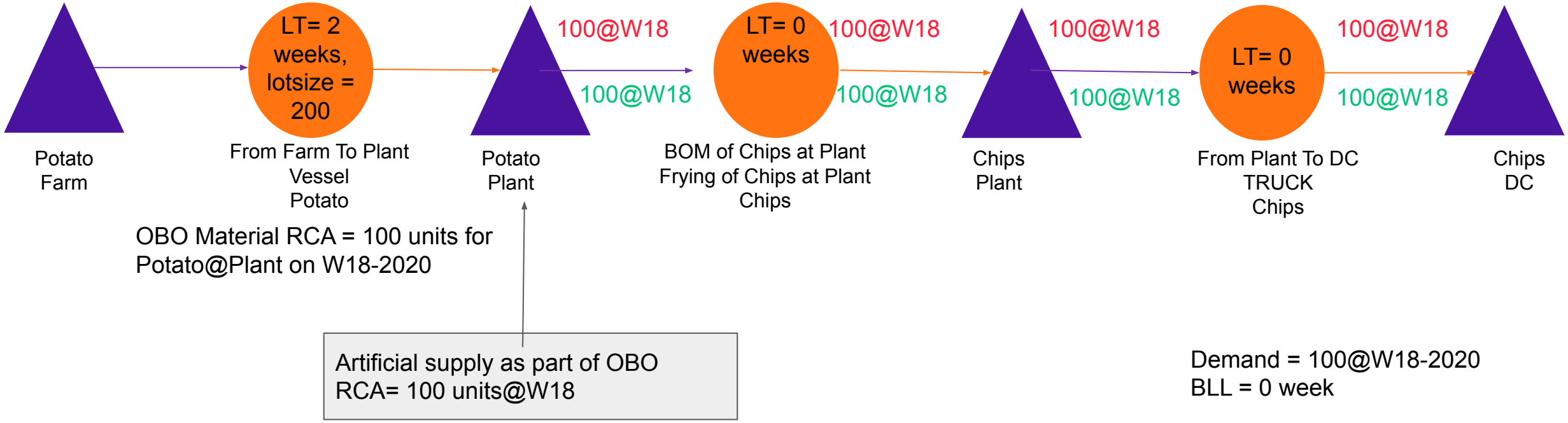
OBO RCA Replanning - Order_1

While planning for the demand on W18-2020 i.e. Order_1 it runs into a material constraint of 100 units at W18-2020 for Potato@Plant and that gets recorded as as a constraint. Since the lotsized activity is not accessed by solver due to leadtime constraint, the lotsize is not factored while generating the constraints.



Demand = 100@W18-2020
BLL = 0 week

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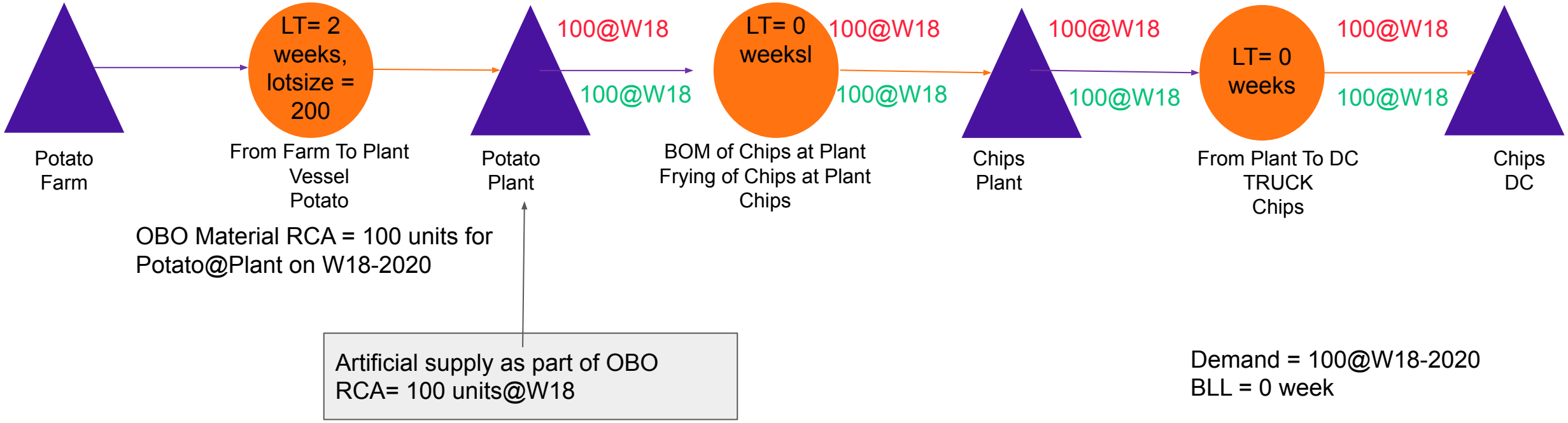
OBO RCA Replanning - Order_2

While planning for the demand on W18-2020 i.e. Order_2 it runs into a material constraint of 100 units at W18-2020 for Potato@Plant and that gets recorded as as a constraint. Since the lotsized activity is not accessed by solver due to leadtime constraint, the lotsize is not factored while generating the constraints.



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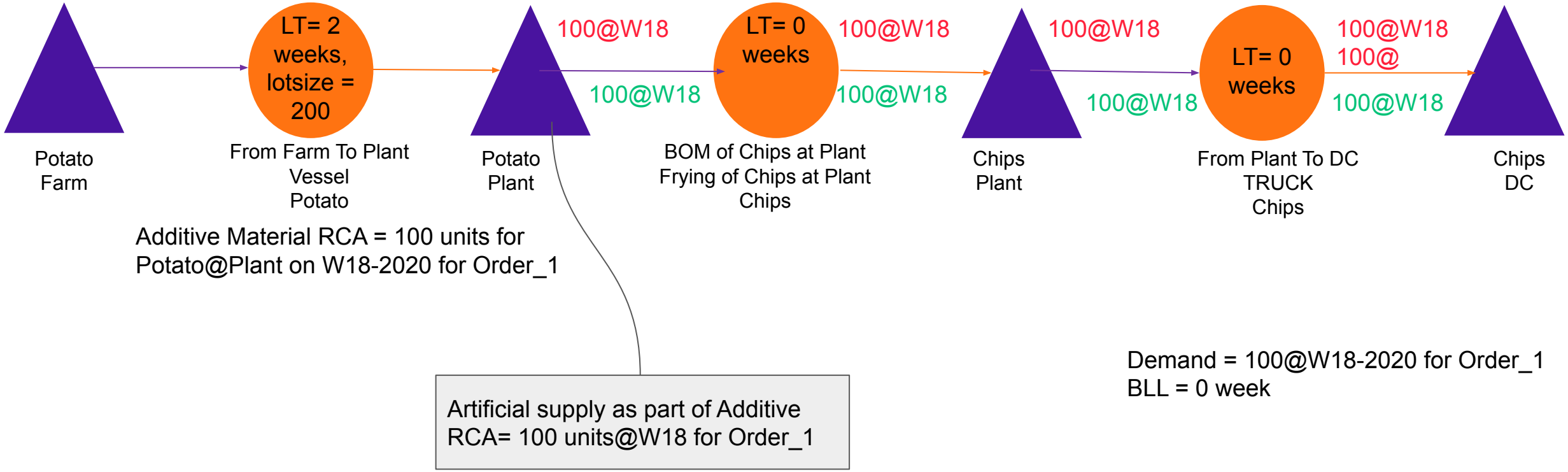
Additive RCA Replanning - Order_1

While planning for the demand on W18-2020 i.e. Order_1 it runs into a material constraint of 100 units at W18-2020 for Potato@Plant and that gets recorded as as a constraint for each demand. Since the lotsized activity is not accessed by solver due to leadtime constraint, the lotsize is not factored while generating the constraints.



Demand = 100@W18-2020 for Order_1
BLL = 0 week

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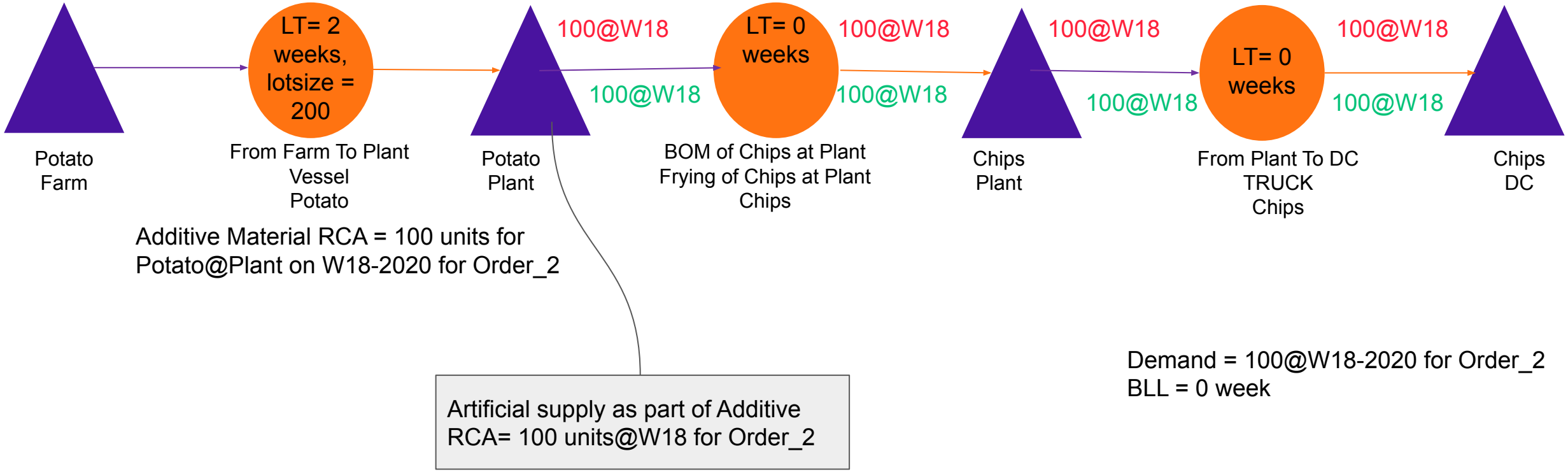
Additive RCA Replanning - Order_2

While planning for the demand on W18-2020 i.e. Order_2 it runs into a material constraint of 100 units at W18-2020 for Potato@Plant and that gets recorded as as a constraint for each demand. Since the lotsized activity is not accessed by solver due to leadtime constraint, the lotsize is not factored while generating the constraints.



Demand = 100@W18-2020 for Order_2
BLL = 0 week

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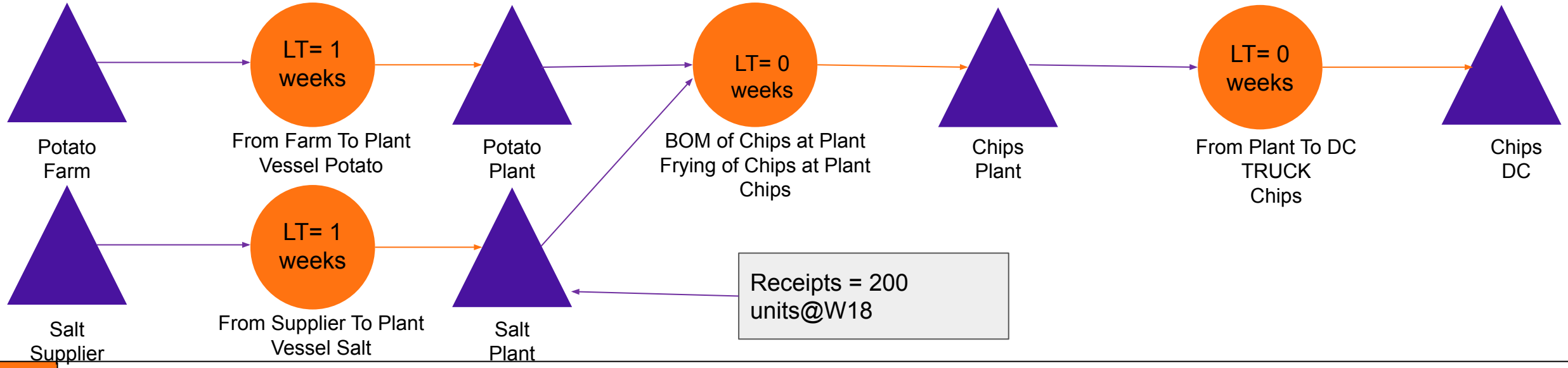
OBO vs Additive RCA - Linked Constraints

Consider the below network with the following details :- Current Bucket = W18

Demand Item	Demand Location	Demand Week	Demand Priority	Demand ID	Customer ID	Demand Quantity	BLL
Chips	DC	W18-2020	1	Order_1	Customer1	100	0
Chips	DC	W18-2020	2	Order_2	Customer1	100	0
Chips	DC	W18-2020	3	Order_3	Customer1	100	0



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Post Plan Delivery Plan Details

Demand item	Demand Location	Demand Week	Customer ID	Demand ID	Demand Priority	Demand Quantity	Met On Time	Late	Short
Chips	DC	W18-2020	Customer1	Order_1	1	100	0	0	100
Chips	DC	W19-2020	Customer1	Order_2	2	100	0	0	100
Chips	DC	W19-2020	Customer1	Order_3	3	100	0	0	100

Upon completion of the solver run we see that all three demands are being short planned.

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RCA Output Data

- OBO RCA output :-

Constraint Item	Constraint Location	Demand Item	Demand Location	Demand Week	Customer	Demand ID	Material Constraint Qty
Potato	Plant1	Chips	DC	W18-2020	Customer_1	Order_1	100
Potato	Plant1	Chips	DC	W18-2020	Customer_1	Order_2	100
Potato	Plant1	Chips	DC	W18-2020	Customer_1	Order_3	100

- Additive RCA Output :-

Constraint Item	Constraint Location	To Item	To Location	To Week	Customer	Demand ID	Material Constraint Qty
Potato	Plant1	Chips	DC	W18-2020	Customer_1	Order_1	100
Potato	Plant1	Chips	DC	W18-2020	Customer_1	Order_2	100
Potato	Plant1	Chips	DC	W18-2020	Customer_1	Order_3	100
Salt	Plant1	Chips	DC	W18-2020	Customer_1	Order_3	100

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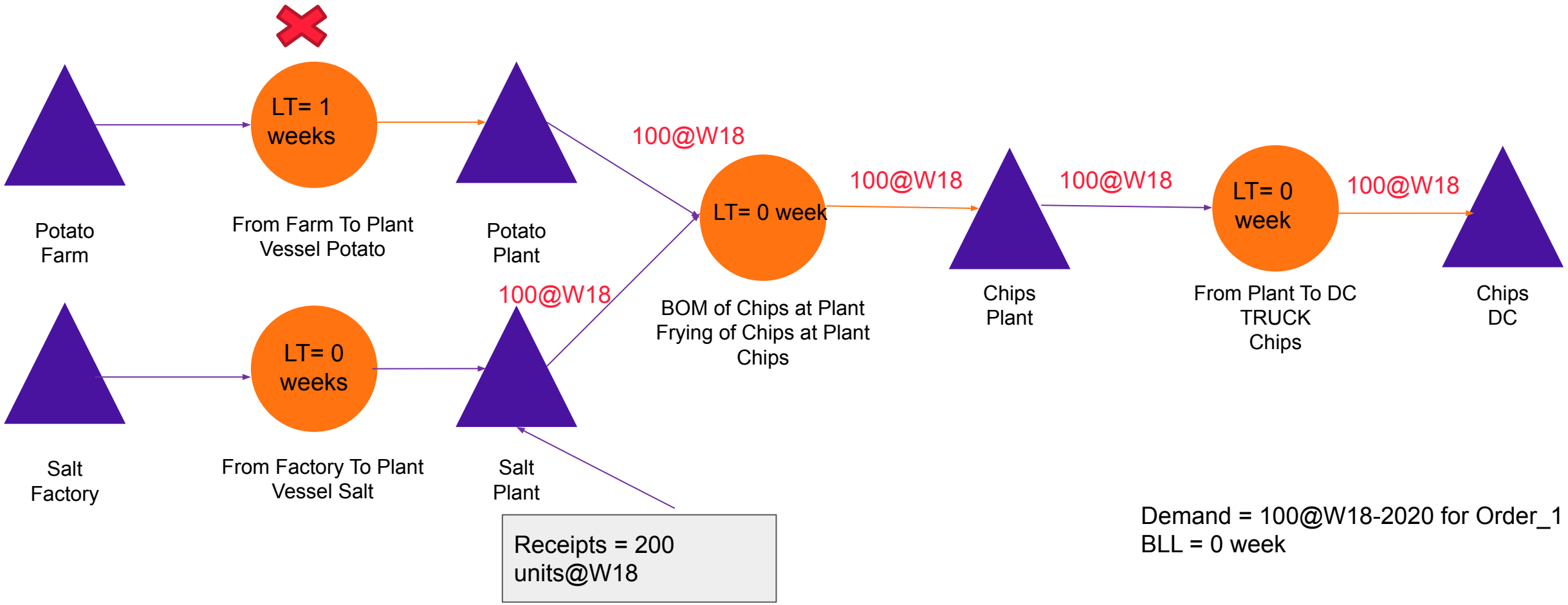
Demand Planning for Order_1

Since the material inventory profile for Potato@Plant is 0 and due to upstream LT constraint



Demand = 100@W18-2020 for Order_1
BLL = 0 week

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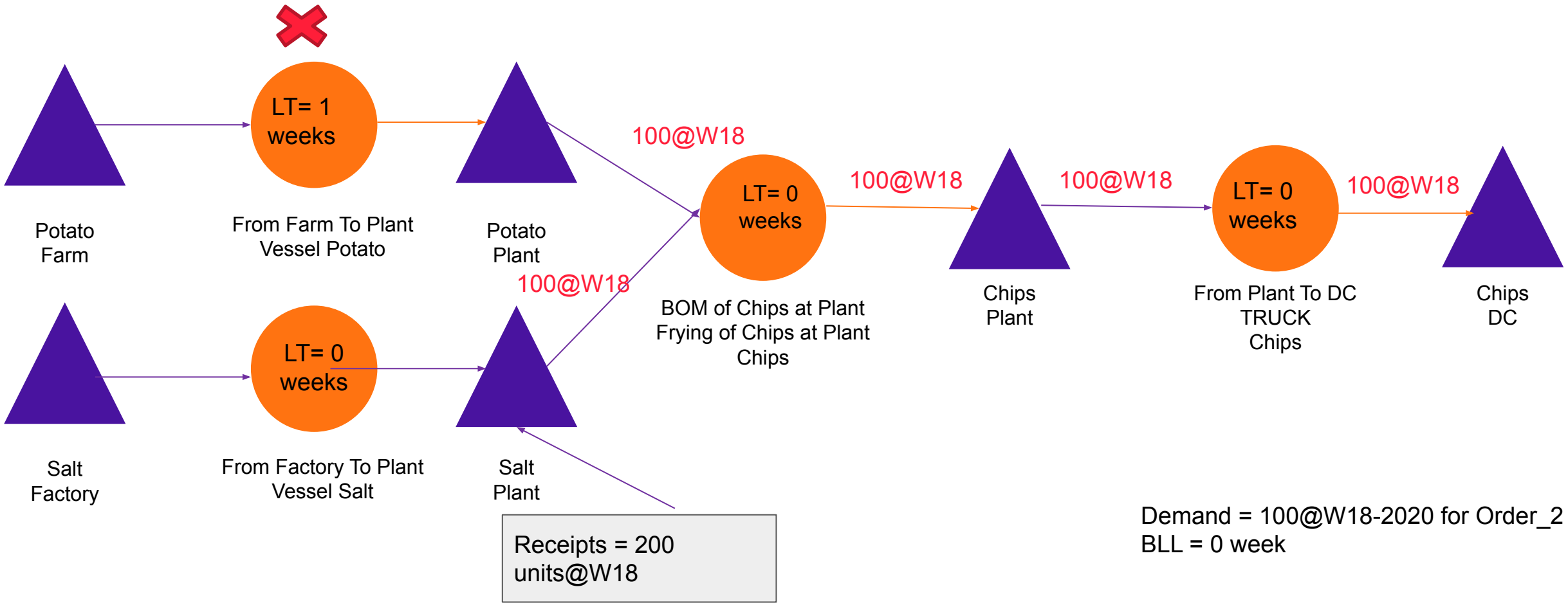
Demand Planning for Order_2

Since the material inventory profile for Potato@Plant is 0 and it cannot get any more stock at that SKU therefore the demand gets shorted.
BLL = 0



Demand = 100@W18-2020 for Order_2
BLL = 0 week

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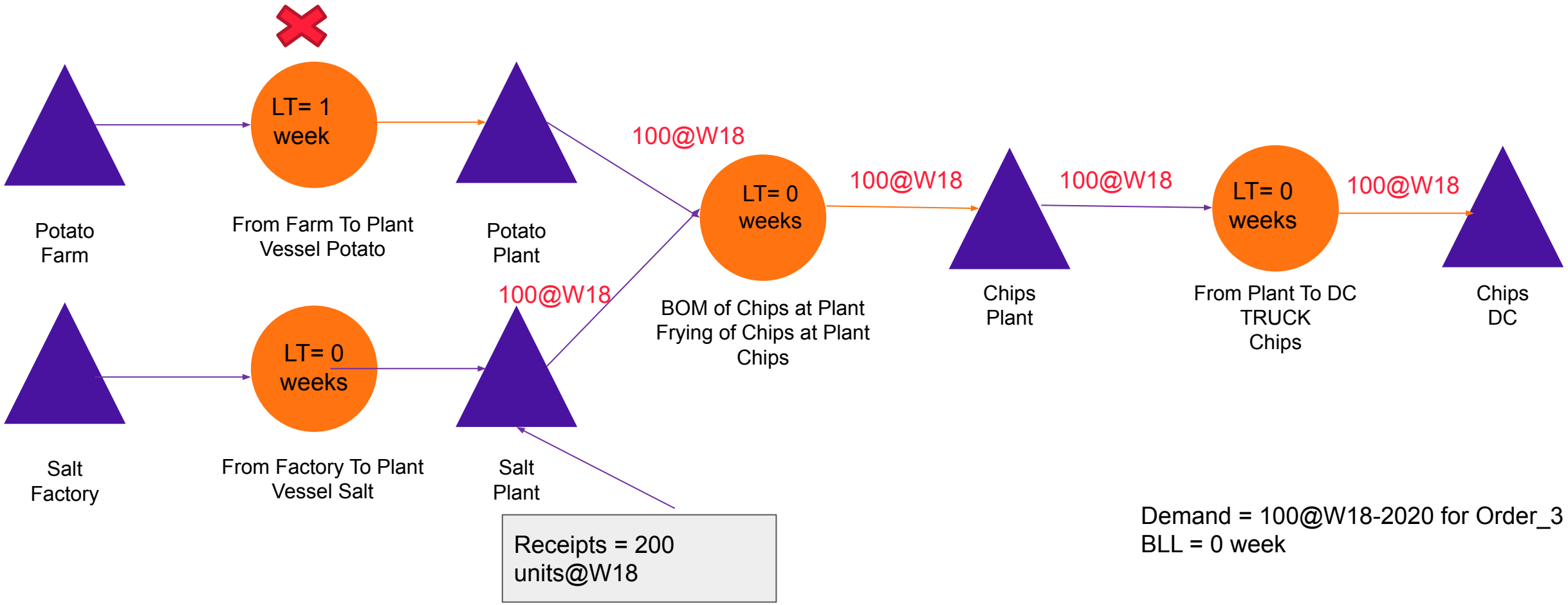
Demand Planning for Order_3

Since the material inventory profile for Potato@Plant is 0 and it cannot get any more stock at that SKU therefore the demand gets shorted.
BLL = 0



Demand = 100@W18-2020 for Order_3
BLL = 0 week

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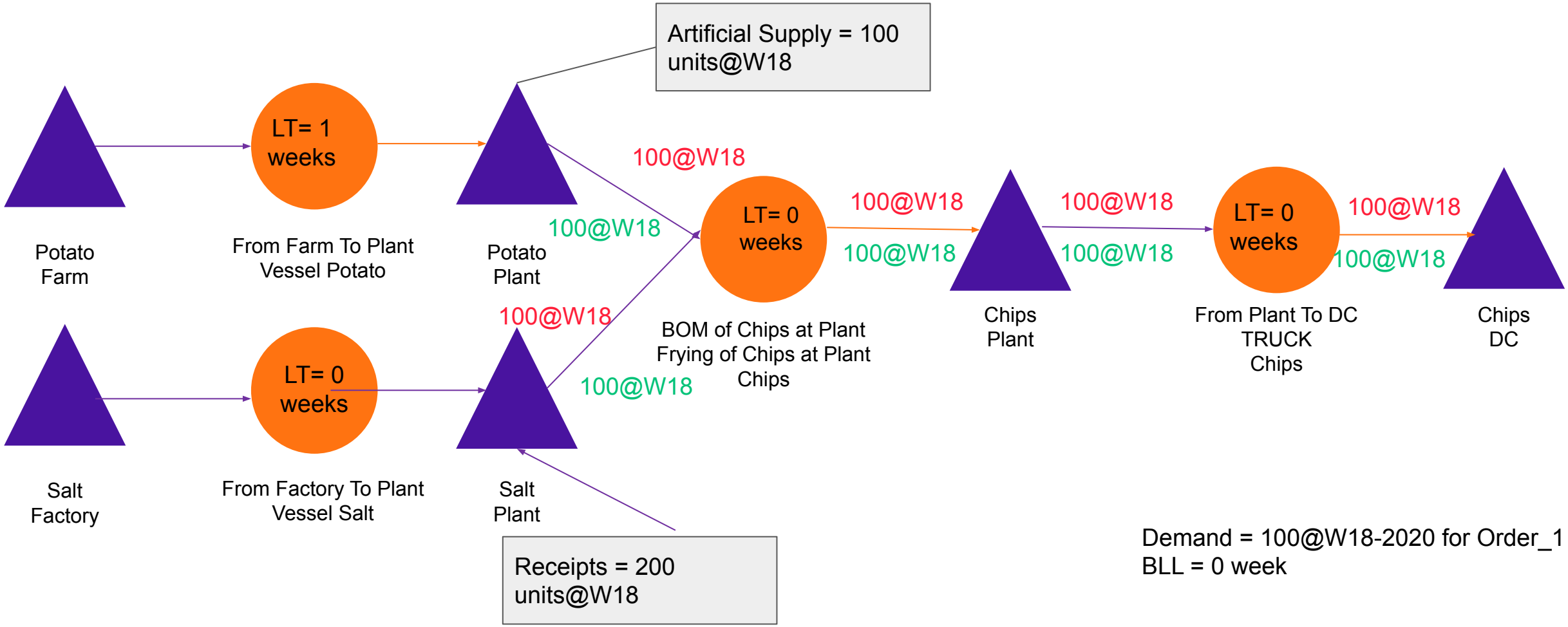
OBO RCA for Order_1

Since the material inventory profile for Potato@Plant is 0, the solver runs into a material constraint at Potato@Plant and this gets recorded as a material constraint during OBO replanning



Demand = 100@W18-2020 for Order_1
BLL = 0 week

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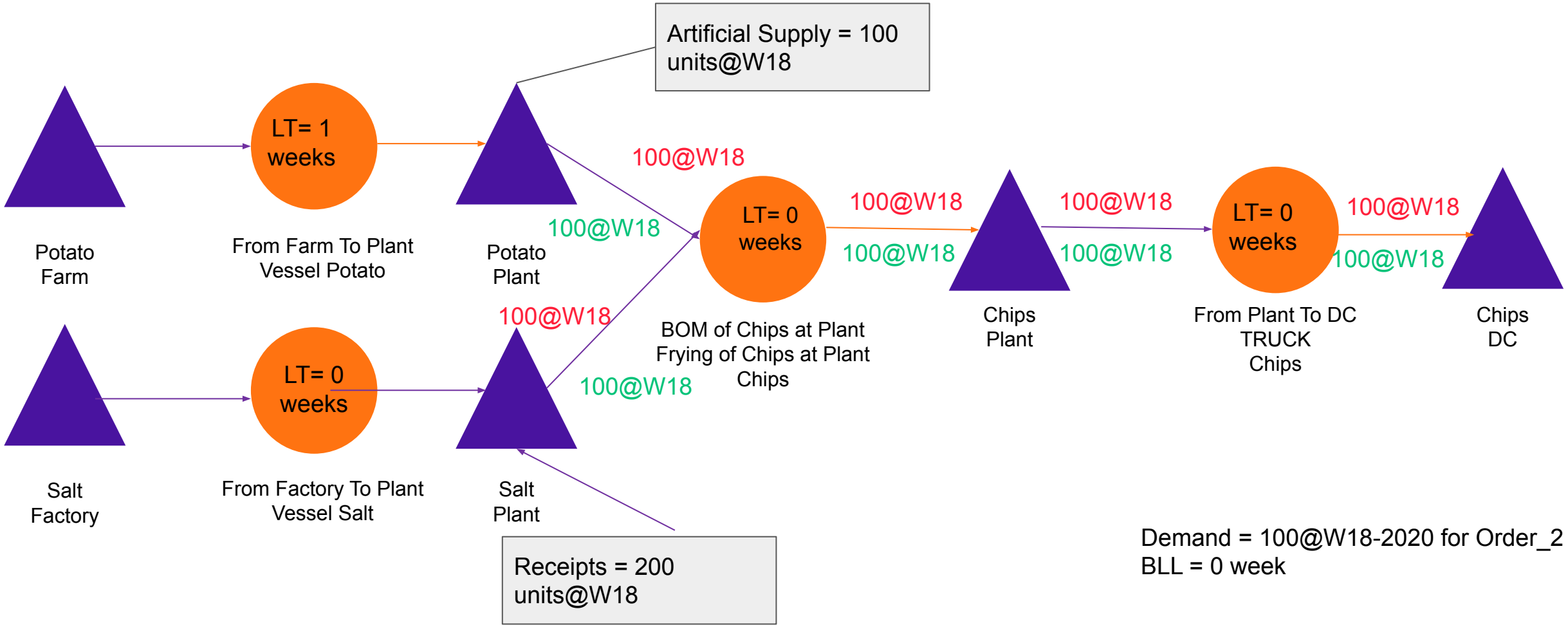
OBO RCA for Order_2

Since the material inventory profile for Potato@Plant is 0, the solver runs into a material constraint at Potato@Plant and this gets recorded as a material constraint during OBO replanning



Demand = 100@W18-2020 for Order_2
BLL = 0 week

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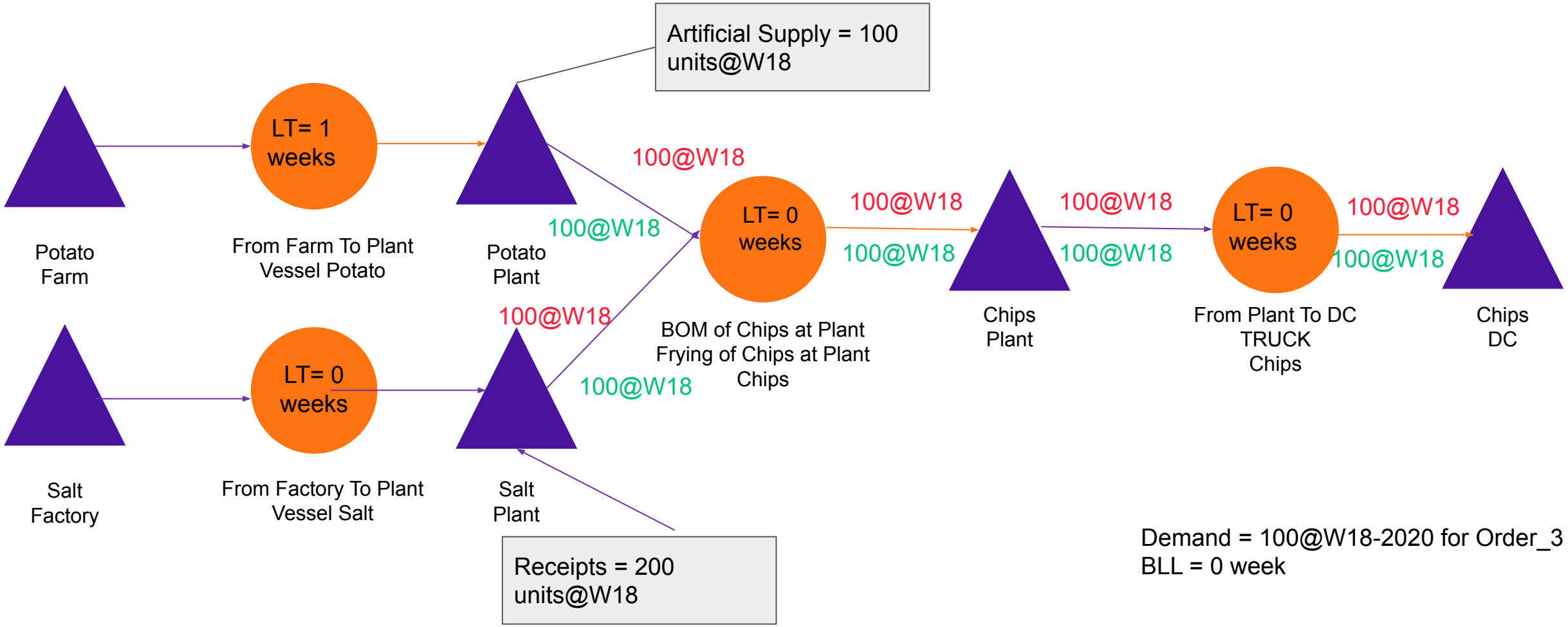
OBO RCA for Order_3

Since the material inventory profile for Potato@Plant is 0, the solver runs into a material constraint at Potato@Plant and this gets recorded as a material constraint during OBO replanning



Demand = 100@W18-2020 for Order_3
BLL = 0 week

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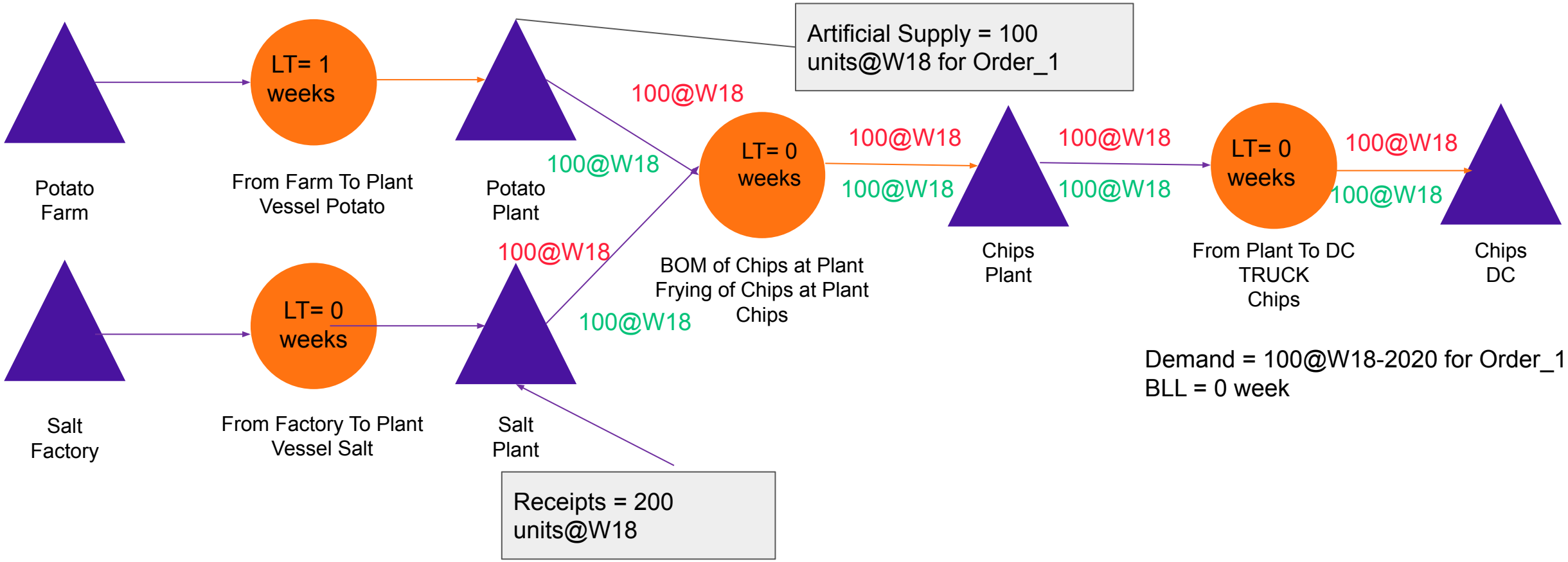
Additive RCA for Order_1

While planning for Order_1, the solver faces a material constraint for Potato@Plant and generates an artificial supply of 100 units and records a material constraint of 100 units.



Demand = 100@W18-2020 for Order_1
BLL = 0 week

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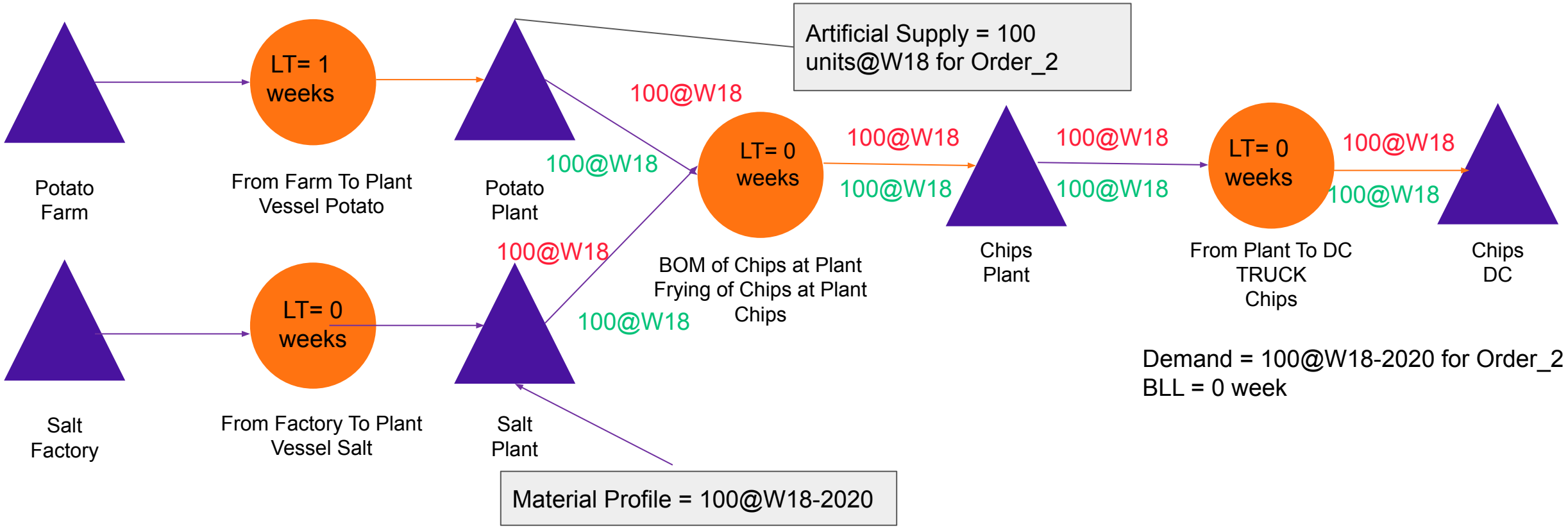
Additive RCA for Order_2

Since the additive RCA maintains the temporary flow plans, the material profile for Salt@Plant is now 100 units. While planning for Order_2, the solver faces a material constraint for Potato@Plant and generates an artificial supply of 100 units and records a material constraint of 100 units.



Demand = 100@W18-2020 for Order_2
BLL = 0 week

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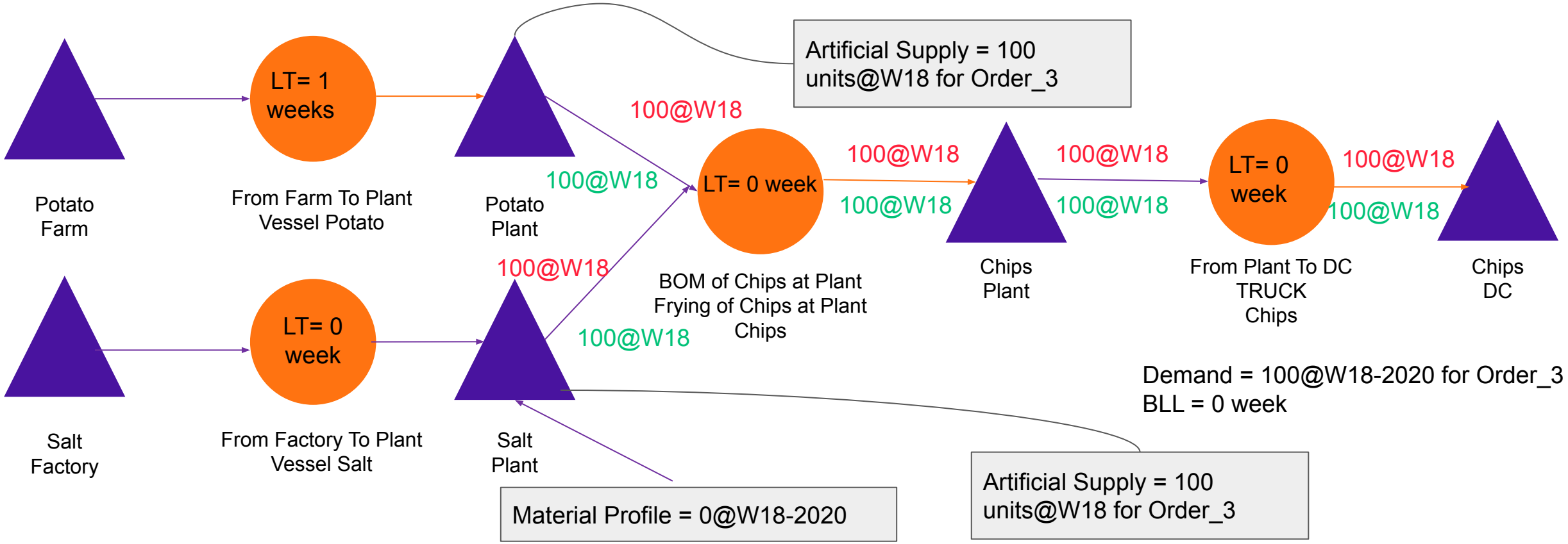
Additive RCA for Order_3

Since the additive RCA maintains the temporary flow plans, the material profile for Salt@Plant is now 0 units.While planning for Order_3, the solver faces a material constraint for Potato@Plant and Salt@Plant and generates an artificial supply of 100 units for both SKUs and records a material constraint of 100 units for both.



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Additive RCA vs OBO RCA explanation :-

The OBO RCA does not retain the temporary flow plans while the Additive RCA does. This behaviour causes the difference in the OBO & Additive RCA values seen here.

During the OBO RCA, the solver performs the following steps:-

1. While replaning for Order_1, the solver encounters a material constraint for Potato@Plant since the material profile is 0 and the solver plans for an artificial supply of 100 units and records a material constraint.
2. While replaning for Order_2, the solver encounters a material constraint for Potato@Plant since the material profile is 0 and the solver plans for an artificial supply of 100 units and records a material constraint. Since temporary flow plans are not retained in OBO RCA, no constraint is recorded for Salt@Plant.
3. While replaning for Order_3, the solver encounters a material constraint for Potato@Plant since the material profile is 0 and the solver plans for an artificial supply of 100 units and records a material constraint. Since temporary flow plans are not retained in OBO RCA, no constraint is recorded for Salt@Plant.

During the Additive RCA, the solver performs the following steps:-

1. While replaning for Order_1, the solver encounters a material constraint for Potato@Plant since the material profile is 0 and the solver plans for an artificial supply of 100 units and records a material constraint..
2. While replaning for Order_2, the solver encounters a material constraint for Potato@Plant since the material profile is 0 and the solver plans for an artificial supply of 100 units and records a material constraint.
3. While replaning for Order_3, the solver encounters a material constraint for Potato@Plant since the material profile is 0 and the solver plans for an artificial supply of 100 units and records a material constraint. By the time Order_3 is planned, the material profile for Salt@Plant is reduced to 0, so te solver plans for an artificial supply of 100 units and records a material constraint, which causes the difference in OBO & Additive RCA.

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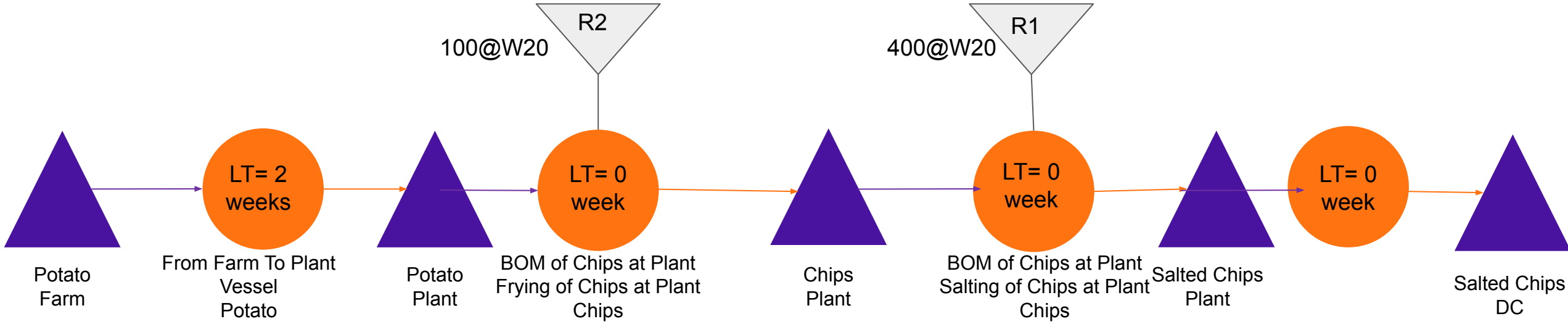
OBO vs Additive RCA - Linked Constraints 2

Demand Item	Demand Location	Demand Week	Demand ID	Customer ID	Demand Priority	Demand Quantity	BLL
Salted Chips	DC	W20-2020	Order_1	Customer1	1	100	0
Salted Chips	DC	W20-2020	Order_2	Customer1	2	100	0
Salted Chips	DC	W20-2020	Order_3	Customer1	3	100	0
Salted Chips	DC	W20-2020	Order_4	Customer1	4	100	0
Salted Chips	DC	W20-2020	Order_5	Customer1	5	100	0



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OBO vs Additive RCA - Linked Constraints 2

Demand item	Demand Location	Demand Week	Customer ID	Demand ID	Demand Priority	Demand Quantity	Demand On Time	Demand Late	Demand Short
Salted Chips	DC	W20-2020	Customer1	Order_1	1	100	100	0	0
Salted Chips	DC	W20-2020	Customer1	Order_2	2	100	0	0	100
Salted Chips	DC	W20-2020	Customer1	Order_2	3	100	0	0	100
Salted Chips	DC	W20-2020	Customer1	Order_2	4	100	0	0	100
Salted Chips	DC	W20-2020	Customer1	Order_2	5	100	0	0	100

Upon completion of the solver run we see that the first demand is planned on time and the remaining four are being short planned.

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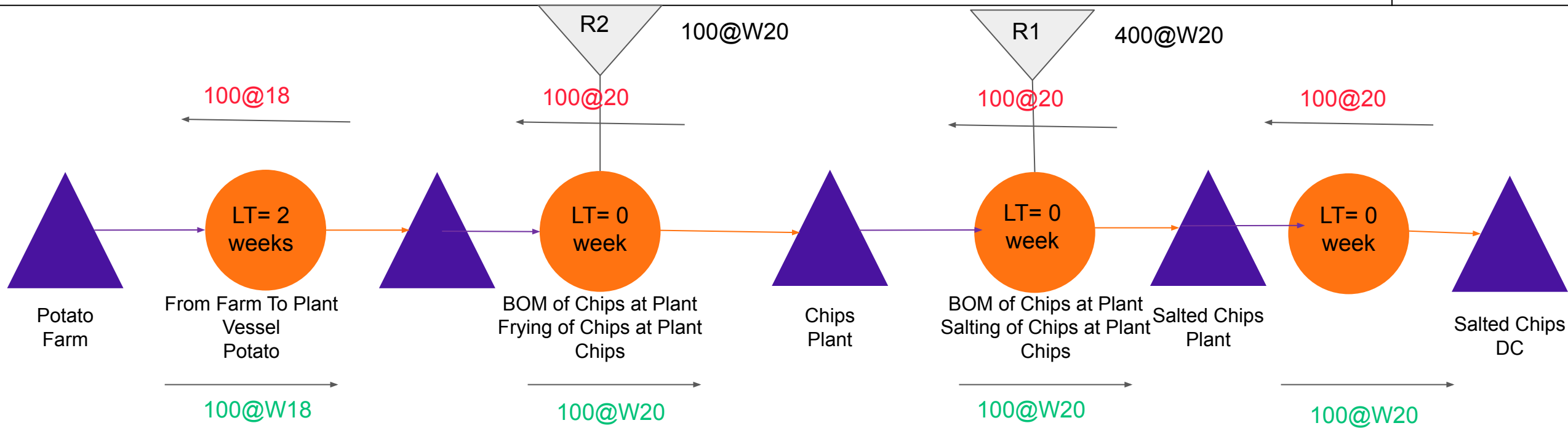
Demand Planning for Order_1



While planning for the first demand there is enough capacity available for both the demands in the JIT bucket, hence the first demand gets met on time.

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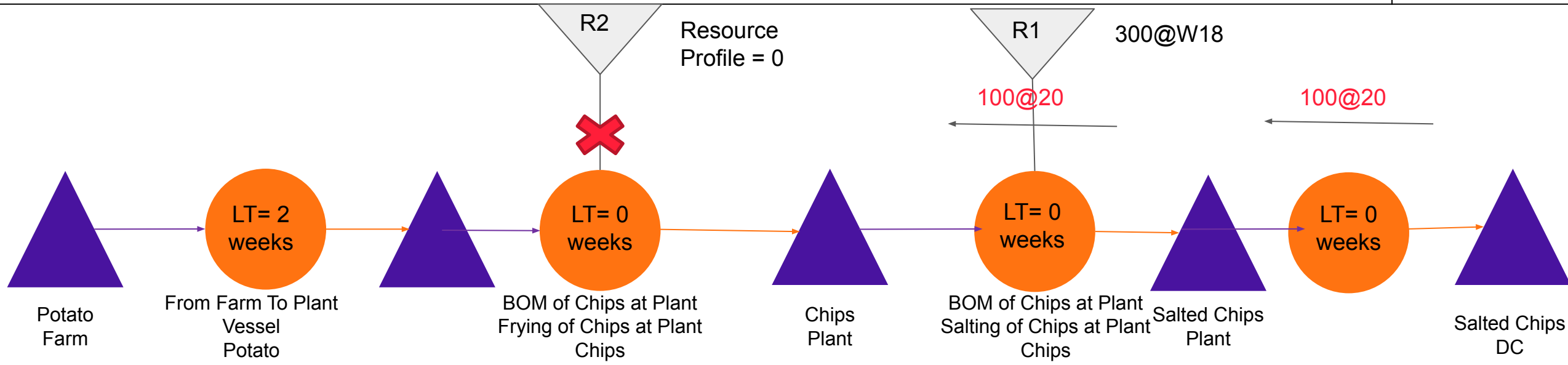
Demand Planning for Order_2



While planning for the 2nd demand there is enough capacity at R1@Plant to meet the demand at JIT bucket but the resource profile at R2@Plant is 0. Since BLL = 0 weeks therefore the demand gets shorted once the solver fails to meet at at the JIT bucket.

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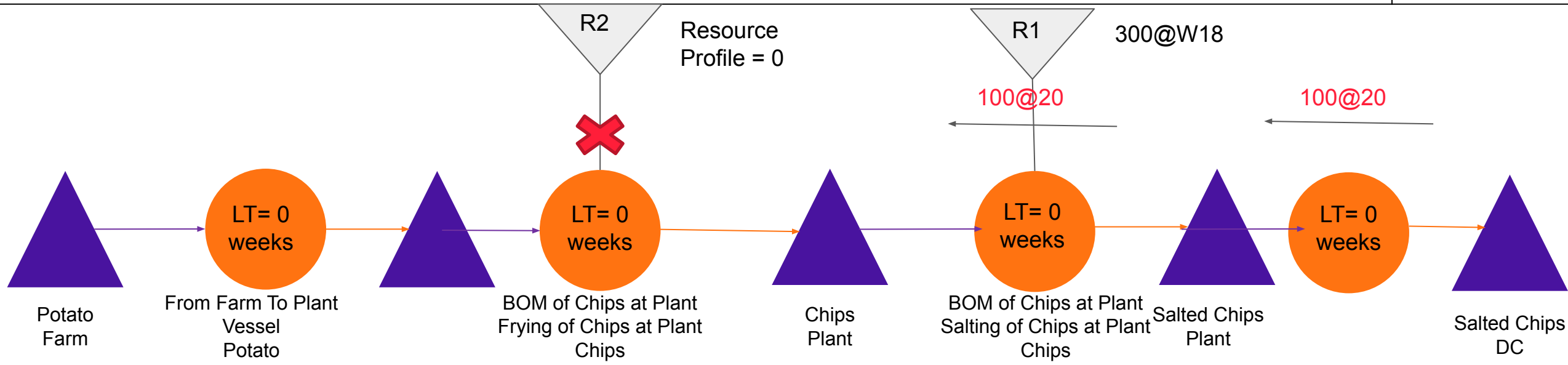
Demand Planning for Order_3



While planning for the 3rd demand there is enough capacity at R1@Plant to meet the demand at JIT bucket but the resource profile at R2@Plant is 0. Since BLL = 0 weeks therefore the demand gets shorted once the solver fails to meet at at the JIT bucket.

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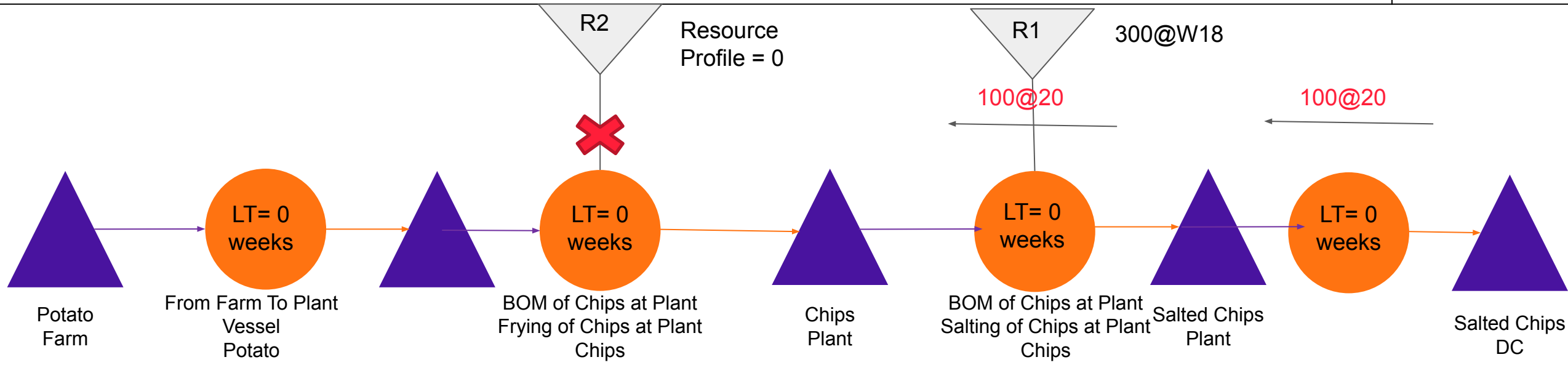
Demand Planning for Order_4



While planning for the 4th demand there is enough capacity at R1@Plant to meet the demand at JIT bucket but the resource profile at R2@Plant is 0. Since BLL = 0 weeks therefore the demand gets shorted once the solver fails to meet at at the JIT bucket.

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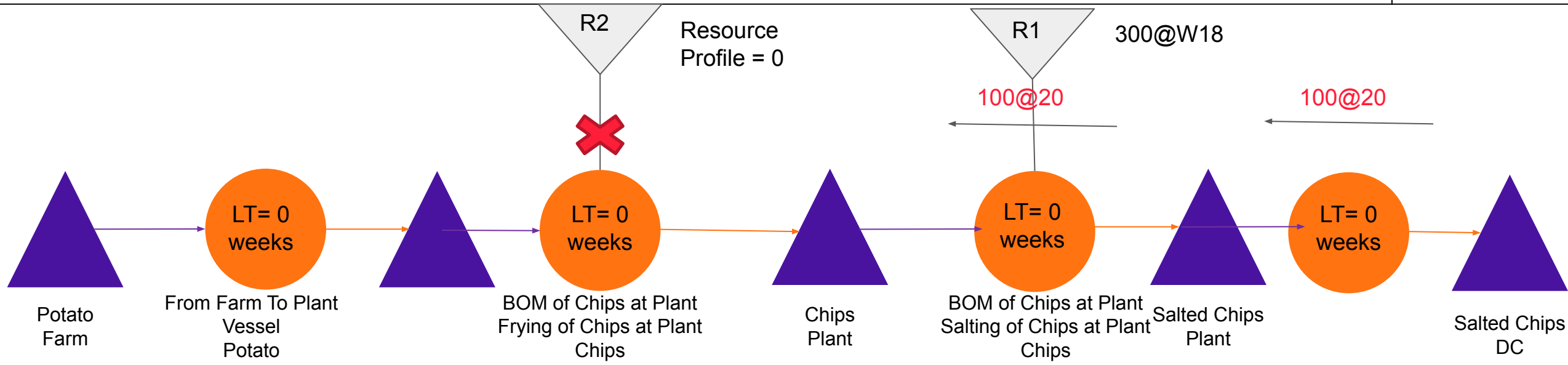
Demand Planning for Order_5



While planning for the 5th demand there is enough capacity at R1@Plant to meet the demand at JIT bucket but the resource profile at R2@Plant is 0. Since BLL = 0 weeks therefore the demand gets shorted once the solver fails to meet at at the JIT bucket.

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RCA Output Data

- OBO RCA output :-

From Resource	From Location	FromWeek	To Item	To Location	To Week	Customer	Demand ID	Resource Constraint Qty
R1	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_2	100
R1	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_3	100
R1	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_4	100
R1	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_5	100

- Additive RCA Output :-

From Resource	From Location	FromWeek	To Item	To Location	To Week	Customer	Demand ID	Resource Constraint Qty
R1	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_2	100
R1	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_3	100
R1	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_4	100
R1	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_5	100
R2	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_5	100

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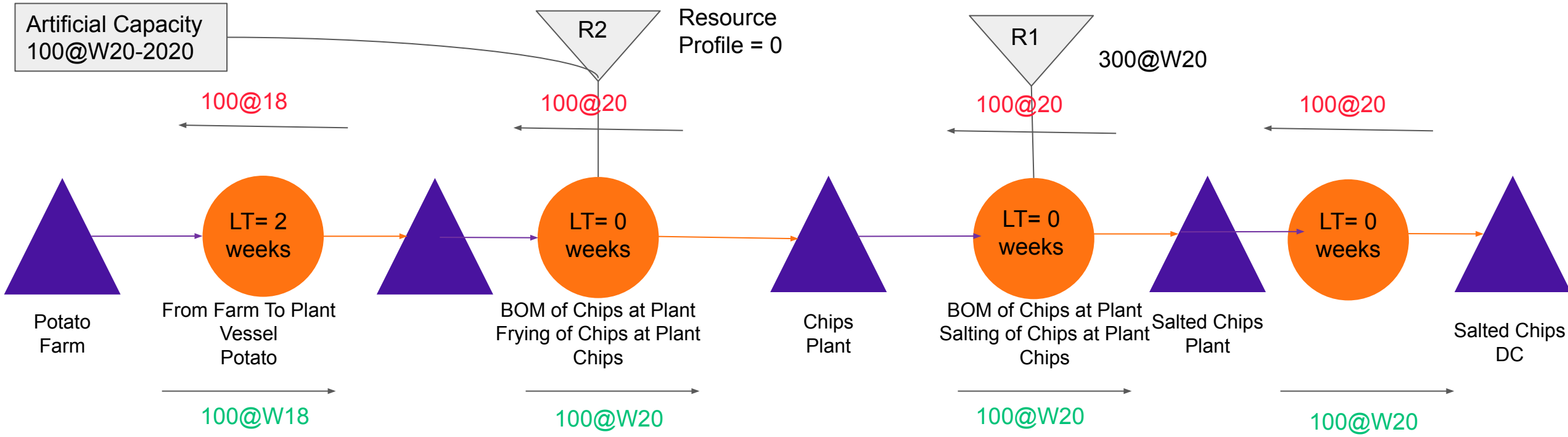
OBO RCA Planning for Order_2



Since the resource profile for R2@Plant is 0 and BLL = 0, the solver generates an artificial capacity of 100 for R2@Plant at W20-2020 to plan Order_2 on the JIT bucket, and records a capacity constraint for R2@Plant on W20-2020

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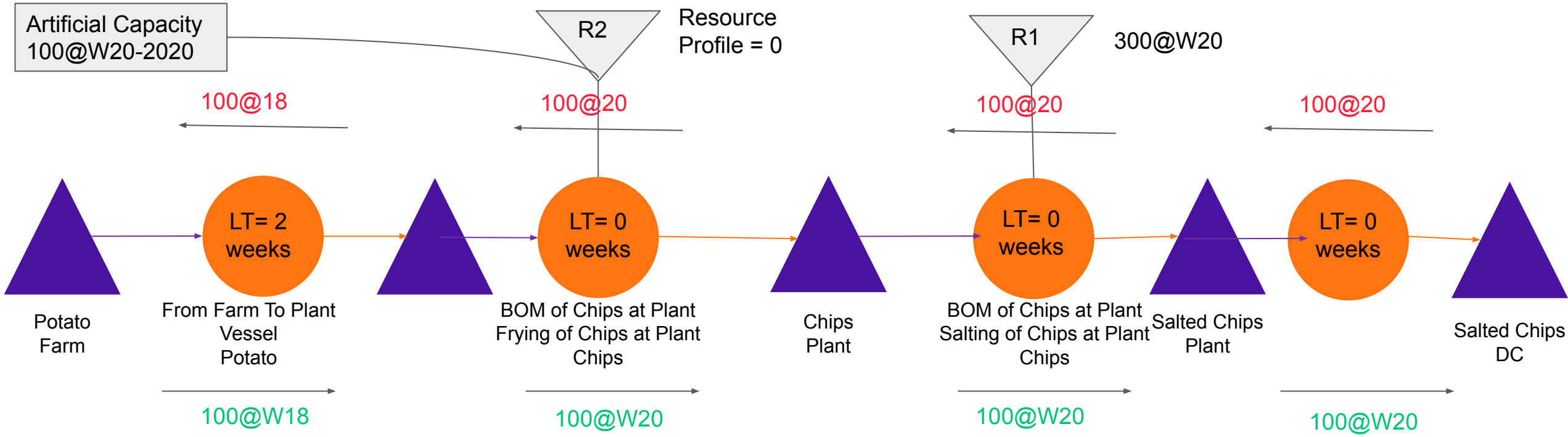
OBO RCA Planning for Order_3



Since the resource profile for R2@Plant is 0 and BLL = 0, the solver generates an artificial capacity of 100 for R2@Plant at W20-2020 to plan Order_3 on the JIT bucket, and records a capacity constraint for R2@Plant on W20-2020

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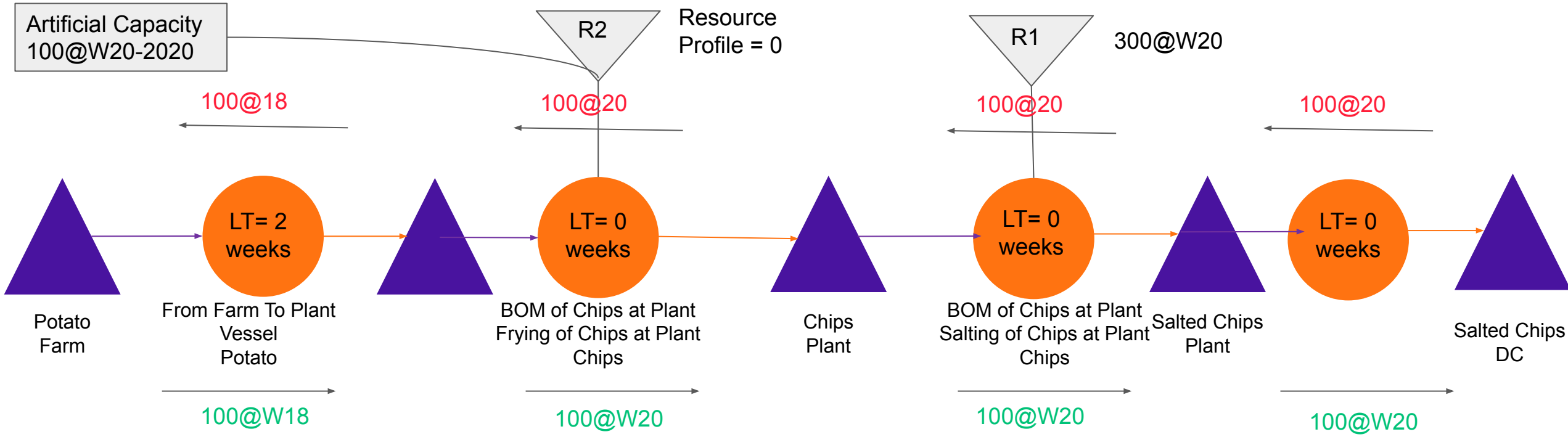
OBO RCA Planning for Order_4



Since the resource profile for R2@Plant is 0 and BLL = 0, the solver generates an artificial capacity of 100 for R2@Plant at W20-2020 to plan Order_4 on the JIT bucket, and records a capacity constraint for R2@Plant on W20-2020

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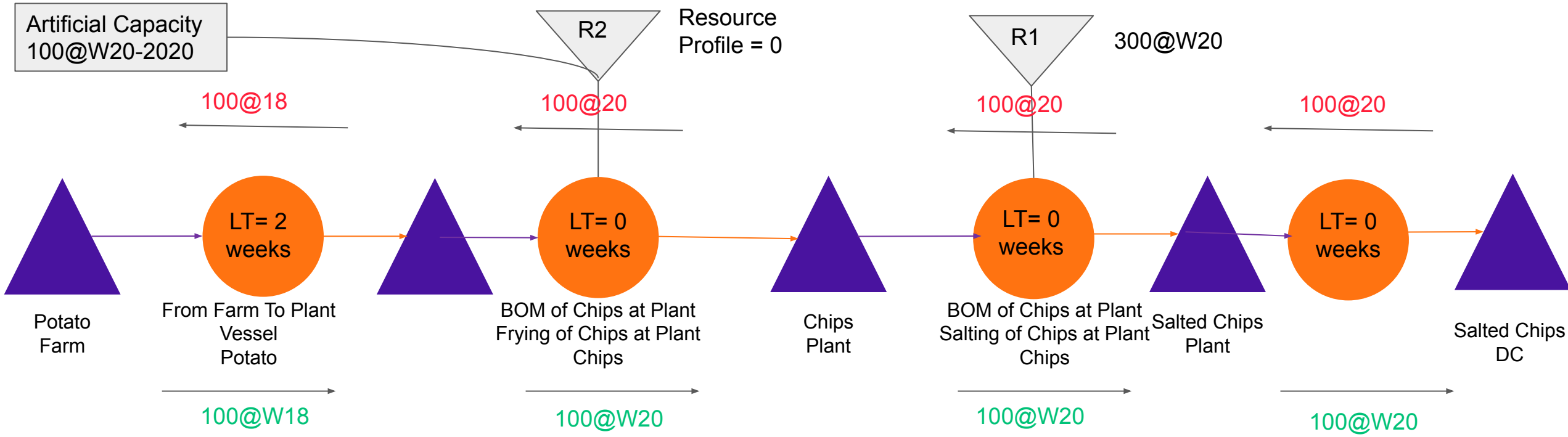
OBO RCA Planning for Order_5



Since the resource profile for R2@Plant is 0 and BLL = 0, the solver generates an artificial capacity of 100 for R2@Plant at W20-2020 to plan Order_5 on the JIT bucket, and records a capacity constraint for R2@Plant on W20-2020

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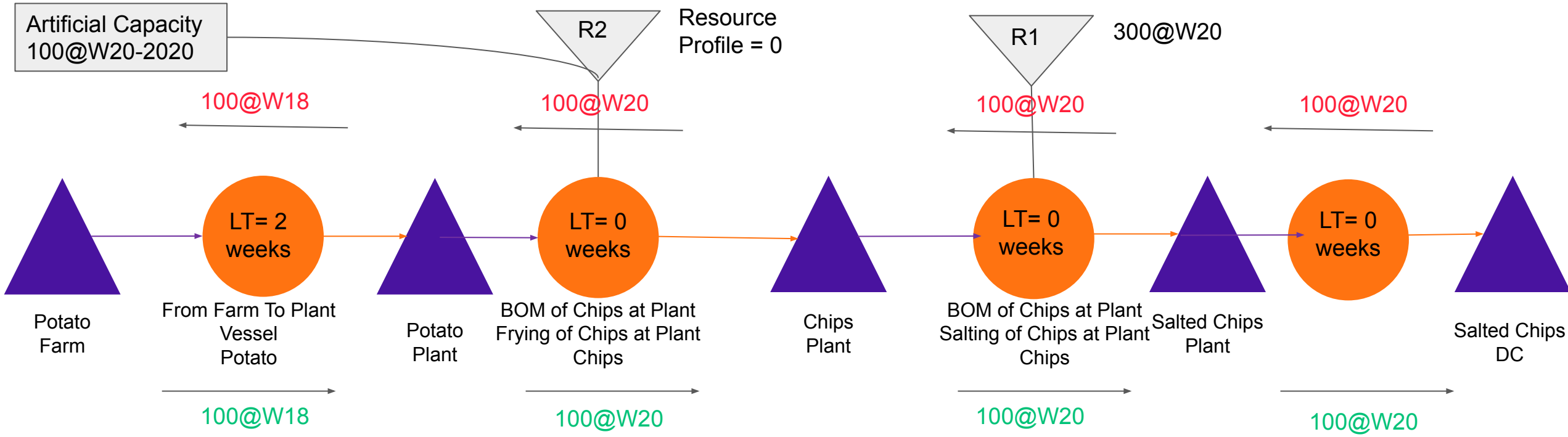
Additive RCA Planning for Order_2



While replanning for Order_2 as part of the Additive RCA, the solver runs into a resource constraint for R2@Plant. This creates an artificial capacity at R2@Plant for W18-2020 and records a resource constraint of 100 at R2@Plant for W18-2020 for Order_2.

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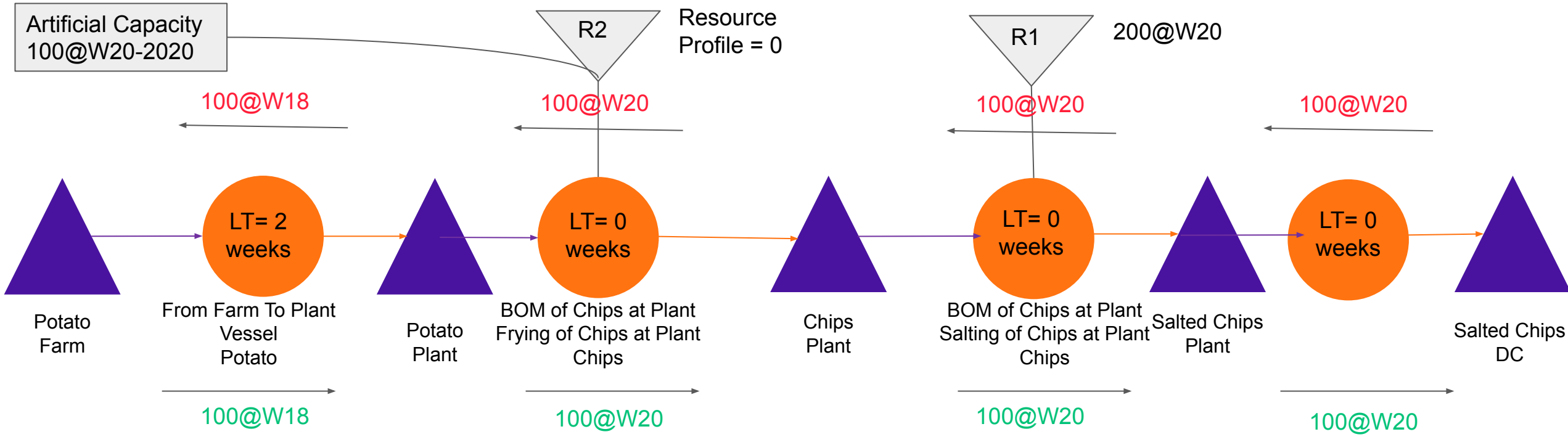
Additive RCA Planning for Order_3



Since Additive RCA retains the temporary flow plans, the resource profile for R1@Plant is 200 units on W20-2020. While replanning for Order_3 as part of the Additive RCA, the solver runs into a resource constraint for R2@Plant. This creates an artificial capacity at R2@Plant for W18-2020 and records a resource constraint of 100 at R2@Plant for W18-2020 for Order_3.

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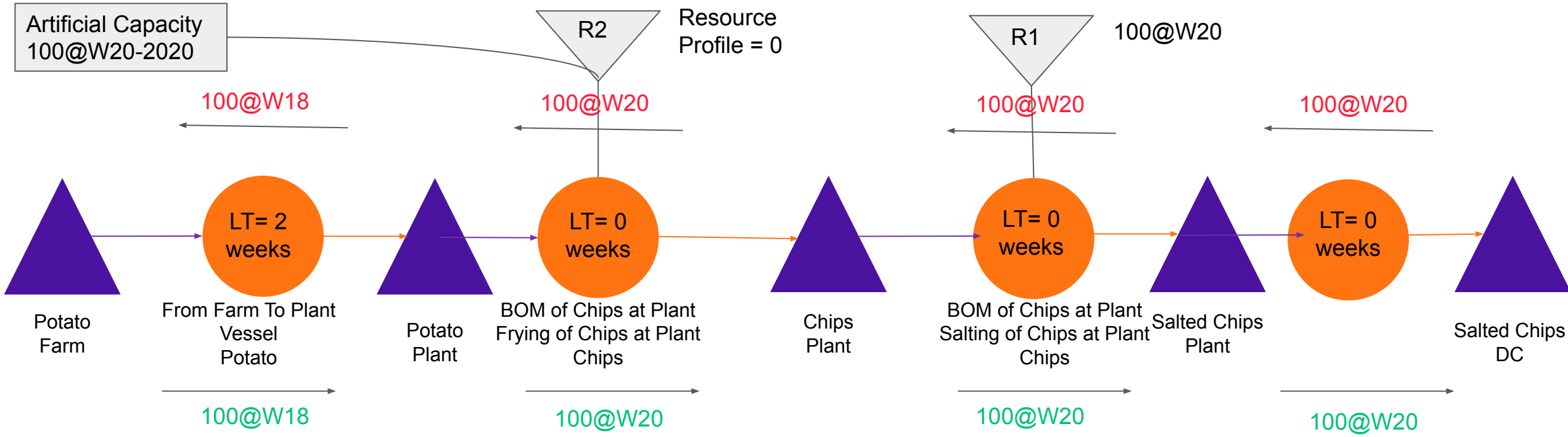
Additive RCA Planning for Order_4



Since Additive RCA retains the temporary flow plans, the resource profile for R1@Plant is 100 units on W20-2020. While replanning for Order_4 as part of the Additive RCA, the solver runs into a resource constraint for R2@Plant. This creates an artificial capacity at R2@Plant for W18-2020 and records a resource constraint of 100 at R2@Plant for W18-2020 for Order_4.

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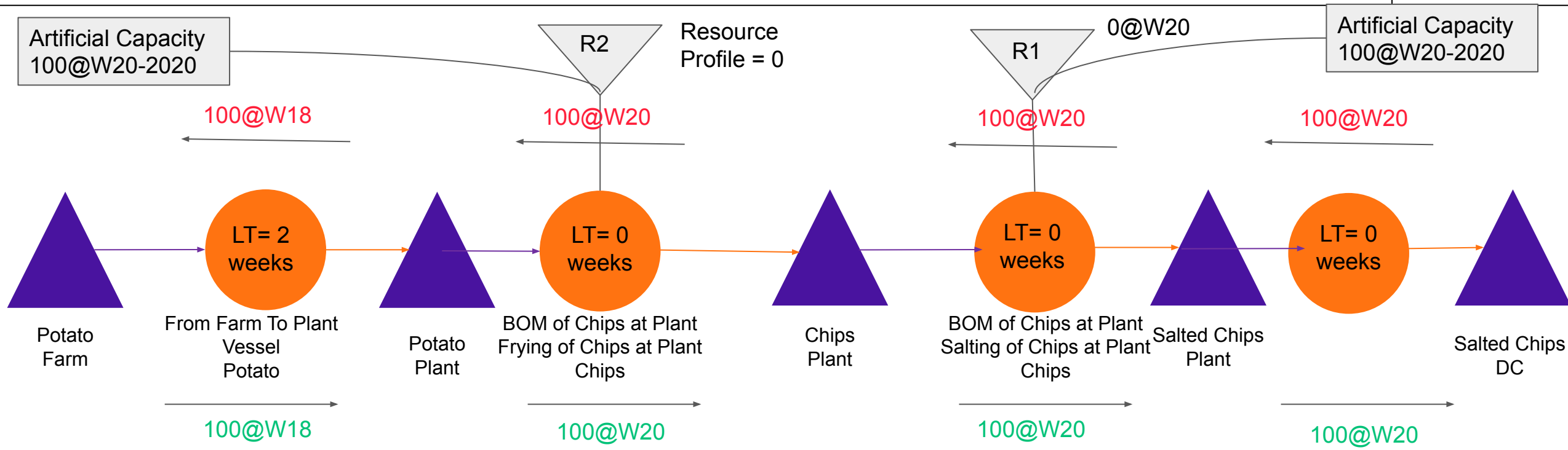
Additive RCA Planning for Order_5



Since Additive RCA retains the temporary flow plans, the resource profile for R1@Plant is 0 units on W20-2020. While replanning for Order_5 as part of the Additive RCA, the solver runs into a resource constraint for R2@Plant. This creates an artificial capacity at R2@Plant for W18-2020 and records a resource constraint of 100 at R2@Plant for W18-2020 for Order_4. Similarly since the resource profile for R1@Plant is also 0, the solver generates an artificial capacity for R1@Plant of 100 units and records a capacity constraint of 100 units on W20-2020 for Order_5.

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Additive RCA vs OBO RCA explanation :-

The OBO RCA does not retain the temporary flow plans while the Additive RCA does. This behaviour causes the difference in the OBO & Additive RCA values seen here.

During the OBO RCA, the solver performs the following steps:-

1. While replanning for the orders Order_2, Order_3, Order_4 & Order_5; the solver encounters a constraint for the resource R1 and creates an artificial capacity of 100 for each demand and records a constraint.

During the Additive RCA, the solver performs the following steps:-

1. While replaning for Order_2, the solver encounters a constraint for the resource R1 and creates an artificial capacity of 100 for each demand and records a constraint. Since there is sufficient capacity at the resource R2 there is no capacity constrained recorded here.
2. While replaning for Order_3 & Order_4, the solver encounters a constraint for the resource R1 and creates an artificial capacity of 100 for each demand and records a constraint. Since there is sufficient capacity at the resource R2 there is no capacity constrained recorded here, however since additive RCA retains the temporary flow plans the resource profile keeps decreasing for R2.
3. While replaning for Order_5, the solver encounters a constraint for the resource R1 and creates an artificial capacity of 100 for each demand and records a constraint. Since the resource profile for R2 is now 0, the solver creates an artificial capacity of 100 for each demand and records a resource constraint. This causes the difference in the OBO & Additive RCA reports.

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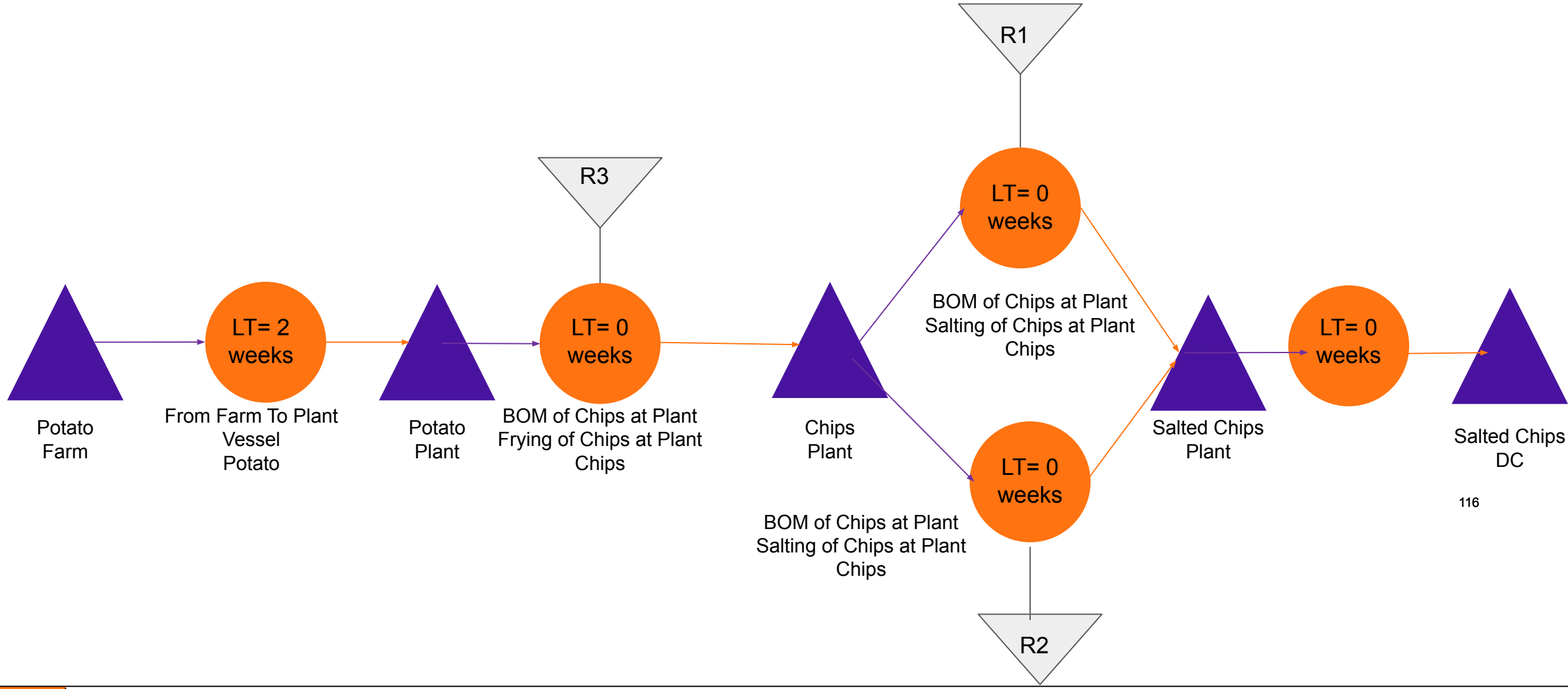
OBO vs Additive RCA - Late RCA



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Demand Details :-

Demand Item	Demand Location	Demand Week	Demand ID	Customer ID	Demand Priority	Demand Quantity	BLL
Salted Chips	DC	W20-2020	Order_1	Customer1	1	100	0
Salted Chips	DC	W20-2020	Order_2	Customer1	2	100	0
Salted Chips	DC	W20-2020	Order_3	Customer1	3	100	0

Demand Item	Demand Location	Demand Week	Demand ID	Customer ID	Demand Priority	Demand Quantity	BLL	Met On Time	Met Late	Met Short
Salted Chips	DC	W20-2020	Order_1	Customer1	1	100	0	100	0	0
Salted Chips	DC	W20-2020	Order_2	Customer1	2	100	0	100	0	0
Salted Chips	DC	W20-2020	Order_3	Customer1	3	100	0	0	0	100

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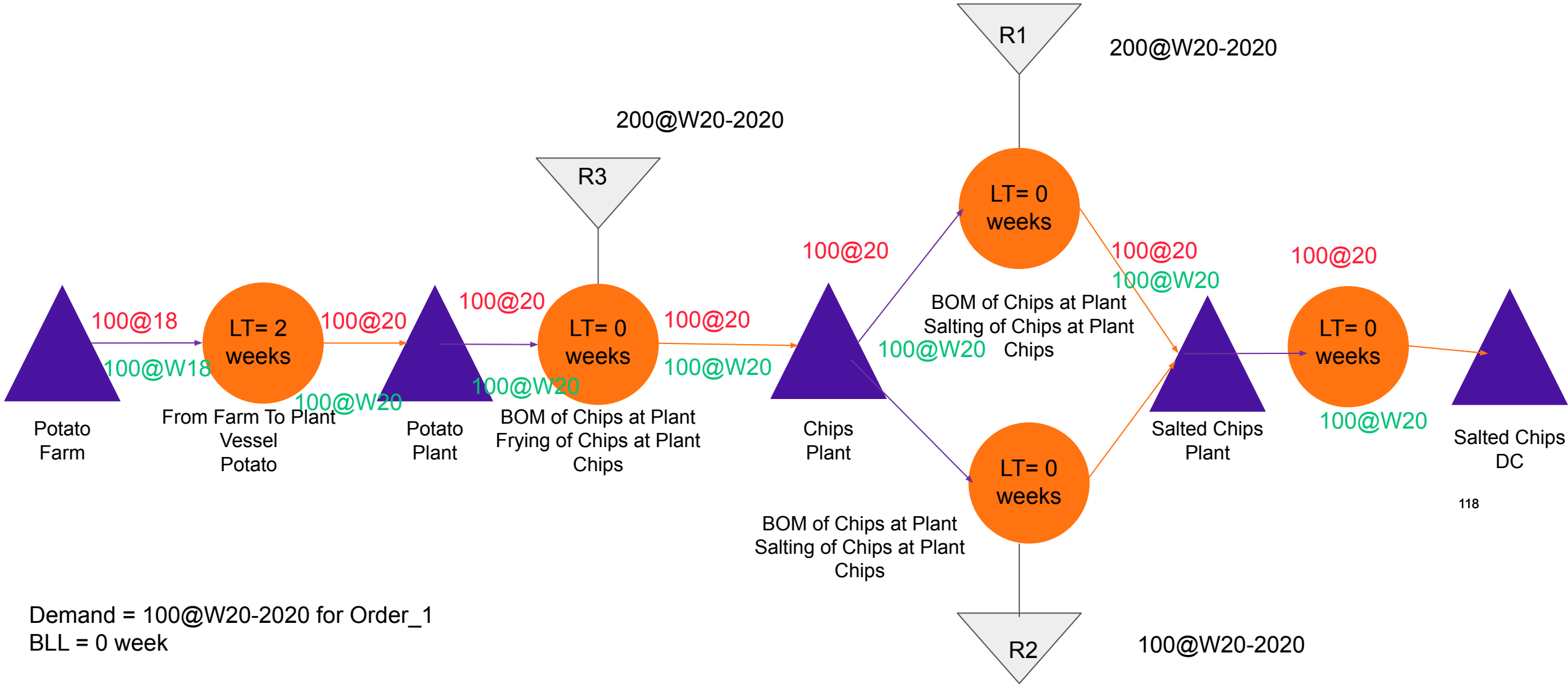
Demand Planning for Order_1



Demand = 100@W20-2020 for Order_1
BLL = 0 week

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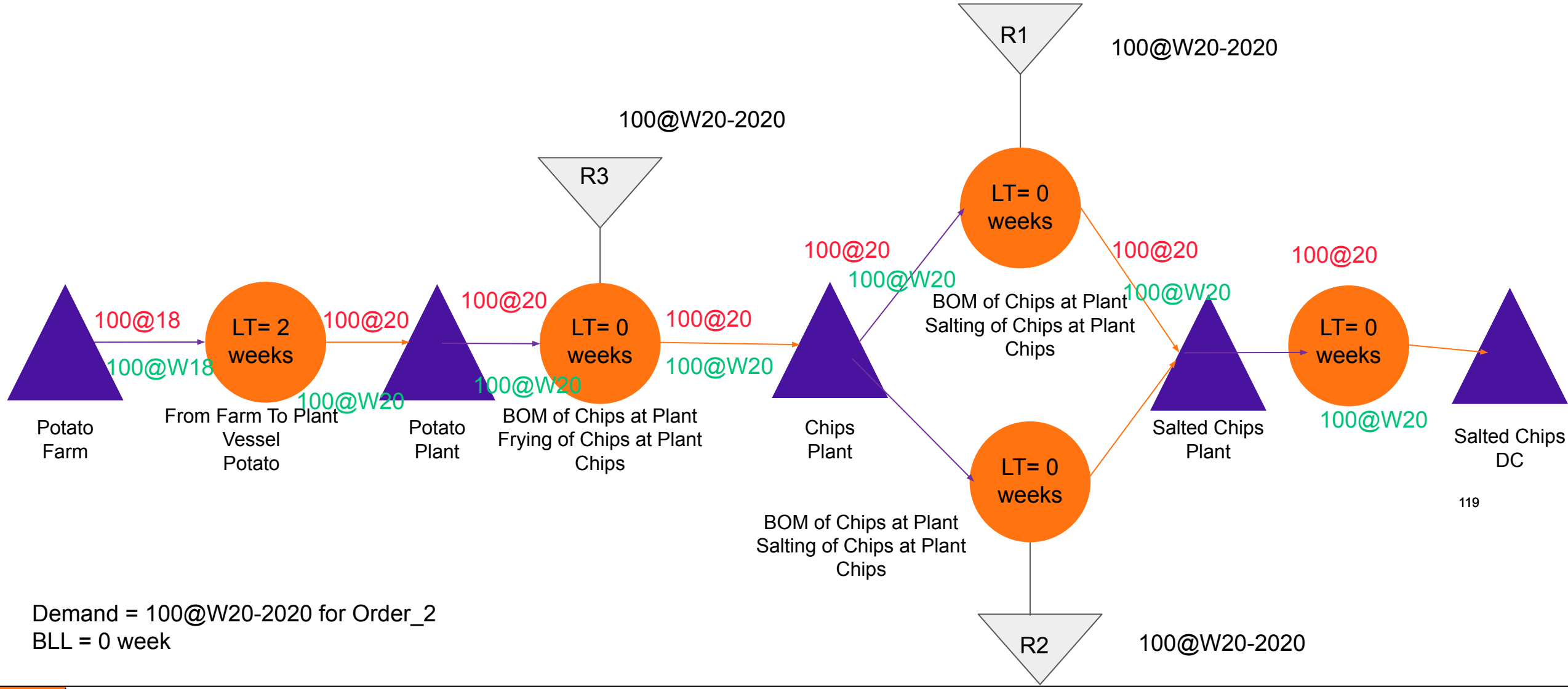
Demand Planning for Order_2



Demand = 100@W20-2020 for Order_2
BLL = 0 week

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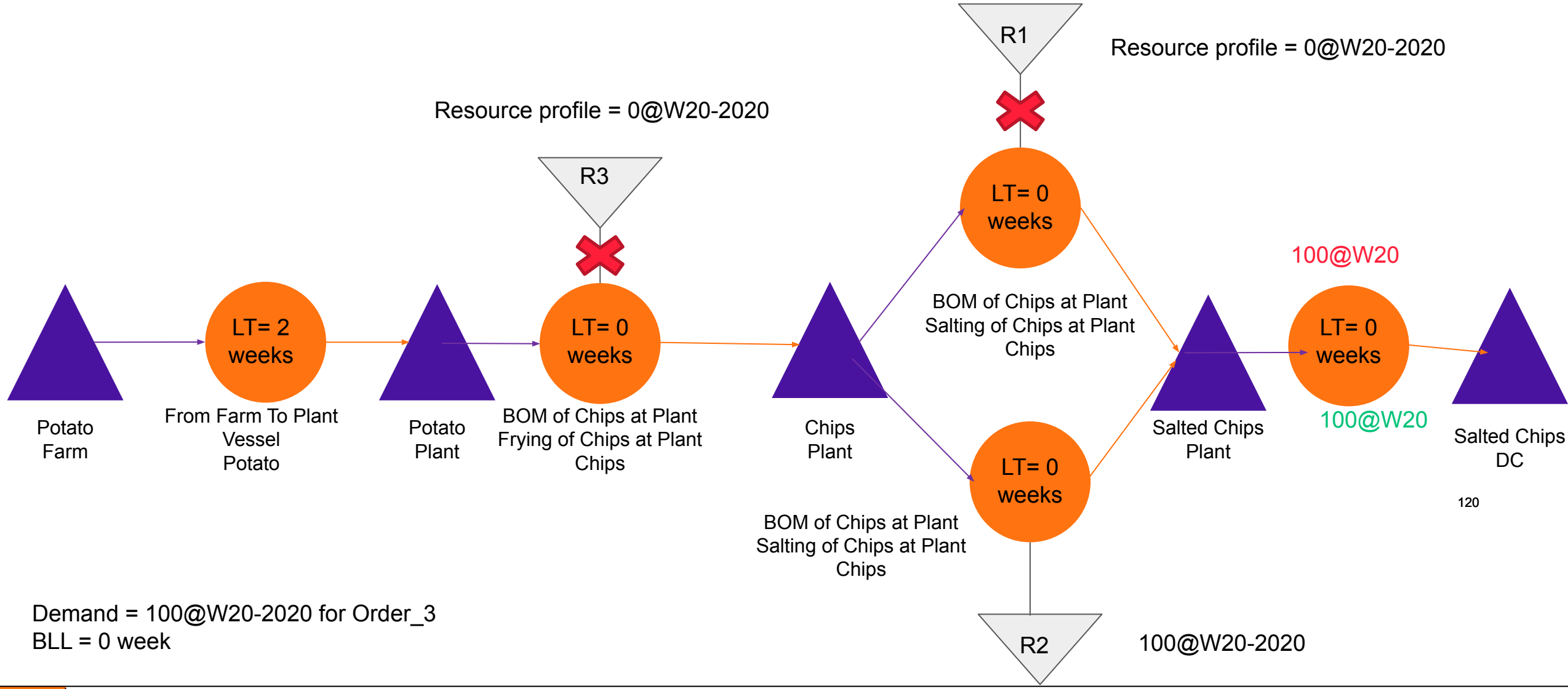
Demand Planning for Order_3



Demand = 100@W20-2020 for Order_3
BLL = 0 week

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Demand Planning Phase

1. Order_1 -> The resource R1 gets consumed for salting activity and R3 gets consumed for frying activity.
2. Order_2 -> The resource R1 gets consumed for salting activity and R3 gets consumed for frying activity.
3. Order_3 -> R1 and R3 both have a resource profile of 0 in W20-2020. Since BLL = 0, the demand is shorted.

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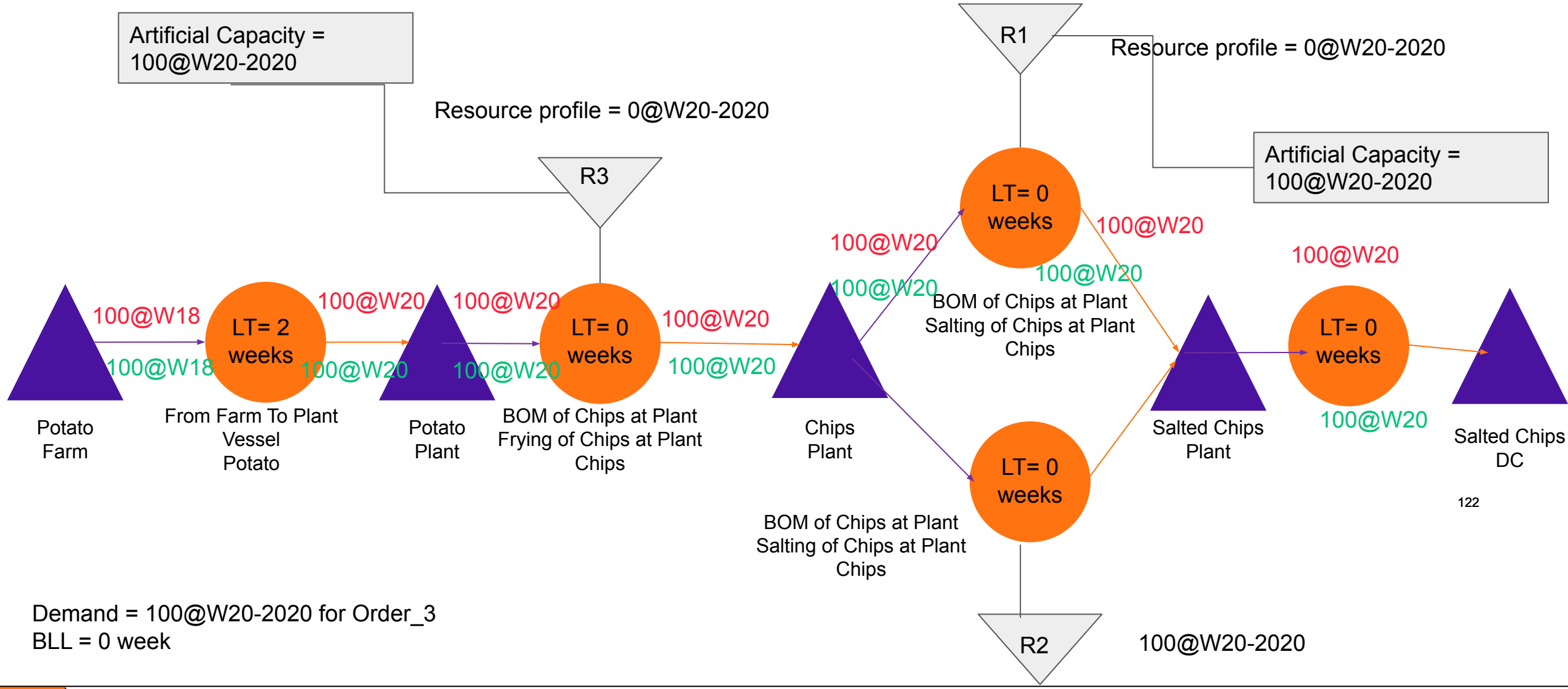
OBO RCA for Order_3



Demand = 100@W20-2020 for Order_3
BLL = 0 week

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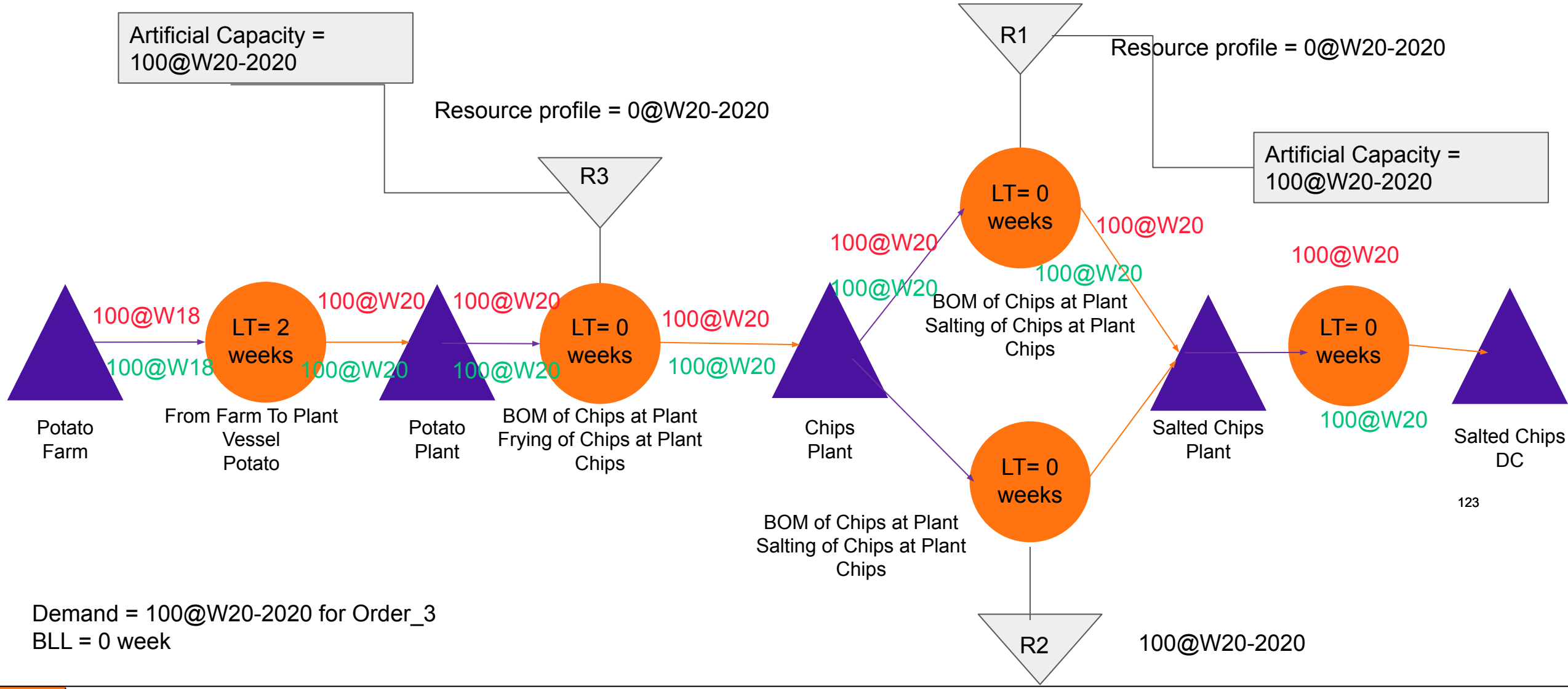
Additive RCA for Order_3



Demand = 100@W20-2020 for Order_3
BLL = 0 week

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RCA explanation for Order_3

1. The resource profile for R1 during the salting operation is 0 for W20-2020. Since BLL = 0, the solver creates an artificial capacity of 100 units for R1 in W20-2020 and records a capacity constraint against it.
2. The resource profile for R3 during the frying operation is 0 for W20-2020. Since BLL = 0, the solver creates an artificial capacity of 100 units for R3 in W20-2020 and records a capacity constraint against it.
3. Since there are no linked constraints involved, OBO & Additive RCA is same for both.

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RCA Output Data

- OBO RCA output :-

From Resource	From Location	FromWeek	To Item	To Location	To Week	Customer	Demand ID	Resource Constraint Qty
R1	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_3	100
R3	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_3	100

- Additive RCA Output :-

From Resource	From Location	FromWeek	To Item	To Location	To Week	Customer	Demand ID	Resource Constraint Qty
R1	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_3	100
R3	Plant1	W20-2020	Chips	DC	W20-2020	Customer_1	Order_3	100

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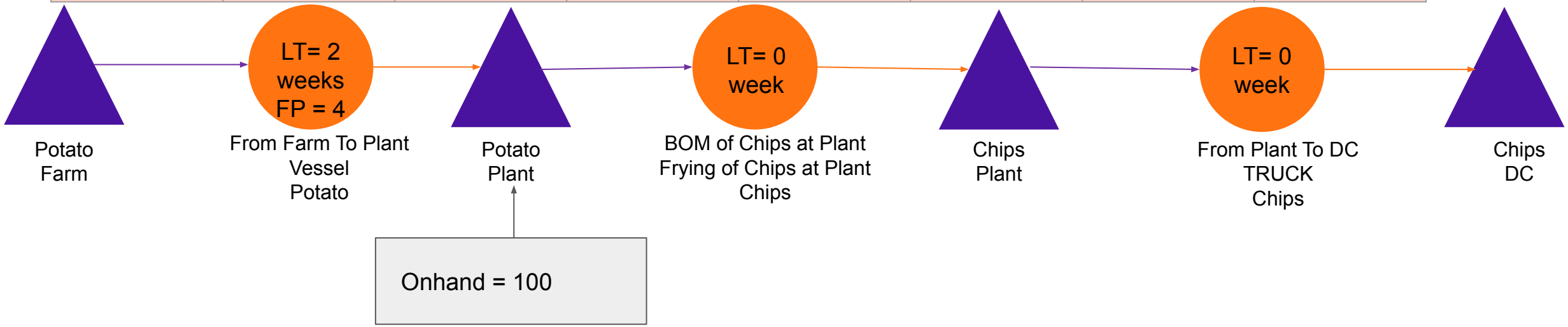
Additive RCA - profile based multiple Problem Types and Constrain buckets

Consider the below network with the following details :- Current bucket = W18

Demand Item	Demand Location	Demand Week	Demand ID	Customer ID	Demand Quantity	Demand Priority	BLL
Chips	DC	W20-2020	Order_1	Customer1	30	1	0
Chips	DC	W19-2020	Order_2	Customer1	70	2	0
Chips	DC	W18-2020	Order_3	Customer1	100	3	0



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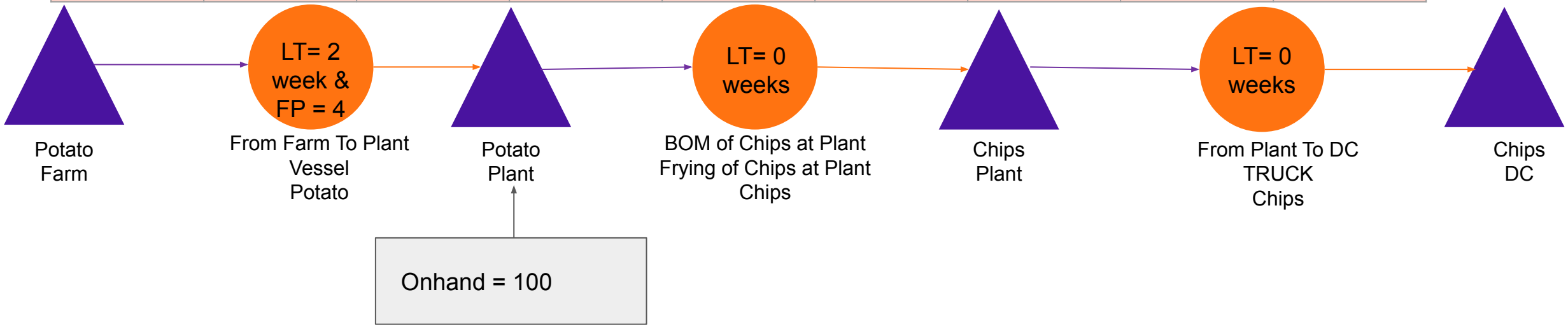
Additive RCA - profile based multiple Problem Types and Constrain buckets

The Demand Supportability picture after the plan run will be

Demand Item	Demand Location	Demand Week	Demand ID	Customer ID	Demand Quantity	Demand Priority	Met On Time	Short
Chips	DC	W20-2020	Order_1	Customer1	30	1	30	
Chips	DC	W19-2020	Order_2	Customer1	70	2	70	
Chips	DC	W18-2020	Order_3	Customer1	100	3		100



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Additive RCA - profile based multiple Problem Types and Constrain buckets

- The inventory plan picture for Potato@Plant will be

Item	Location	Data	W18	W19	W20	W21
Potato	Plant	BOH	100	100	30	0
		Demand	0	70	30	0
		Supply	0	0	0	0
		EOH	100	30	0	0
		RCA		70	30	

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Additive RCA - profile based multiple Problem Types and Constrain buckets

- The inventory profile during the additive RCA phase

Item	Location	Data	W18	W19	W20	W21
Potato	Plant	BOH	100	0	-70	0
		Demand	100	70	30	0
		Supply	0	0	0	0
		EOH	0	-70	-30	0
		RCA		70(LT)	30(FP)	

Frozen period and LTs are added together.

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RCA Output Data

- Additive RCA Output :-

Constraint Item	Constraint Location	Constraint Week	Problem Type	To Item	To Location	To Week	Customer	Demand ID	Material Constraint Qty
Potato	Plant	W19	Lead Time	Chips	DC	W18-2020	Customer_1	Order_3	70
Potato	Plant	W20	Frozen Period	Chips	DC	W18-2020	Customer_1	Order_3	30

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